HOR 101
FUNDAMENTALS OF HORTICULTURE

Lecture-wise Theory Study Material

All 33 Lectures, in Syllabus Order

B.Sc. (Hons) Agriculture
First Year • First Semester
2025 Syllabus • Tamil Nadu Agricultural University

This book follows the official HOR 101 lecture sequence. Each lecture is a self-contained chapter with headed sections, comparison tables, worked numerical examples, labelled diagrams and an Exam Focus block. Sources are listed in plain prose at the end of the book.

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How to Use This Book

This book is organised strictly lecture by lecture, matching the HOR 101 syllabus order. Each lecture is a self-contained chapter. The layout of every chapter is the same, so the reader knows where to look.

- **Learning outcomes** at the start of each lecture state what the reader should be able to do after studying it.
- **Headed sections** (1.1, 1.2 and so on) expand each sub-topic of the syllabus line.
- **Tables** compare crops, methods, systems and materials for quick revision.
- **Diagrams** (box art) show structures, layouts and cycles.
- **Worked examples** show numerical calculations step by step - plant population, fertilizer dose, ppm preparation, density and economics.
- **Exam focus** at the end of each lecture gives key points, short-answer questions with model answers, and long-answer questions with answer outlines.
- **Glossary and reference list** at the end explain terms and list standard texts.

How to Prepare for the Examination

1. Read the learning outcomes first, then the sections, then the tables and diagrams.
2. Practise the worked examples with the numbers changed.
3. Write the short-answer model answers from memory.
4. Use the long-answer outlines to structure full answers.
5. Revise the key points of all lectures in the last week.

A Note on Figures

Area, production, export and storage figures change every year and vary with variety and source. Where a precise value is not certain, a standard range has been given. For examinations and for field use, always quote the latest official statistics with the year and the source. Variety names follow those released by TNAU, ICAR and the State Department of Horticulture.

Syllabus at a Glance

No.	Lecture topic	No.	Lecture topic
1	Definition, history, scope, branches, nutritive value	18	Training in fruit crops
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8	Dormancy, invigoration and seed treatment	25	Mulching
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10	Cutting and layering	27	Bearing habits of fruit trees
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12	Specialized plant parts in propagation	29	Fertilization and parthenocarpy

No.	Lecture topic	No.	Lecture topic
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14	Micropropagation	31	Maturity and methods of judging maturity
15	Vegetable gardens and special systems	32	Post-harvest: harvesting, grading, packing
16	Orchard planning, layout and planting systems	33	Transport and storage
17	High density and ultra high density planting		

The 33 lectures of HOR 101 in order

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Lecture 1. Definition, History, Scope and Importance, Branches of Horticulture and Nutritive Value of Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define horticulture and explain its origin and meaning.
- List and describe the main branches of horticulture.
- Trace the history of horticulture in India in its four phases.
- Explain the scope and importance of horticulture with examples.
- State the nutritive value of major fruits and vegetables.

Horticulture is the science and art of cultivating fruits, vegetables, flowers, spices, plantation crops and ornamental plants. The word comes from the Latin *hortus* (garden) and *cultura* (cultivation), so horticulture literally means garden cultivation. It is an intensive form of agriculture in which crops of high value are grown on relatively small areas with careful management, and in which produce is often perishable and needs special handling after harvest.

1.1 Definition and Meaning

Horticulture is defined as the branch of agriculture concerned with the production, processing, marketing and utilisation of fruits, vegetables, flowers, ornamental plants, spices, medicinal and aromatic plants, and plantation crops. In the classical sense it means the cultivation of plants in a protected or intensively managed garden.

- Latin origin: *hortus* = garden, *cultura* = cultivation.
- It is an intensive system: high inputs, high returns, small holdings.
- It deals largely with perishable produce, so pre- and post-harvest care is central.
- It combines science (physiology, genetics, breeding, pathology) with art (design, layout, aesthetics).

1.2 Branches of Horticulture

Horticulture is divided into several branches based on the type of crop and the purpose of cultivation. The main branches are given below.

Branch	Crop group dealt with	Examples
Pomology	Fruit science: production of fruit crops	Mango, banana, citrus, guava, apple
Olericulture	Production of vegetables	Tomato, brinjal, bhendi, cabbage, onion
Floriculture	Commercial flower production	Rose, jasmine, chrysanthemum, marigold
Arboriculture	Culture of trees and shrubs (often woody ornamentals)	Avenue trees, Ficus, ornamental shrubs
Landscape horticulture	Design and beautification of outdoor and indoor spaces	Gardens, lawns, parks, indoor plants
Plantation crops	Crops grown on large estates, mostly perennials	Coconut, tea, coffee, rubber, cashew
Spices and condiments	Aromatic crops used for flavouring	Pepper, cardamom, turmeric, chilli

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Branch	Crop group dealt with	Examples
Medicinal and aromatic plants	Plants for drugs, perfumes and essential oils	Isabgol, senna, lemon grass, palmarosa
Post-harvest technology	Handling, processing and preservation of produce	Grading, packaging, drying, canning

Table 1.1 Branches of horticulture and their scope

1.3 History of Horticulture

Horticulture is as old as settled agriculture. Its development in India can be traced through four broad phases.

- **Ancient and Vedic period:** Cultivation of mango, banana, date, coconut and many vegetables is described in Sanskrit texts and in the works of Kautilya (Arthashastra). Gardens and orchards were part of temple and royal life.
- **Medieval period:** Mughal emperors developed formal gardens; the Shalimar and Nishat gardens of Kashmir and the Taj Mahal garden are examples. Fruit introduction and grafting practices spread widely.
- **Colonial period:** Botanical gardens were established (for example, the Indian Botanic Garden, Howrah, 1787). Tea, coffee and rubber plantations began; organised fruit and vegetable research started in the early twentieth century.
- **Post-independence period:** Systematic research and development through ICAR institutes, State Agricultural Universities and Horticulture Departments; the National Horticulture Mission and later the Mission for Integrated Development of Horticulture (MIDH) gave a strong push after 2000.

1.4 Scope and Importance

Horticulture occupies a small share of the cropped area but contributes a large share of the value of agricultural output. Its importance is economic, nutritional and social.

- **High returns per unit area:** Fruits, vegetables and flowers give far higher gross income per hectare than most cereals.
- **Employment generation:** Horticulture is labour-intensive in planting, harvesting, grading and packing; it supports rural employment.
- **Nutritional security:** Fruits and vegetables supply vitamins, minerals and fibre that cereals lack.
- **Raw material:** Supplies the food-processing, canning, beverage and cosmetic industries.
- **Foreign exchange:** Fruits, vegetables, spices, cashew, tea and coffee are important exports.
- **Land use:** Suits small and marginal farms, wastelands, hilly and rain-fed areas, and kitchen gardens.
- **Environmental and aesthetic value:** Plantation crops and landscaping improve environment and quality of life.

In Tamil Nadu, horticulture is a leading sector: the state is a major producer of banana, mango, coconut, tapioca, tomato, onion and flowers. Districts such as Krishnagiri (mango), Theni and Dindigul (banana, vegetables), the Nilgiris (vegetables and flowers) and Erode (turmeric, coconut) are well known.

1.5 Nutritive Value of Horticultural Crops

Fruits and vegetables are called protective foods because they protect the body against deficiency diseases. Their main contributions are vitamins (especially A and C), minerals, dietary fibre, antioxidants and small amounts of energy. The table gives the approximate composition of important crops; values are typical ranges and vary with variety, soil and maturity.

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Crop	Energy (kcal)	Carbohydrate (g)	Protein (g)	Vitamin C (mg)	Carotene / Vitamin A	Notable mineral
Mango	~60-70	~16-17	0.6	~20-30	High (beta-carotene)	Potassium
Banana	~90-110	~23-27	1.1	~8-10	Low-moderate	Potassium, magnesium
Guava	~50-70	~11-14	0.9	Very high (~200+)	Moderate	Vitamin C richest fruit
Citrus (orange)	~45-50	~11	0.9	~50	Low-moderate	Potassium, calcium
Papaya	~40-45	~10	0.6	~55-60	High (beta-carotene)	Potassium
Apple	~50-60	~13	0.3	~4-5	Low	Potassium
Tomato	~20-25	~4	0.9	~20-25	High (lycopene, carotene)	Potassium
Carrot	~35-45	~9-10	0.9	~5-8	Very high (carotene)	Calcium
Green leafy vegetables	~25-50	~3-6	2-4	Moderate-high	Very high	Iron, calcium
Potato	~70-80	~17-20	1.6	~15-20	Very low	Potassium
Onion	~30-40	~9-11	1.1	~5-10	Low	Phosphorus, calcium
Bitter gourd / bhendi	~20-40	~4-7	1.5-2.0	Moderate-high	Moderate	Iron, calcium

Table 1.2 Approximate nutritive value of major fruits and vegetables (per 100 g edible portion)

Points to remember from the table: guava and amla are outstanding sources of vitamin C; mango, papaya and carrot are rich in carotene (provitamin A); green leafy vegetables supply iron and calcium; banana and potato supply energy and potassium. This is why the Indian dietary guidelines recommend at least 300-400 g of fruits and vegetables per person per day.

1.6 Horticulture and Agriculture - How They Differ

Horticulture is a part of agriculture, but it differs in intensity, produce and management. The table makes the distinction clear.

Point	Horticulture	Agriculture (agronomy)
Crops	Fruits, vegetables, flowers, plantation, spices	Cereals, millets, pulses, oilseeds
Intensity	Intensive; high inputs per unit area	Extensive; lower inputs per unit area
Plot size	Small to medium	Large
Produce	Mostly perishable	Mostly non-perishable (grains)
Post-harvest	Special handling, cold chain needed	Simple drying and storage
Returns	High value per unit area	Lower value per unit area
Skill	High; skilled operations (grafting, pruning)	Moderate

Table 1.3 Horticulture compared with agriculture (agronomy)

1.7 Present Status of Horticulture in India

- India is the second largest producer of fruits and vegetables in the world.

- Leading states in fruits: Andhra Pradesh, Maharashtra, Karnataka, Tamil Nadu, Gujarat.
- Leading states in vegetables: West Bengal, Uttar Pradesh, Madhya Pradesh, Bihar, Maharashtra.
- Leading states in flowers: Tamil Nadu, Karnataka, West Bengal, Andhra Pradesh.
- Horticulture gives a large share of agricultural output value from a small share of cropped area.
- The sector supports nutrition, employment, processing and exports.

1.8 Career and Enterprise Opportunities in Horticulture

- Orchard and nursery enterprise, floriculture unit and protected cultivation.
- Horticultural officer and extension roles in the State Department of Horticulture.
- Food processing, grading, packing and cold-chain enterprises.
- Landscape design and maintenance services.
- Seed and planting-material production and certification.
- Teaching, research and farm advisory services.

EXAM FOCUS

Key points to remember

- Horticulture = *hortus* (garden) + *cultura* (cultivation); intensive cultivation of fruits, vegetables, flowers and plantation crops.
- Main branches: pomology, olericulture, floriculture, arboriculture, landscape, plantation crops, spices, medicinal and aromatic plants, post-harvest technology.
- Four historical phases: ancient/Vedic, medieval (Mughal gardens), colonial (botanical gardens, plantations), post-independence (ICAR, SAUs, MIDH).
- Fruits and vegetables are protective foods rich in vitamins A and C, minerals and fibre.
- Guava and amla are the richest common sources of vitamin C; mango, papaya and carrot are rich in carotene.
- Horticulture gives high returns per hectare, high employment and important export earnings.
- Tamil Nadu is a leading horticulture state: banana, mango, coconut, tapioca, tomato, onion and flowers.

Short-answer questions

Q1. Define horticulture and state its origin of the word.

Model answer: Horticulture is the science and art of cultivating fruits, vegetables, flowers, ornamental and plantation crops intensively. The word is derived from Latin *hortus* (garden) and *cultura* (cultivation).

Q2. Name any four branches of horticulture with one example each.

Model answer: Pomology (mango), olericulture (tomato), floriculture (jasmine), plantation crops (coconut).

Q3. Why are fruits and vegetables called protective foods?

Model answer: Because they are rich in vitamins (A, C), minerals and fibre that protect the body against deficiency diseases, though they supply little energy.

Long-answer questions

Q1. Discuss the scope and importance of horticulture in India with reference to Tamil Nadu.

Answer outline:

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1. Meaning and branches of horticulture in one or two lines.
2. Economic importance: high returns per hectare, employment, processing industry.
3. Nutritional importance: protective foods, vitamins and minerals.
4. Export importance: fruits, vegetables, spices, cashew, tea, coffee.
5. Land use: small farms, hilly areas, kitchen gardens.
6. Tamil Nadu examples: Krishnagiri mango, Nilgiris vegetables and flowers, Theni and Dindigul banana, Erode turmeric.
7. Conclusion: horticulture is a growth engine for income and nutrition.

Q2. Explain the nutritive value of horticultural crops with suitable examples.

Answer outline:

1. Definition of protective foods.
2. Energy and carbohydrate contribution: banana, potato.
3. Vitamin C: guava, amla, citrus.
4. Provitamin A (carotene): mango, papaya, carrot, green leafy vegetables.
5. Minerals: iron and calcium in leafy vegetables.
6. Dietary guidelines: 300-400 g fruits and vegetables per day.
7. Conclusion: role in combating hidden hunger.

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Lecture 2. Classification of Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Explain why horticultural crops are classified.
- Classify crops botanically, by climate, by life cycle, by edible part and by use.
- Give family names and examples for important crops.
- Differentiate loose flowers from cut flowers.

Classification means grouping crops that share common characters so that they can be studied, managed and recommended together. Horticultural crops are classified in several ways: botanically, by climatic requirement, by life cycle, by the edible part used, by cultural requirement and by use. No single system is complete; the systems are used together.

2.1 Botanical Classification

Botanical classification places crops in families, genera and species based on flower and fruit structure. It is the most scientific system and is important for breeding, grafting compatibility and disease management.

Family	Common name of family	Important horticultural crops
Anacardiaceae	Cashew family	Mango, cashew
Rutaceae	Citrus family	Orange, lemon, lime, mandarin, acid lime
Musaceae	Banana family	Banana, plantain
Myrtaceae	Myrtle family	Guava, clove
Sapotaceae	Sapota family	Sapota, star apple
Rosaceae	Rose family	Apple, pear, peach, plum, rose, strawberry
Solanaceae	Nightshade family	Tomato, brinjal, chilli, potato
Cucurbitaceae	Gourd family	Cucumber, pumpkin, bitter gourd, watermelon
Fabaceae	Legume family	Pea, bean, cowpea
Malvaceae	Mallow family	Bhendi (okra) and cotton relatives
Brassicaceae	Mustard family	Cabbage, cauliflower, radish, knol-khol
Amaryllidaceae / Alliaceae	Onion family	Onion, garlic
Asteraceae	Sunflower family	Chrysanthemum, marigold, lettuce
Arecaceae	Palm family	Coconut, arecanut, oil palm, date palm
Zingiberaceae	Ginger family	Turmeric, ginger, cardamom

Table 2.1 Botanical classification of important horticultural crops

2.2 Classification Based on Climatic Requirement

This is the most useful classification for deciding where a crop can be grown.

Group	Climate	Characteristics	Examples
Temperate fruits	Cool winters with chilling below about 7 deg C; distinct seasons	Deciduous; need winter chilling to break dormancy	Apple, pear, peach, plum, apricot, cherry, strawberry, walnut

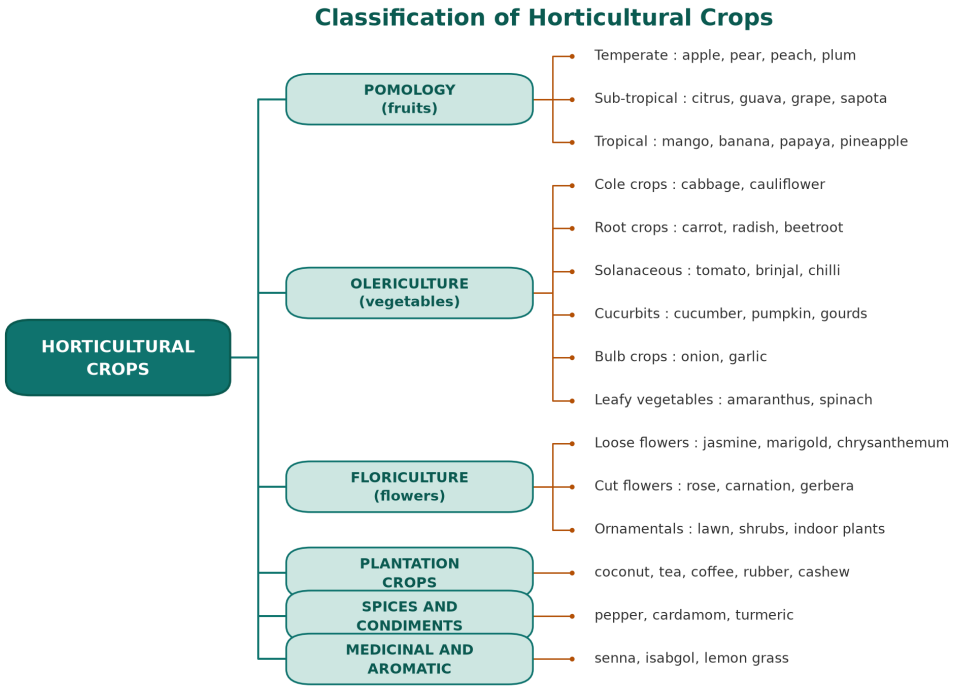


Figure 2.1 Classification of horticultural crops

Tree diagram: fruits, vegetables, flowers, spices, plantation and medicinal crops.

Group	Climate	Characteristics	Examples
Sub-tropical fruits	Mild winters, warm summers, no severe frost	Mostly evergreen; tolerate light cold	Citrus, guava, grape, sapota, fig, pomegranate, litchi
Tropical fruits	Hot humid or hot dry climate; frost-free	Evergreen; sensitive to frost and low temperature	Mango, banana, papaya, pineapple, coconut, cashew, jackfruit, avocado

Table 2.2 Temperate, sub-tropical and tropical fruits

2.3 Classification Based on Life Cycle

- **Annuals:** Complete life cycle in one season - most vegetables (tomato, brinjal, bhendi), some flowers (marigold).
- **Biennials:** Complete life cycle in two seasons, usually storing food in the first year - cabbage, cauliflower, carrot, beetroot, onion.
- **Perennials:** Live for many years - fruit trees (mango, coconut, guava), plantation crops, and perennial ornamentals.

2.4 Classification Based on Edible Part Used

Edible part	Vegetables
Root	Carrot, radish, beetroot, turnip, tapioca
Stem / tuber	Potato, sweet potato, colocasia, yam, ginger, turmeric
Bulb	Onion, garlic, leek
Leaf	Amaranthus, spinach, palak, coriander, methi, cabbage
Flower / flower bud	Cauliflower, broccoli, banana flower
Fruit (immature)	Tomato, brinjal, bhendi, cucumber, beans, drumstick
Fruit (mature)	Pumpkin, ash gourd, bitter gourd, watermelon, muskmelon
Seed / pod	Pea, bean, cowpea

Table 2.3 Classification of vegetables by edible part

2.5 Classification of Flowers

- **Loose flowers:** Sold by weight for garlands, decoration and worship - jasmine, chrysanthemum, marigold, crossandra, tube rose.
- **Cut flowers:** Sold by number with a stem, for bouquets and arrangements - rose, carnation, gerbera, gladiolus, anthurium, orchid.
- **Cut foliage:** Ferns and decorative leaves used in arrangements.
- **Ornamentals for garden:** Lawn grasses, hedge plants, shrubs, trees, climbers.

2.6 Classification Tree

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HORTICULTURAL CROPS
|
+-- POMOLOGY (fruits)
|     +-- Temperate : apple, pear, peach, plum
|     +-- Sub-tropical : citrus, guava, grape, sapota
|     +-- Tropical : mango, banana, papaya, pineapple
|
+-- OLERICULTURE (vegetables)
|     +-- Cole crops : cabbage, cauliflower
|     +-- Root crops : carrot, radish, beetroot
|     +-- Solanaceous : tomato, brinjal, chilli
|     +-- Cucurbits : cucumber, pumpkin, gourds
|     +-- Bulb crops : onion, garlic
|     +-- Leafy vegetables : amaranthus, spinach
|
+-- FLORICULTURE (flowers)
|     +-- Loose flowers : jasmine, marigold, chrysanthemum
|     +-- Cut flowers : rose, carnation, gerbera
|     +-- Ornamentals : lawn, shrubs, indoor plants
|
+-- PLANTATION CROPS : coconut, tea, coffee, rubber, cashew
|
+-- SPICES AND CONDIMENTS : pepper, cardamom, turmeric
|
+-- MEDICINAL AND AROMATIC : senna, isabgol, lemon grass
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Fig. 2.1 A simple classification tree for horticultural crops.

2.7 Classification Based on Use and Purpose

Use	Crops
Fresh consumption	Mango, banana, tomato, guava
Processing	Tomato (sauce), mango (pulp), citrus (juice), potato (chips), turmeric
Export	Mango, banana, grapes, onion, cut flowers, spices
Medicinal and aromatic	Senna, isabgol, lemon grass, palmarosa, aloe
Ornamental and landscape	Rose, jasmine, lawn grasses, shrubs, avenue trees
Spices and condiments	Pepper, cardamom, chilli, turmeric, ginger

Table 2.4 Classification of horticultural crops by use

2.8 Classification Based on Cultural Requirement

- **Irrigated horticulture:** Crops grown with assured irrigation - most fruits and vegetables.
- **Rain-fed horticulture:** Crops that tolerate dry conditions - ber, custard apple, aonla, some mango.
- **Protected cultivation:** Crops grown under polyhouse, shade net or greenhouse - capsicum, cucumber, exotic vegetables, cut flowers.
- **Plantation/estate crops:** Crops managed as large estates - tea, coffee, rubber, coconut, cashew.

2.9 Why Classification Matters

- Helps in crop planning for a given climate and soil.
- Guides breeding, as related species cross more easily.
- Helps in choosing rootstocks for grafting, based on family relationships.
- Helps in organising research, extension and statistics.
- Helps the grower choose crops with similar management needs.

EXAM FOCUS

Key points to remember

- Classification groups crops with common characters for study and management.
- Botanical classification uses family, genus and species - useful for breeding and grafting compatibility.
- Climatic classification into temperate, sub-tropical and tropical fruits is the most useful for site selection.
- Life-cycle classification: annual, biennial, perennial.
- Vegetables may be classified by edible part - root, stem, leaf, flower, fruit, seed.
- Flowers are classified as loose flowers and cut flowers.
- No single system is complete; systems are used together.

Short-answer questions

Q1. Classify fruits based on climatic requirement with examples.

Model answer: Temperate (apple, peach), sub-tropical (citrus, guava), tropical (mango, banana).

Q2. Give the botanical name of the family of mango, tomato and banana.

Model answer: Mango - Anacardiaceae; tomato - Solanaceae; banana - Musaceae.

Q3. Differentiate loose flowers from cut flowers.

Model answer: Loose flowers are sold by weight for garlands and worship (jasmine, marigold); cut flowers are sold by number with a stem for bouquets (rose, gerbera).

Long-answer questions

Q1. Explain the various systems of classification of horticultural crops.

Answer outline:

1. Need for classification.
2. Botanical classification with families and examples.
3. Climatic classification: temperate, sub-tropical, tropical.
4. Life-cycle classification: annual, biennial, perennial.
5. Classification of vegetables by edible part.
6. Classification of flowers: loose and cut.
7. Conclusion: systems are complementary.

Q2. Classify vegetables based on the edible part used, with two examples for each.

Answer outline:

1. Root - carrot, radish.
2. Stem/tuber - potato, colocasia.
3. Bulb - onion, garlic.
4. Leaf - amaranthus, cabbage.
5. Flower - cauliflower, broccoli.
6. Fruit - tomato, pumpkin.
7. Seed/pod - pea, bean.

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Lecture 3. Horticultural Zones of India and of Tamil Nadu, Area and Production Statistics

Learning Outcomes

- After studying this lecture the student should be able to:
- Define a horticultural zone.
- Name and describe the eight horticultural zones of India.
- Describe the seven zones of Tamil Nadu and their crops.
- State the approximate area and production of horticultural crops.

India is a large country with wide variation in climate and soil, so it is divided into horticultural zones so that crops and varieties can be matched to the environment. A horticultural zone is a region with broadly similar climate, soil and cropping pattern.

3.1 Horticultural Zones of India

The country is commonly divided into eight agro-climatic / horticultural zones. Each zone is suited to particular fruit, vegetable and plantation crops.

Zone	States covered	Climate	Important crops
1. Temperate / North-Western Himalayan	J&K;, Himachal Pradesh, Uttarakhand hills	Cold, temperate	Apple, pear, peach, plum, walnut, cherry
2. North-Western Plains	Punjab, Haryana, Delhi, Western UP, Rajasthan	Sub-tropical, dry	Citrus, guava, grapes, mango, potato, onion
3. North-Eastern	Assam, Meghalaya, Nagaland, Manipur, Tripura, Arunachal	Humid sub-tropical	Banana, pineapple, citrus, tea, ginger, chilli
4. Eastern / Lower Gangetic Plains	West Bengal, Bihar, Jharkhand, Eastern UP	Sub-tropical humid	Mango, litchi, banana, potato, vegetables
5. Central / Plateau	Madhya Pradesh, Chhattisgarh, parts of Maharashtra	Sub-tropical to tropical	Mango, guava, citrus, onion, garlic, chilli
6. Western / Deccan Plateau	Maharashtra, Karnataka, Telangana, Andhra	Tropical, semi-arid	Grapes, pomegranate, banana, citrus, tomato, onion
7. Southern / Coastal	Tamil Nadu, Kerala, coastal AP, Karnataka coast	Tropical, humid	Coconut, banana, mango, tapioca, spices, cashew
8. Islands	Andaman & Nicobar, Lakshadweep	Tropical, maritime	Coconut, arecanut, banana, spices, cashew

Table 3.1 Horticultural zones of India and their characteristic crops

3.2 Horticultural Zones of Tamil Nadu

Tamil Nadu is divided into seven agro-climatic zones, which serve as the practical horticultural zones of the state. The state's horticulture is diversified from hill vegetables to coastal coconut.

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Zone	Districts (typical)	Climate and soil	Important horticultural crops
North-Eastern	Kancheepuram, Chengalpattu, Vellore, Ranipet, Tiruvannamalai	Semi-arid, red and lateritic soils	Mango, sapota, groundnut-based, vegetables, flowers
North-Western	Dharmapuri, Krishnagiri, Salem, Namakkal	Semi-arid, red loam	Mango, tomato, onion, tapioca, flowers
Western	Erode, Coimbatore, Tiruppur, Nilgiris (part)	Sub-tropical to temperate in hills	Turmeric, coconut, banana, hill vegetables, flowers
Cauvery Delta	Thanjavur, Nagapattinam, Tiruvarur, Mayiladuthurai	Hot, humid, deltaic alluvium	Banana, coconut, vegetables, flowers
Southern	Madurai, Theni, Dindigul, Virudhunagar, Sivagangai	Semi-arid to sub-tropical	Banana, mango, grapes, vegetables, flowers
High Rainfall	Kanyakumari, Tirunelveli, Tenkasi	Humid, high rainfall	Rubber, banana, coconut, spices, tapioca
Hilly	The Nilgiris, Kodaikanal (Dindigul), Yercaud (Salem)	Temperate, cool	Potato, cabbage, carrot, beans, tea, cut flowers

Table 3.2 Horticultural zones of Tamil Nadu and their crops

The Nilgiris is the leading hill zone for temperate vegetables (potato, cabbage, carrot, beans) and cut flowers. Theni and Dindigul are important for banana, mango and grapes. Krishnagiri and Dharmapuri are famous for mango and tomato. Erode and Coimbatore are important for turmeric, coconut and banana.

3.3 Area and Production Statistics

India is the second largest producer of fruits and vegetables in the world. The exact figures change every year; the ranges below show the scale and are given as approximate, since precise year-wise values must be taken from current NHB or Horticulture Department publications.

Crop group	India - area (lakh ha, approx.)	India - production (million tonnes, approx.)	Tamil Nadu - leading crops
Fruits	~65-70	~100-110	Banana, mango, coconut, sapota, grapes
Vegetables	~105-115	~200-210	Tomato, onion, brinjal, bhendi, tapioca
Flowers	~3-4	Loose ~2.5-3 MT; cut ~thousand million stems	Jasmine, chrysanthemum, marigold, rose
Spices	~35-40	~9-11	Turmeric, pepper, cardamom, chilli
Plantation crops	~30-35	Coconut, arecanut, cashew, tea, coffee, rubber	Coconut, tea, coffee, rubber
Total horticulture	~250-280	~330-350	A leading state in fruits and flowers

Table 3.3 Approximate area and production of horticultural crops in India and Tamil Nadu

- India leads the world in the production of mango, banana, sapota and lime/lemon, and is a leading producer of vegetables, coconut and spices.
- Horticulture accounts for a large share of the value of agricultural output from a much smaller share of the cropped area.
- Tamil Nadu is among the top states in banana, coconut, flowers (jasmine and chrysanthemum) and tapioca.

- Figures are indicative; always quote the latest NHB / Department of Horticulture statistics in the exam with the year.

3.4 Leading Horticultural Districts of Tamil Nadu

Crop	Leading districts	Note
Banana	Theni, Dindigul, Trichy, Erode, Tuticorin	Largest fruit area in the state
Mango	Krishnagiri, Dharmapuri, Salem, Vellore	Neelam and Banganapalli are important
Coconut	Coimbatore, Erode, Thanjavur, Tiruppur, Kanyakumari	A leading coconut state
Grapes	Theni, Dindigul, Coimbatore	Panneer Thratchai is a leading variety
Sapota	Dindigul, Tirunelveli, Krishnagiri	PKM-1 is a popular TNAU variety
Tomato	Krishnagiri, Dharmapuri, Salem, Coimbatore	Important for the fresh market
Onion	Perambalur, Tiruchy, Dindigul	Small onion (aggregatum) is famous
Turmeric	Erode, Coimbatore, Dharmapuri	Erode turmeric has a GI tag
Hill vegetables and cut flowers	The Nilgiris	Potato, carrot, cabbage, rose, carnation
Jasmine and flowers	Madurai, Dindigul, Trichy, Erode	Loose flowers for garlands

Table 3.4 Leading districts of Tamil Nadu for important crops

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Key points to remember

- A horticultural zone is a region of broadly similar climate, soil and cropping pattern.
- India has eight horticultural zones from the temperate Himalaya to the islands.
- Tamil Nadu has seven agro-climatic (horticultural) zones.
- The Nilgiris is the hill zone for temperate vegetables and cut flowers.
- India is the second largest producer of fruits and vegetables in the world.
- India is a world leader in mango, banana, sapota and lime/lemon.
- Statistics change yearly; quote the year and source.

Short-answer questions

Q1. Name the eight horticultural zones of India.

Model answer: Temperate/NW Himalayan, NW Plains, North-Eastern, Eastern/Lower Gangetic Plains, Central/Plateau, Western/Deccan Plateau, Southern/Coastal, Islands.

Q2. Which zone of Tamil Nadu grows hill vegetables and cut flowers?

Model answer: The Hilly zone (The Nilgiris, Kodaikanal and Yercaud).

Q3. Name three crops in which India leads world production.

Model answer: Mango, banana and sapota (also lime/lemon).

Long-answer questions

Q1. Describe the horticultural zones of Tamil Nadu and the crops grown in each.

Answer outline:

1. Concept of a horticultural zone.
2. Seven zones with typical districts.
3. Crops of each zone with examples.
4. Special features: Nilgiris hill vegetables and flowers, Theni and Dindigul banana and grapes, Krishnagiri mango.
5. Importance of zoning for crop planning.

Q2. Write a note on the area and production of horticultural crops in India.

Answer outline:

1. India's rank in world production.
2. Approximate area and production of fruits and vegetables.
3. Flowers, spices and plantation crops.
4. Share of horticulture in agricultural value.
5. Note that figures are yearly and should be quoted with source.

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Lecture 4. Exports and Imports of Horticultural Crops in India and Central and State Agencies Involved in Horticultural Research and Development

Learning Outcomes

- After studying this lecture the student should be able to:
- List the major horticultural exports and imports of India.
- Name the central research and development agencies and their roles.
- Name the Tamil Nadu state agencies and their functions.
- Explain the role of APEDA, NHB, TNAU and Uzhavar Sandhai.

Horticultural produce is a major source of agricultural export earnings for India. Fruits, vegetables, spices, cashew, tea and coffee together contribute a large part of the export basket. Imports are smaller and consist mainly of temperate fruits, some processed products and planting material.

4.1 Exports of Horticultural Crops

- **Major export fruits:** mango, banana, grapes, pomegranate, citrus, sapota (fresh and processed).
- **Major export vegetables:** onion, tomato, green chilli, gherkin, drumstick, bitter gourd, and value-added products.
- **Spices:** pepper, cardamom, chilli, turmeric, cumin, ginger - India is a leading exporter.
- **Plantation:** tea, coffee, cashew (kernel) and coconut products.
- **Flowers:** fresh cut flowers, especially rose, carnation and gerbera, and loose flowers to nearby markets.
- **Important destinations:** Middle East (UAE, Saudi Arabia), Bangladesh, Nepal, Sri Lanka, Malaysia, Singapore, the USA and European Union.
- **Value-added exports:** fruit pulp, juices, pickles, dried and frozen vegetables, spice oils and oleoresins.

4.2 Imports of Horticultural Crops

- Temperate fruits such as apple, pear, kiwi and some citrus are imported during the off-season.
- Processed products: fruit juices, concentrates, wine and some canned goods.
- Planting material: improved varieties, rootstocks, hybrid seeds and tissue-culture plants.
- Dry fruits: almond, walnut, pistachio and date.
- Imports are regulated by plant quarantine rules to prevent entry of pests and diseases.

Group	Export position	Import position
Fruits	Mango, banana, grapes, pomegranate, citrus	Apple, pear, kiwi, some citrus
Vegetables	Onion, tomato, chilli, gherkin	Small; mainly processed / off-season
Spices	Pepper, chilli, turmeric, cardamom	Very small
Plantation	Tea, coffee, cashew, coconut	Cashew raw nuts for processing
Flowers	Cut flowers and loose flowers	Planting material, some cut flowers
Planting material	Tissue-culture plants, hybrid seed	Improved varieties, rootstocks, hybrid seed

Table 4.1 Indicative export and import position of horticulture

Export figures change every year and depend on crop, season and policy. For examination purposes, remember the leading crops and destinations rather than fixed values.

4.3 Central Agencies

Agency	Role
ICAR (Indian Council of Agricultural Research)	Apex body for agricultural research and education; funds and coordinates research
IIHR, Bengaluru (Indian Institute of Horticultural Research)	National institute for fruit, vegetable and flower research
CISH, Lucknow (Central Institute for Subtropical Horticulture)	Research on sub-tropical fruits
CIAH, Bikaner (Central Institute for Arid Horticulture)	Research on arid-zone horticulture
NRCs (National Research Centres)	Crop-specific centres - banana (Tiruchirappalli), grapes (Pune), pomegranate (Solapur), citrus (Nagpur), litchi (Muzaffarpur), onion and garlic (Rajgurunagar), seed spices, orchid, cashew, arecanut
IARI, New Delhi	Premier institute; division of horticulture
IIVR, Varanasi (Indian Institute of Vegetable Research)	Vegetable research
NHB (National Horticulture Board)	Promotion of horticulture, database and statistics, development schemes
MIDH / NHM (Mission for Integrated Development of Horticulture)	Flagship scheme for area expansion, protected cultivation, post-harvest management
APEDA (Agricultural and Processed Food Products Export Development Authority)	Export promotion, infrastructure and market development
NABARD	Credit and refinance for horticulture projects
Directorate of Marketing and Inspection / FSSAI	Grading, standards and food safety
DARE / Ministry of Agriculture and Farmers Welfare	Policy and administration

Table 4.2 Central agencies in horticultural research and development

4.4 State Agencies (Tamil Nadu)

- **Tamil Nadu Agricultural University (TNAU), Coimbatore:** Teaching, research and extension; releases varieties and packages of practices for horticultural crops.
- **Department of Horticulture and Plantation Crops, Tamil Nadu:** Implements state schemes, subsidises orchards, protected cultivation, drip irrigation and post-harvest infrastructure.
- **Tamil Nadu Horticultural Producers' Cooperative Marketing Federation (TANHODA / HORTICORP):** Procurement and marketing support.
- **Uzhavar Sandhai:** Direct farmer-to-consumer markets run by the state to give farmers a better price and consumers fresh produce.
- **Horticultural College and Research Institutes (HC&RI;):** At Coimbatore, Periyakulam and other centres; teaching and crop research.
- **Regional research stations and Krishi Vigyan Kendras (KVKs):** Local testing and transfer of technology.
- **Tamil Nadu State Agricultural Marketing Board and regulated markets:** Marketing infrastructure.

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4.5 Export Promotion and Infrastructure

- **APEDA:** Registers exporters, promotes markets, funds infrastructure such as pack-houses and cold stores.
- **Agri Export Zones (AEZs):** Region-specific export zones for crops such as mango, grapes and flowers.
- **Pack-houses and pre-cooling units:** Set up near production areas to prepare export consignments.
- **Air cargo and sea reefer services:** Essential for perishable exports; major airports handle cut flowers and fruits.
- **Plant quarantine and phytosanitary certification:** Required for every export consignment to meet importing-country rules.
- **MIDH and state schemes:** Provide subsidy for cold storage, pack-houses and reefer vehicles.

EXAM FOCUS

Key points to remember

- Horticulture is a major source of India's agricultural export earnings.
- Leading exports: mango, banana, grapes, pomegranate, onion, spices, tea, coffee and cashew.
- Imports are small: temperate fruits, processed products, planting material and dry fruits.
- ICAR is the apex research body; IIHR, CISH, CIAH, IIVR and NRCs are specialised institutes.
- NHB and MIDH are the main development agencies; APEDA promotes exports.
- TNAU and the TN Department of Horticulture are the main state agencies.
- Uzhavar Sandhai gives direct farmer-to-consumer marketing.

Short-answer questions

Q1. Name any four horticultural export crops of India.

Model answer: Mango, banana, onion and spices (pepper/turmeric).

Q2. What is the role of APEDA?

Model answer: It promotes exports of agricultural and processed food products, including horticultural crops, and develops export infrastructure and markets.

Q3. What is Uzhavar Sandhai?

Model answer: A direct farmer-to-consumer market run by the Tamil Nadu government to ensure a fair price to farmers and fresh produce to consumers.

Long-answer questions

Q1. Describe the central and state agencies involved in horticultural research and development.

Answer outline:

1. Central research: ICAR and its institutes (IIHR, CISH, CIAH, IIVR, NRCs).
2. Central development: NHB, MIDH/NHM.
3. Export and credit: APEDA, NABARD.
4. State research and education: TNAU and HC&RI.;
5. State development: Department of Horticulture, TANHODA/HORTICORP.
6. Marketing: Uzhavar Sandhai, regulated markets.
7. Conclusion: a coordinated research-development-marketing chain.

Q2. Write a note on the export and import of horticultural crops in India.

Answer outline:

1. Importance of horticultural exports.
2. Major export crops and destinations.
3. Value-added exports.
4. Major imports and their reasons.
5. Role of plant quarantine.
6. Role of APEDA and NHB in export promotion.
7. Conclusion: export-led growth potential.

Lecture 5. Soil and Climate Requirements of Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- State the soil requirements of horticultural crops.
- State the climate requirements of horticultural crops.
- Match important crops to their soil and climate needs.
- Explain how soil problems are managed.

The success of a horticultural crop depends on how well the soil and climate of a site match the crop's requirement. Soil supplies water, nutrients and anchorage; climate supplies heat, light and moisture. A crop grown outside its suitable range gives poor yield and quality however well it is managed.

5.1 Soil Requirements

- **Depth:** Fruit trees need deep soil (at least 1-1.5 m) for good root development; shallow soils suit only shallow-rooted crops.
- **Drainage:** Good internal drainage is essential; waterlogging kills roots of most fruit trees.
- **Texture:** Medium-textured loam and sandy loam are ideal; heavy clay is poor for most fruits.
- **pH:** Most crops prefer slightly acidic to neutral (pH 6.0-7.5). Acid lime tolerates higher pH; blueberry needs very acid soil.
- **Organic matter:** Improves structure, water-holding capacity and nutrient supply.
- **Salinity and alkalinity:** Most horticultural crops are sensitive; banana, mango and citrus are moderately sensitive, while date palm and pomegranate tolerate more.
- **Free lime:** High free lime causes micronutrient (iron, zinc) deficiency, common in many Indian soils.

5.2 Climate Requirements

- **Temperature:** Determines which crops can be grown. Temperate fruits need winter chilling; tropical fruits are injured by frost.
- **Rainfall:** Total rainfall and its distribution matter. Most fruit crops need 600-1500 mm well distributed.
- **Humidity:** High humidity favours fungal diseases; low humidity with high temperature causes flower drop.
- **Light:** Bright sunshine is needed for photosynthesis, flowering and fruit colour.
- **Wind:** Strong wind causes lodging, fruit drop and poor pollination; windbreaks help.
- **Altitude:** In the tropics, altitude creates a cooler climate suitable for temperate crops, as in the Nilgiris.

Crop	Soil	pH range	Climate	Special note
Mango	Deep well-drained loam	5.5-7.5	Tropical/sub-tropical, dry before flowering	Avoid waterlogging; needs dry spell for flowering
Banana	Deep rich loam, good drainage	6.0-7.5	Warm humid tropical	Very sensitive to waterlogging and wind
Citrus (acid lime, orange)	Light loam, well-drained	5.5-7.5	Sub-tropical to tropical	Sensitive to salinity and free lime

Crop	Soil	pH range	Climate	Special note
Guava	Wide range, tolerates poor soil	5.5-8.2	Tropical/sub-tropical	Hardy; tolerates drought and salinity to some extent
Grapes	Well-drained sandy loam	6.5-7.5	Dry warm climate	Rain at ripening causes berry cracking
Sapota	Deep well-drained soil	6.0-8.0	Tropical, warm	Tolerates salinity; sensitive to waterlogging
Apple	Well-drained loam	5.5-6.5	Temperate, winter chilling needed	Grown at high altitude in India
Tomato	Well-drained sandy loam rich in organic matter	6.0-7.0	Warm, 20-27 deg C	High temperature reduces fruit set
Onion	Sandy loam, well-drained	6.0-7.0	Cool during growth, warm and dry at maturity	Long day length for bulb formation in some types
Potato	Loose sandy loam, acidic to neutral	5.2-6.5	Cool, 15-20 deg C	Tubers form best in cool weather
Carrot	Deep loose sandy loam	6.0-6.8	Cool climate	Stony soil gives forked roots
Cabbage / cauliflower	Rich loam, good moisture	6.0-7.0	Cool, 15-20 deg C	High temperature causes loose curd/head
Coconut	Deep sandy loam, coastal	5.2-8.0	Tropical, humid	Tolerates salinity; needs good drainage
Tea	Deep acidic loam	4.5-5.5	Cool humid, hilly	Requires high rainfall and drainage

Table 5.1 Soil and climate requirements of important horticultural crops

General rule: match the crop to the soil and climate of the site before planting; corrections after planting are costly and often impossible. Where a soil problem exists (drainage, salinity, low pH), choose a tolerant crop rather than force a sensitive one.

5.3 Managing Soil Problems for Horticulture

- **Poor drainage:** Provide drains and raised beds; choose tolerant crops; avoid planting in low spots.
- **Acidity (low pH):** Apply lime to raise pH; acid-loving crops such as tea are grown on acid soils.
- **Alkalinity and free lime:** Apply gypsum and organic matter; correct iron and zinc deficiency by foliar sprays.
- **Salinity:** Leach salts with good-quality water, improve drainage and add organic matter.
- **Shallow soil:** Grow shallow-rooted crops, or deepen the soil with organic matter and terracing.
- **Erosion on slopes:** Contour planting, terracing and cover crops.

5.4 Effect of Climate on Yield and Quality

- Temperature at flowering decides fruit set in mango and tomato.
- Rain at ripening causes cracking in grape and splitting in tomato.
- High humidity increases fungal disease; low humidity causes flower drop.
- Bright sunshine improves fruit colour and TSS.
- Strong wind causes fruit drop and lodging; windbreaks reduce loss.

5.5 Soil Preparation for Horticultural Crops

- Deep ploughing and levelling to improve the root zone.

- Addition of organic matter (FYM, compost) to improve structure and fertility.
- Making raised beds in areas with heavy rain or poor drainage.
- Digging and filling of pits for fruit trees before planting.
- Soil and water testing before planting to correct pH and nutrient status.

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Key points to remember

- Soil supplies water, nutrients and anchorage; climate supplies heat, light and moisture.
- Fruit trees need deep, well-drained soil; waterlogging is fatal to most.
- Most crops prefer pH 6.0-7.5; free lime causes micronutrient deficiency.
- Temperature decides the crop group; temperate fruits need winter chilling.
- Rainfall amount and distribution both matter.
- Altitude creates a cooler climate in the tropics (Nilgiris).
- Match crop to site; correcting soil problems after planting is costly.

Short-answer questions

Q1. Why is good drainage essential for fruit crops?

Model answer: Because waterlogging excludes air from the root zone and kills roots; most fruit trees cannot tolerate standing water.

Q2. What is the ideal pH range for most horticultural crops?

Model answer: About pH 6.0 to 7.5 (slightly acidic to neutral).

Q3. Name two crops that tolerate salinity better than others.

Model answer: Coconut and sapota (also pomegranate and date palm).

Long-answer questions

Q1. Discuss the soil and climate requirements of horticultural crops with examples.

Answer outline:

1. Role of soil and climate.
2. Soil factors: depth, drainage, texture, pH, organic matter, salinity.
3. Climate factors: temperature, rainfall, humidity, light, wind, altitude.
4. Crop examples from the table.
5. Importance of matching crop to site.
6. Conclusion.

Q2. Explain the climate requirements of fruit crops with reference to tropical, sub-tropical and temperate groups.

Answer outline:

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1. Temperature ranges for each group.
2. Chilling requirement of temperate fruits.
3. Rainfall and humidity needs.
4. Frost sensitivity of tropical fruits.
5. Altitude effect (Nilgiris).
6. Conclusion.

Lecture 6. Types of Plant Propagation and Plant Propagation Structures

Learning Outcomes

- After studying this lecture the student should be able to:
- Differentiate sexual from asexual propagation.
- List the methods of asexual propagation.
- Describe the important propagation structures and their uses.
- Explain the criteria for choosing a structure.

Propagation is the multiplication of plants. Horticultural plants are propagated either by seed (sexual) or by vegetative parts (asexual). The choice depends on the crop, the need for true-to-type plants, and the availability of material. Both methods need suitable structures to give the young plant a protected start.

6.1 Types of Plant Propagation

Basis	Sexual (seed) propagation	Asexual (vegetative) propagation
Material used	Seed	Vegetative parts - stem, root, leaf, bud
Offspring type	Segregating; not identical to parent (unless inbred/apomictic)	True to type; identical clone
Use	Breeding, rootstocks, annuals, vegetables, plantation crops	Fruit trees, ornamentals, varieties with sterile seed
Time to bear	Late (long juvenile phase)	Early (short juvenile phase)
Life span	Generally long-lived, hardy	May be shorter; some viruses transmitted
Examples	Papaya, tomato, most vegetables, rootstocks	Mango (grafting), banana (sucker), rose (cutting)

Table 6.1 Two broad types of propagation

6.2 Methods of Asexual Propagation (Overview)

- **Cutting:** A part of the plant is separated and rooted - stem cutting (rose, grape), root cutting, leaf cutting (Bryophyllum).
- **Layering:** Roots are formed on a stem while it is still attached to the parent - air layering (mango, guava), ground layering, mound layering.
- **Grafting:** A scion is joined to a rootstock so they unite - used in mango, citrus, sapota.
- **Budding:** A single bud is inserted into the rootstock - T-budding in citrus and rose.
- **Specialised parts:** Suckers (banana), offsets (pineapple, date), bulbs (onion), corms, tubers, rhizomes (turmeric, ginger), runners (strawberry).
- **Micropropagation:** Tissue culture for mass multiplication of disease-free plants (banana).

6.3 Plant Propagation Structures

Propagation structures are built to control temperature, humidity and light so that seed germinates and cuttings or grafts root or unite successfully. They range from simple frames to modern polyhouses.

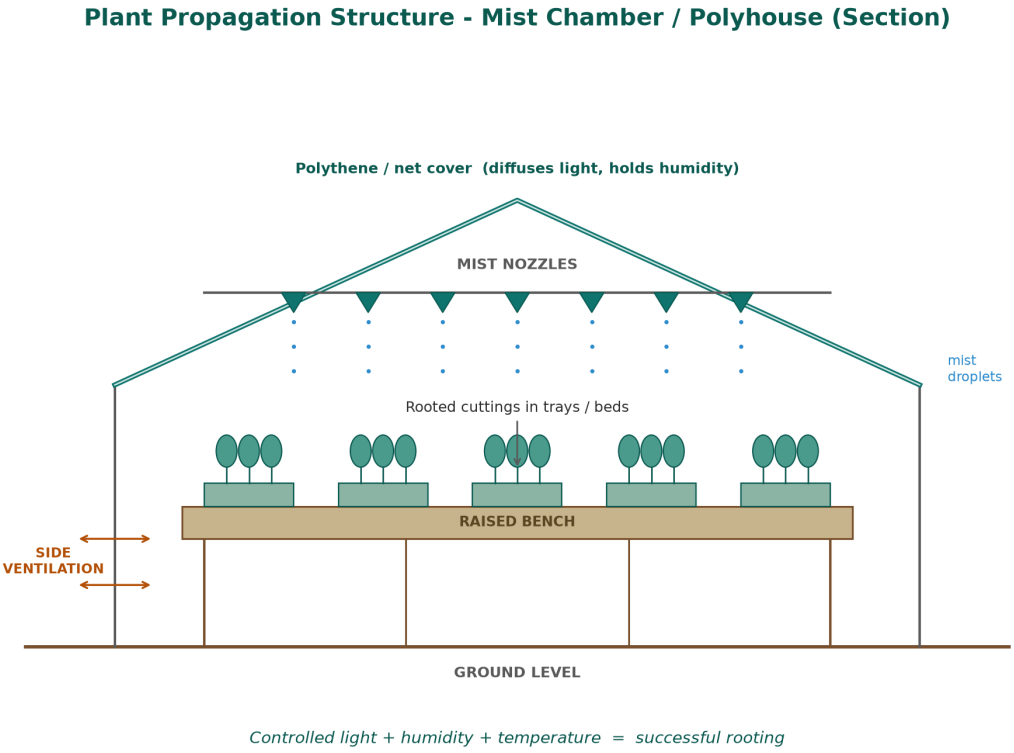


Figure 6.1 Plant propagation structures

Labelled cross-section of a mist chamber / polyhouse with its components named.

Structure	Description	Use
Cold frame	Low box with a glass/polythene cover, no artificial heat	Hardening of seedlings, rooting of easy cuttings
Hot bed	Frame with a heating source (manure or electric cable) below	Rooting of difficult cuttings in cold weather
Propagation frame	Covered frame with lath or glass, controlled shade	Rooting of cuttings and germination of seed
Mist chamber / mist house	Enclosure with intermittent mist over cuttings	Rooting of leafy stem cuttings (hard-to-root species)
Lath house	Semi-shade structure of wooden or bamboo laths	Hardening and growing of shade-loving plants
Polyhouse / greenhouse	Structure covered with polythene or net	Protected cultivation, off-season crops, high-value nursery
Shade net house	Covered with shade net of different percentages	Nursery raising and hardening in hot climate
Glasshouse	Structure covered with glass	Research, year-round propagation and display
Screen house / insect-proof house	Fine mesh screen	Raising virus-free and disease-free nursery stock

Table 6.2 Plant propagation structures



Fig. 6.1 A simple propagation structure (mist chamber / polyhouse) - section view.

Selection of the structure depends on the species, the difficulty of rooting, the season and the cost. A mist chamber gives the highest success for leafy cuttings; a simple shade net house is enough for hardening hardened plants. In Tamil Nadu, polyhouses and shade net houses are widely subsidised under state horticulture schemes, especially for the Nilgiris floriculture and vegetable nursery trade.

6.4 Criteria for Selecting a Propagation Structure

- Species and ease of rooting of the crop.
- Season and climate of the location.
- Number of plants to be produced.
- Cost of construction and maintenance.
- Availability of water, power and labour.
- Purpose - nursery trade, research or own planting.

6.5 Nursery Management

- Select a site with good drainage, water and light.
- Prepare raised beds of a convenient size with a fine, well-manured medium.

- Sow seed or plant cuttings/grafts at the right spacing and season.
- Provide shade, watering and plant protection as needed.
- Harden the plants before lifting; lift with a soil ball to avoid root damage.
- Grade the nursery plants and keep records of variety and source.

6.6 Modern Protected Structures in Tamil Nadu

- Naturally ventilated polyhouses for capsicum, cucumber and cut flowers in the Nilgiris.
- Shade net houses for nursery raising and hardening in the plains.
- Insect-proof screen houses for virus-free nursery production.
- These structures are subsidised under MIDH and state horticulture schemes.

EXAM FOCUS

Key points to remember

- Propagation is multiplication of plants - sexual (seed) or asexual (vegetative).
- Seed propagation gives segregating offspring; vegetative propagation gives true-to-type clones.
- Vegetative methods: cutting, layering, grafting, budding, specialised parts, micropropagation.
- Propagation structures control temperature, humidity and light.
- Cold frame has no heat; hot bed has a heat source.
- Mist chamber is best for rooting leafy stem cuttings.
- Polyhouses and shade net houses are widely used and subsidised in Tamil Nadu.

Short-answer questions

Q1. Differentiate sexual from asexual propagation.

Model answer: Sexual uses seed and gives variable offspring; asexual uses vegetative parts and gives true-to-type plants.

Q2. Name four propagation structures.

Model answer: Cold frame, hot bed, mist chamber, polyhouse/shade net house.

Q3. Which structure is used for rooting leafy stem cuttings?

Model answer: The mist chamber (mist house), because it maintains high humidity around the leaves.

Long-answer questions

Q1. Explain the types of plant propagation and the structures used.

Answer outline:

1. Meaning of propagation.
2. Sexual vs asexual - table of differences.
3. Methods of asexual propagation in brief.
4. Propagation structures - cold frame, hot bed, mist chamber, lath house, polyhouse, screen house.
5. Factors in choosing a structure.
6. Conclusion.

Q2. Describe any five propagation structures with their uses.

Answer outline:

1. Cold frame.
2. Hot bed.
3. Mist chamber.
4. Lath house / shade net house.
5. Polyhouse / glasshouse.
6. Their role in successful propagation.

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Lecture 7. Sexual / Seed Propagation - Importance, Advantages and Disadvantages

Learning Outcomes

- After studying this lecture the student should be able to:
- Explain the importance of seed propagation.
- List its advantages and disadvantages.
- State the seed viability and germination of important crops.
- Describe seed selection and nursery management.

Seed propagation is the multiplication of plants from seed. It is the natural method and is the main method for annual crops, for raising rootstocks, and in plant breeding. A seed is a mature fertilised ovule containing an embryo, stored food and a protective seed coat.

7.1 Importance of Seed Propagation

- It is the cheapest and easiest method for large-scale multiplication.
- Seed can be stored, transported and handled easily.
- It is essential in plant breeding to create variation and new varieties.
- Seedlings are generally vigorous, deep-rooted and long-lived.
- It is the only practical method for annuals and biennials (most vegetables).
- It avoids the transmission of many viruses that spread through vegetative parts.
- It is used to raise rootstocks for later grafting or budding.

7.2 Advantages of Seed Propagation

1. Simple, cheap and rapid multiplication of a large number of plants.
2. Seed is easy to store and transport over long distances.
3. Seedlings are usually free from many viruses and soil-borne pathogens of the parent.
4. Produces new variation, useful in breeding and selection.
5. Seedlings are hardy, vigorous and have a long life.
6. No special skill (as in grafting) is required.

7.3 Disadvantages of Seed Propagation

1. Offspring are not true to type; they do not resemble the parent closely.
2. Trees raised from seed have a long juvenile phase and bear late.
3. Fruit quality and yield are variable and often inferior.
4. Seed of some crops is sterile, scarce or expensive (many fruit hybrids).
5. Some seeds have dormancy and germinate irregularly.
6. Seed may lose viability in storage; seed-borne diseases can be carried.
7. Cannot maintain the characters of a superior clone.

7.4 Seed Quality and Viability

Good seed should be true to type, free from disease, of high germination percentage and of good vigour. Seed viability is the period for which seed remains capable of germination; it varies widely between crops.

Crop	Approximate seed viability in storage	Approximate germination % (fresh good seed)
Tomato	3-5 years	85-95
Brinjal	3-5 years	80-90
Bhendi	2-3 years	70-85
Chilli	3-4 years	70-85
Onion	1-2 years (poor storer)	70-85
Cabbage / cauliflower	3-5 years	80-90
Carrot	2-3 years	70-80
Cucurbits	4-6 years	80-90
Papaya	1-2 years	70-85
Mango (stone)	Short; loses viability quickly	Variable

Table 7.1 Approximate seed viability and germination of important crops

Viability values are typical ranges and depend on storage conditions. Seed stored cool and dry keeps longer; onion and papaya seed lose viability quickly and should be sown fresh.

7.5 Seed Selection, Sowing and Nursery Management

- Use certified or good-quality seed of the recommended variety.
- Test germination before sowing to decide the seed rate.
- Treat seed against pests and diseases before sowing.
- Sow at the correct depth and spacing in a well-prepared nursery bed.
- Provide shade and regular light watering until germination.
- Thin and prick out seedlings to avoid crowding.
- Harden seedlings before transplanting to the main field.

7.6 Seed Rate and Its Calculation

The seed rate depends on the recommended plant population, the germination percentage, the purity and the seedling survival. A simple working formula is: seed rate = (plant population x 100) / (germination % x purity %). Higher-quality seed needs a lower seed rate.

EXAM FOCUS

Key points to remember

- Seed propagation is multiplication from seed; the cheapest and easiest method.
- It is the main method for annuals and biennials and for raising rootstocks.
- Advantages: cheap, easy, easy storage, vigorous seedlings, useful in breeding.
- Disadvantages: not true to type, late bearing, variable quality, dormancy problems.
- Seedlings are often free from many viruses carried in vegetative parts.
- Seed viability varies with crop and storage; onion and papaya seed are short-lived.
- Good seed is true to type, disease-free, high in germination and vigour.

Short-answer questions

Q1. Why is seed propagation not suitable for commercial fruit growing?

Model answer: Because seedling trees are not true to type, bear late and give variable fruit quality.

Q2. State two advantages of seed propagation.

Model answer: It is cheap and easy for large-scale multiplication; seed is easy to store and transport.

Q3. Which seeds lose viability quickly?

Model answer: Onion and papaya (also many tropical fruit seeds); they should be sown fresh.

Long-answer questions

Q1. Discuss the importance, advantages and disadvantages of seed propagation.

Answer outline:

1. Meaning of seed propagation.
2. Importance: cheapest method, breeding, rootstocks, annuals.
3. Advantages - six points.
4. Disadvantages - not true to type, late bearing, variability, dormancy.
5. Comparison with vegetative propagation.
6. Conclusion: seed for annuals and rootstocks, vegetative for superior fruit varieties.

Q2. Explain seed viability and the factors that affect it.

Answer outline:

1. Definition of viability.
2. Typical viability of different crops (table).
3. Factors: storage temperature, moisture, seed coat, pests.
4. Methods to maintain viability: cool dry storage, proper containers.
5. Conclusion.

Lecture 8. Types of Dormancies - Seed Invigoration - Seed Treatments

Learning Outcomes

- After studying this lecture the student should be able to:
- Define dormancy and list its types.
- Describe the methods of breaking dormancy.
- Explain seed invigoration and its techniques.
- Describe common seed treatments before sowing.

Many seeds do not germinate even when given water, air and suitable temperature because they are dormant. Dormancy is a resting condition in which a viable seed fails to germinate under favourable conditions. It is a survival mechanism, but in cultivation it delays and uneven germination, so it must be broken.

8.1 Types of Dormancy

Type	Cause	Examples
Hard seed coat (physical) / seed coat dormancy	Impermeable seed coat prevents water and air entry	Legumes, ber, many forest species, custard apple
Mechanical dormancy	Very hard seed coat physically resists embryo expansion	Some nuts and stone fruits
Physiological dormancy	Immature embryo or inhibiting chemicals (ABA) present	Apple, pear, many temperate fruits
Morphological dormancy	Embryo is small and immature at shedding	Some palms, celery
Double (combined) dormancy	Both seed coat and physiological factors	Some Rosaceae
Secondary dormancy	Dormancy induced after shedding by unfavourable conditions	Many species under stress

Table 8.1 Types of seed dormancy and their causes

8.2 Methods of Breaking Dormancy (Seed Treatments)

Method	How it works	Suitable for
Scarification - mechanical	Seed coat is scratched or nicked with sandpaper, file or knife	Hard-coated legumes, ber
Scarification - acid	Concentrated sulphuric acid dissolves part of the coat	Hard seeds; used with care
Scarification - hot water	Hot water softens the coat	Some legumes and vegetables
Stratification / chilling	Seeds kept moist at low temperature (about 0-7 deg C) for weeks	Apple, pear, peach, temperate fruits
Warm stratification	Moist warm period followed by cool period	Some temperate and palm species
Soaking / water treatment	Soaking in water leaches inhibitors, softens coat	Many vegetables
Chemical treatment	Gibberellic acid (GA3), potassium nitrate, thiourea applied	Physiological dormancy; GA3 widely used
Light treatment	Exposure to light for light-requiring seeds	Lettuce (some cultivars)

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Method	How it works	Suitable for
Removal of seed coat / endosperm	Coat removed by hand	Research and difficult species

Table 8.2 Dormancy-breaking methods

8.3 Seed Invigoration

Seed invigoration (seed enhancement) is any treatment that improves germination speed, uniformity and seedling vigour beyond what the raw seed gives. It is important for costly hybrid seed and for raising a uniform nursery.

- **Seed priming:** Seed is soaked in water or a solution (for example, potassium nitrate, PEG) under controlled conditions, then dried back. It starts the germination process without allowing radicle emergence, giving faster and more uniform germination.
- **Hydropriming:** Soaking in plain water for a fixed time, then drying.
- **Osmopriming:** Soaking in an osmotic solution (for example, polyethylene glycol) to control water uptake.
- **Halopriming:** Soaking in salt solutions (for example, KNO3, NaCl).
- **Seed hardening:** Soaking and drying to improve performance under stress; widely used in dryland crops.
- **Seed coating / pelleting:** Applying a coating of inert material with nutrients or protectants.
- **Bio-priming:** Treating seed with beneficial microbes (for example, *Trichoderma*, *Pseudomonas*, rhizobia).

8.4 Seed Treatments Before Sowing

Apart from dormancy breaking, seed is treated to protect it against pests and diseases and to improve establishment.

- **Fungicidal treatment:** Thiram, captan or carbendazim to control seed-borne and soil-borne fungi.
- **Insecticidal treatment:** For storage pests and soil insects.
- **Bio-agent treatment:** *Trichoderma viride*, *Pseudomonas fluorescens* for disease control; rhizobium and *Azospirillum* for nitrogen.
- **Seed hardening:** For rain-fed crops.
- **Nutrient treatment:** Soaking in micronutrient solutions to correct deficiencies at establishment.

8.5 Seed Testing and Germination Percentage

Germination percentage is the proportion of seeds that produce normal seedlings under standard conditions. It is used to decide the seed rate and to judge seed quality.

Worked example. If 200 seeds are sown and 170 produce normal seedlings, germination % = (170 / 200) x 100 = **85%**. To raise 10,000 healthy seedlings at 85% germination, sow about 10,000 / 0.85 = **11,765 seeds** (about 11,800, allowing a small margin for field losses).

- Seed testing also checks purity, moisture content and freedom from pests and disease.
- Good seed has high germination, high purity, low moisture and good vigour.

8.6 Storage of Seed

- Store seed cool and dry; moisture is the main enemy of seed viability.
- Use airtight containers and keep them in a dry, shaded place.

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- Label the seed with variety, date and germination.
- Onion and papaya seed are short-lived and should be sown fresh.
- Fumigate or treat stored seed against storage pests.

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Key points to remember

- Dormancy is the failure of a viable seed to germinate under favourable conditions.
- Types: seed coat (physical), mechanical, physiological, morphological, double, secondary.
- Hard seed coat prevents water and air entry; physiological dormancy involves immature embryo or inhibitors.
- Breaking methods: scarification (mechanical, acid, hot water), stratification (chilling), soaking, chemicals, light.
- GA3 is widely used for physiological dormancy.
- Seed invigoration: priming, hardening, coating, bio-priming.
- Seed treatment also includes fungicides, insecticides and bio-agents.

Short-answer questions

Q1. Define seed dormancy.

Model answer: It is the condition in which a viable seed does not germinate even when given favourable conditions of water, air and temperature.

Q2. Name three methods of breaking hard seed coat dormancy.

Model answer: Mechanical scarification, acid scarification and hot water treatment.

Q3. What is seed priming?

Model answer: A seed invigoration treatment in which seed is soaked under controlled conditions to start germination, then dried back, giving faster and more uniform germination.

Long-answer questions

Q1. Explain the types of dormancy and the methods of breaking them.

Answer outline:

1. Definition of dormancy.
2. Types with causes and examples.
3. Methods of breaking - scarification, stratification, soaking, chemicals, light.
4. Suitability of each method.
5. Conclusion.

Q2. What is seed invigoration? Describe the important invigoration techniques.

Answer outline:

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1. Definition of invigoration.
2. Priming, hydropriming, osmopriming, halopriming.
3. Seed hardening.
4. Seed coating and pelleting.
5. Bio-priming.
6. Advantages in nursery and hybrid seed.

Lecture 9. Asexual / Vegetative Propagation - Importance, Advantages and Disadvantages

Learning Outcomes

- After studying this lecture the student should be able to:
- Define vegetative propagation and the term clone.
- Explain its importance, advantages and disadvantages.
- Compare sexual and asexual propagation.
- State when vegetative propagation is essential.

Asexual or vegetative propagation is the multiplication of plants from vegetative parts - stem, root, leaf or bud - without the fusion of male and female gametes. The new plant is genetically identical to the parent and is called a clone. It is the main method for commercial fruit growing, for many ornamentals and for crops that do not set viable seed.

9.1 Importance of Vegetative Propagation

- Produces true-to-type plants that retain all characters of a superior variety.
- Shortens the juvenile phase, so plants bear fruit early.
- Fixes and maintains hybrid vigour in perennial crops.
- Multiplies crops that are seedless or set little viable seed (banana, many citrus, some grapes).
- Allows the use of rootstocks that give disease resistance, vigour control or soil adaptation.
- Helps in rapid multiplication of new and rare varieties.
- Allows repair of damaged trees and changing the variety of an established orchard (top working).

9.2 Advantages of Vegetative Propagation

1. Plants are true to type and uniform; the orchard is even and easy to manage.
2. Early bearing: grafted mango or budded citrus comes to bearing much earlier than seedlings.
3. Desirable characters (fruit quality, size, colour) are retained exactly.
4. Rootstock characters (disease resistance, drought tolerance, dwarfing) can be combined with scion quality.
5. Useful for crops with poor or sterile seed.
6. Large numbers of plants can be produced from a small amount of material.
7. Shortens breeding and selection cycles.

9.3 Disadvantages of Vegetative Propagation

1. Costly and requires skilled labour and special structures.
2. Diseases and viruses present in the parent are transmitted to the progeny.
3. No genetic variation, so the crop is vulnerable to a new pest or disease.
4. The propagated plants may be shorter-lived and less deep-rooted than seedlings.
5. Some methods fail in certain species (for example, difficult-to-root cuttings).
6. Graft and bud unions can fail or show incompatibility.
7. Large-scale multiplication is slower than seed for some crops.

9.4 Comparison of Sexual and Asexual Propagation

Character	Sexual (seed)	Asexual (vegetative)
Nature of offspring	Variable; not true to type	Uniform; true to type (clone)
Juvenile phase	Long; late bearing	Short; early bearing
Cost and skill	Cheap, little skill	Costly, needs skill
Disease transmission	Mostly free from parent viruses	Transmits parent diseases/viruses
Life span and vigour	Long-lived, hardy, deep-rooted	May be shorter; depends on rootstock
Use	Annuals, rootstocks, breeding	Fruit trees, ornamentals, sterile crops

Table 9.1 Sexual versus asexual propagation

9.5 Situations Where Vegetative Propagation Is Essential

- When the crop does not produce viable seed (banana, many citrus and grapes).
- When a superior variety must be maintained exactly (all commercial fruit varieties).
- When the seed produces plants of inferior quality or unknown type.
- When early bearing is required for quick returns.
- When a specific rootstock character (disease resistance, dwarfing) is needed.
- When the crop is a hybrid perennial that cannot be fixed by seed.

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Key points to remember

- Vegetative propagation uses vegetative parts and produces genetically identical clones.
- It gives true-to-type, uniform plants that bear early.
- It is essential for seedless or poor-seeding crops such as banana and many citrus.
- Rootstock can add disease resistance, dwarfing or soil adaptation.
- Disadvantages: costly, needs skill, transmits viruses, no genetic variation.
- Top working allows changing the variety of an established tree.
- Choice between seed and vegetative propagation depends on crop and purpose.

Short-answer questions

Q1. Define vegetative propagation.

Model answer: It is the multiplication of plants from vegetative parts such as stem, root, leaf or bud, without fusion of gametes, producing true-to-type clones.

Q2. Give two advantages of vegetative propagation.

Model answer: Plants are true to type and bear early; desirable rootstock characters can be combined with scion quality.

Q3. Why is vegetative propagation risky for a crop?

Model answer: Because there is no genetic variation, so the whole crop is vulnerable to a new pest or disease, and parent viruses are transmitted.

Long-answer questions

Q1. Discuss the importance, advantages and disadvantages of vegetative propagation.

Answer outline:

1. Definition and meaning of clone.
2. Importance - true-to-type, early bearing, sterile crops.
3. Advantages - six points.
4. Disadvantages - cost, virus transmission, no variation.
5. Comparison with seed propagation.
6. Conclusion.

Q2. Compare sexual and asexual propagation with a table.

Answer outline:

1. Offspring nature.
2. Juvenile phase and bearing.
3. Cost and skill.
4. Disease transmission.
5. Life span and vigour.
6. Use in horticulture.

Lecture 10. Vegetative Propagation - Cutting and Layering Methods

Learning Outcomes

- After studying this lecture the student should be able to:
- Define a cutting and list its types.
- Explain the factors affecting rooting of cuttings.
- Describe the types of layering with examples.
- Explain the care of cuttings and layers.

Cutting and layering are the two simplest methods of vegetative propagation. In cutting, a part is detached from the parent and made to form roots and shoots on its own. In layering, roots are formed on a stem while it is still attached to the parent plant, and the rooted part is separated later.

10.1 Cutting - Principle and Types

A cutting is a detached vegetative part which, when placed in a suitable medium, regenerates the missing parts (roots and shoots) and becomes an independent plant. Success depends on the ability of the cutting to form adventitious roots.

Type	Material used	Examples
Stem cutting - hardwood	Mature, dormant woody stem	Grape, fig, pomegranate, rose (hardwood)
Stem cutting - semi-hardwood	Partially mature stem	Rose, hibiscus, croton
Stem cutting - softwood	Young, soft, leafy stem	Chrysanthemum, geranium, many ornamentals
Herbaceous cutting	Soft herbaceous stem	Coleus, carnation
Root cutting	Pieces of root	Breadfruit, sweet potato, some ornamentals
Leaf cutting	Whole leaf or leaf part	Bryophyllum, African violet, sansevieria
Leaf-bud cutting	Leaf with a small piece of stem and bud	Rubber, lemon, some tropicals

Table 10.1 Types of cuttings

10.2 Factors Affecting Rooting of Cuttings

- **Species and variety:** Easy-to-root (grape, fig) versus difficult-to-root species.
- **Age of parent plant:** Cuttings from young plants root better; juvenility favours rooting.
- **Type of wood:** Semi-hardwood and softwood root more readily than very old wood in many species.
- **Season:** The right season (usually the active growing season) improves rooting.
- **Presence of leaves and buds:** Leaves supply food and auxin; buds are necessary in some species.
- **Rooting medium:** Well-aerated, well-drained medium such as sand, vermiculite or a mixture.
- **Environment:** High humidity (mist), moderate temperature, diffused light.
- **Growth regulators:** IBA, NAA, IAA applied to the base improve rooting of difficult species.

10.3 Layering - Principle and Types

Layering is the formation of roots on a stem while it is still attached to the parent plant. Because the stem stays attached, it continues to get water and food, so layering succeeds even in species that are difficult to root as cuttings.

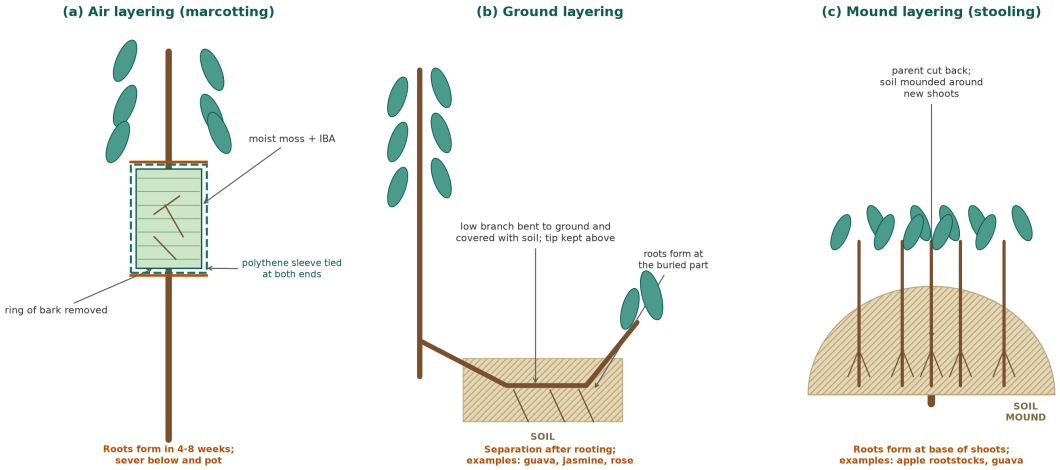


Figure 10.1 Layering methods

Air layering (marcotting), ground layering and mound layering (stooling).

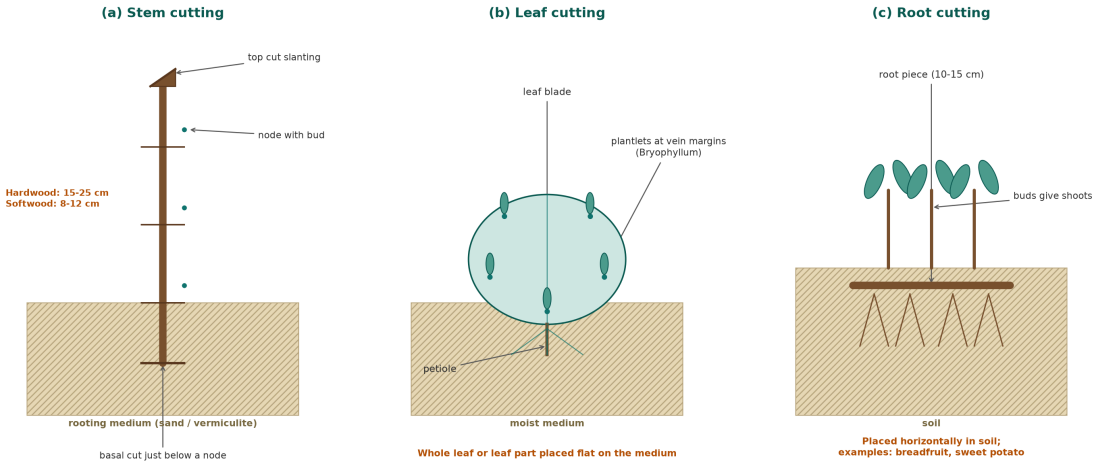


Figure 10.2 Cutting types

Stem, leaf and root cuttings, each drawn in its correct form.

Type	Method	Examples
Simple / ground layering	A low branch is bent to the ground and covered with soil; tip kept above	Guava, jasmine, rose
Tip layering	The tip of a branch is buried in soil	Blackberry, raspberry
Compound / serpentine layering	Stem is alternately buried and exposed at intervals	Grape, clematis
Mound layering / stooling	The plant is cut back and soil mounded around new shoots	Apple rootstocks, guava, pomegranate
Air layering (goo-tee / marcotting)	A ring of bark is removed on an upright branch, wrapped with moist moss and polythene	Mango, guava, litchi, croton, fig
Trench layering	Whole branch laid in a shallow trench and covered	Apple and pear rootstocks

Table 10.2 Types of layering

10.4 Cutting and Layering Diagrams

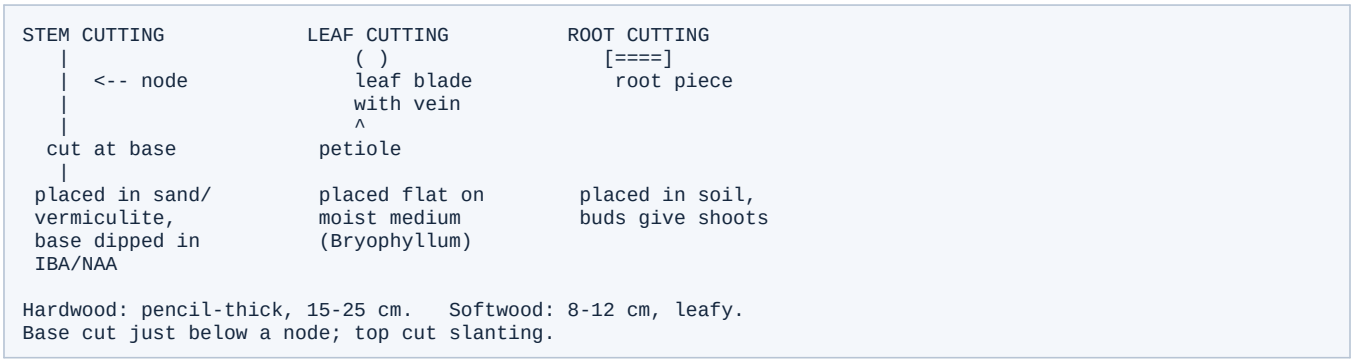


Fig. 10.1 Types of cuttings (diagrammatic).

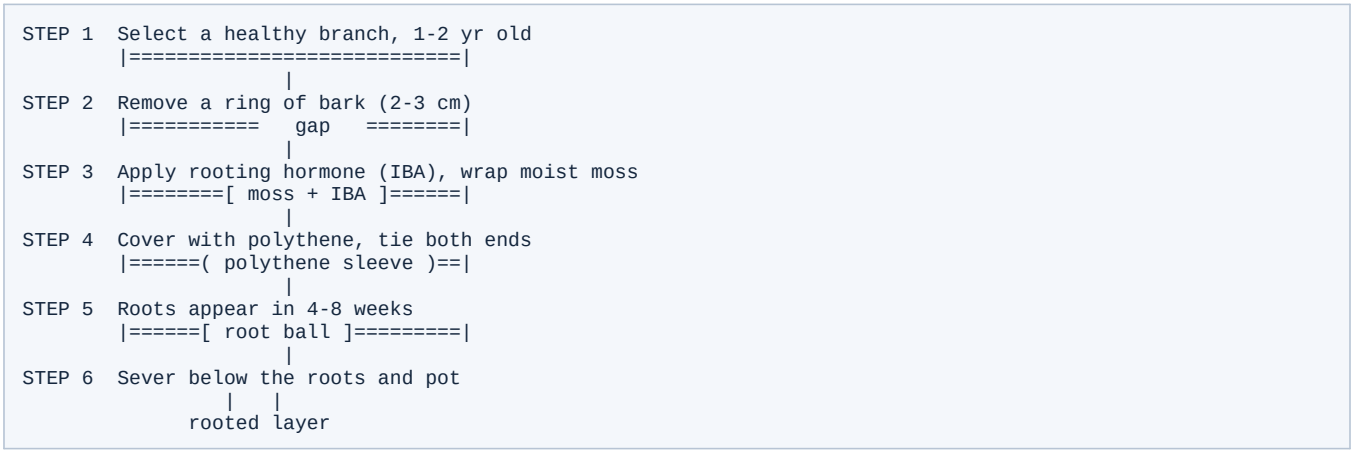


Fig. 10.2 Air layering (marcotting) - steps.

10.5 Cutting versus Layering

Point	Cutting	Layering
Attachment to parent	Detached at the start	Attached until roots form
Rooting success	Lower in difficult species	Higher; parent supplies water and food
Number of plants	Many from one parent	Fewer; limited by branches
Skill and space	Simple; needs a rooting structure	Simple; needs space to bend branches
Examples	Rose, grape, hibiscus	Mango, guava, litchi (air layering)

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Table 10.3 Cutting compared with layering

10.6 Care of Cuttings and Layers

- Keep the rooting medium moist but not waterlogged.
- Provide shade and high humidity (mist) for leafy cuttings.
- Avoid disturbing the cuttings while roots form.
- Harden rooted cuttings before potting or field planting.
- For layers, keep the moss moist and separate only after good root formation.

10.7 Rooting Hormones for Cuttings

- IBA is the most widely used rooting hormone; NAA is also common.
- Quick dip (concentrated, a few seconds) or soak (dilute, several hours) methods are used.
- Indicative concentrations: 500-5,000 ppm quick dip; 100-500 ppm for soaking.
- Treat only the base of the cutting; avoid contaminating the leaves.

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Key points to remember

- A cutting is a detached vegetative part that regenerates roots and shoots.
- Types: hardwood, semi-hardwood, softwood, herbaceous, root, leaf and leaf-bud cuttings.
- Rooting depends on species, age of parent, type of wood, season, leaves/buds, medium, environment and hormones.
- Layering forms roots on a stem still attached to the parent.
- Types: simple, tip, compound, mound (stooling), air layering and trench layering.
- Air layering (marcotting) is used in mango, guava and litchi.
- IBA and NAA are the common rooting hormones.

Short-answer questions

Q1. What is a cutting? Name two types.

Model answer: A cutting is a detached vegetative part that forms roots and shoots on its own. Types include hardwood cutting and leaf cutting.

Q2. Why does layering succeed where cuttings fail?

Model answer: Because the layered stem remains attached to the parent and continues to receive water and nutrients while roots form.

Q3. Which layering method is used in mango and litchi?

Model answer: Air layering (marcotting).

Long-answer questions

Q1. Describe the types of cuttings and the factors affecting their rooting.

Answer outline:

1. Definition of cutting.
2. Types of cuttings with examples.
3. Factors: species, age, wood type, season, leaves/buds, medium, environment, hormones.
4. Role of IBA/NAA.
5. Conclusion.

Q2. Explain the different methods of layering with examples.

Answer outline:

1. Principle of layering.
2. Simple, tip, compound, mound (stooling), air and trench layering.
3. Suitability and examples of each.
4. Advantages of layering.
5. Conclusion.

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Lecture 11. Vegetative Propagation - Grafting and Budding Methods

Learning Outcomes

- After studying this lecture the student should be able to:
- Explain the principle of grafting and budding.
- Describe the important methods of grafting.
- Describe the important methods of budding.
- Explain the factors affecting success and after-care.

Grafting and budding are methods in which a part of one plant (the scion) is joined to another plant (the rootstock) so that they unite and grow as one plant. Grafting uses a scion with one or more buds (usually a shoot piece); budding uses a single bud. These methods are the standard way of propagating most fruit trees.

11.1 Grafting - Principle and Requirements

- The cambium layers of scion and rootstock must be brought into close contact so that they unite.
- Scion and rootstock should be compatible (usually of the same species, or closely related genera).
- The union must be kept moist and protected from drying until it heals.
- Dormant scion wood of the right maturity gives the best results.
- The operation should be done in the right season (active growth for the rootstock).

11.2 Methods of Grafting

Method	Description	Examples
Whip / tongue grafting	Slanting cuts on scion and stock of equal size, interlocked	Apple, pear, temperate fruits
Cleft grafting	Stock split and scion wedges inserted; for thick stocks	Mango top working, old trees
Side grafting	Scion inserted into a side cut on the stock	Mango, jackfruit, citrus
Veneer grafting	A shallow cut on the stock and matching cut on scion	Mango, cashew, sapota
Approach grafting	Two attached plants joined, later separated	Difficult species; mango, guava
Stone / epicotyl grafting	Young seedling stock grafted near the collar	Mango, cashew (quick, cheap)
Bark grafting	Scion inserted under the bark of the stock	Top working of large trees
Inarching	Two independent plants joined while on their own roots	Mango, difficult unions

Table 11.1 Methods of grafting

11.3 Budding - Principle and Methods

Budding is a form of grafting in which a single bud, with a small shield of bark, is inserted into the rootstock. It is quicker and uses less scion material than grafting.

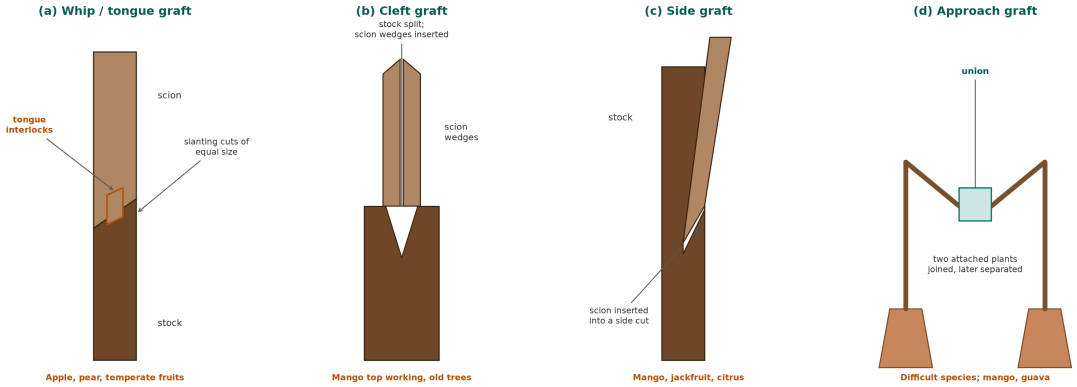


Figure 11.1 Grafting methods

Whip / tongue, cleft, side and approach grafting, showing the cut and the join.

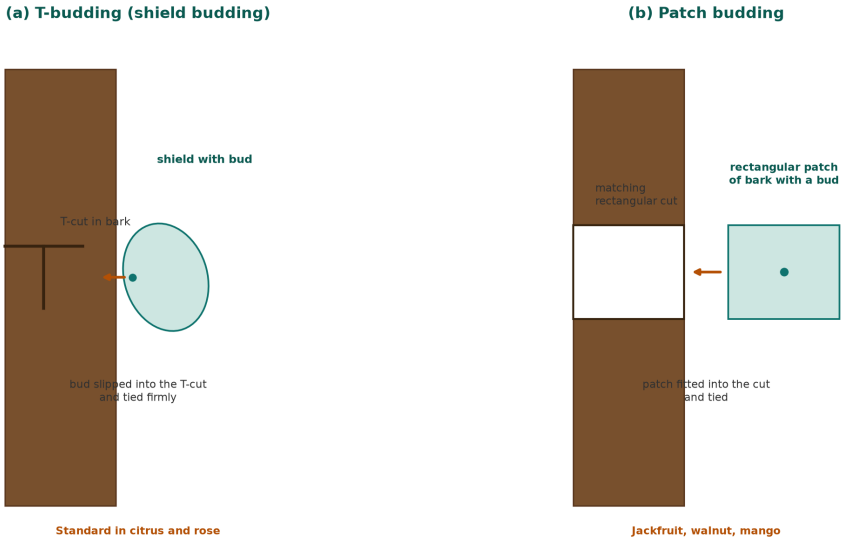


Figure 11.2 Budding methods

T-budding (shield budding) and patch budding.

Method	Description	Examples
T-budding / shield budding	A T-shaped cut in the stock; shield with bud slipped in and tied	Citrus, rose
Inverted T-budding	T-cut is inverted; used where sap flow is heavy	Some citrus
Patch budding	A rectangular patch of bark with bud fitted into a matching cut	Jackfruit, walnut, mango
Ring budding	A complete ring of bark with bud fitted around the stock	Jackfruit, some citrus
Chip budding	A chip of wood with bud inserted into a matching chip cut	Apple, pear, walnut
Forkert budding	Similar to patch budding, used in tropical trees	Mango, cashew

Table 11.2 Methods of budding

11.4 Grafting vs Budding

Point	Grafting	Budding
Scion material	A shoot piece with one or more buds	A single bud with a shield of bark
Material required	More scion wood	Very little scion material
Speed	Slower per plant	Faster; more plants per unit scion
Union	Usually stronger	Depends on method and care
Best season	Dormant/active season as per method	Active sap-flow season for T-budding
Examples	Mango, apple, cashew	Citrus, rose, jackfruit

Table 11.3 Grafting versus budding

11.5 Grafting and Budding Diagrams

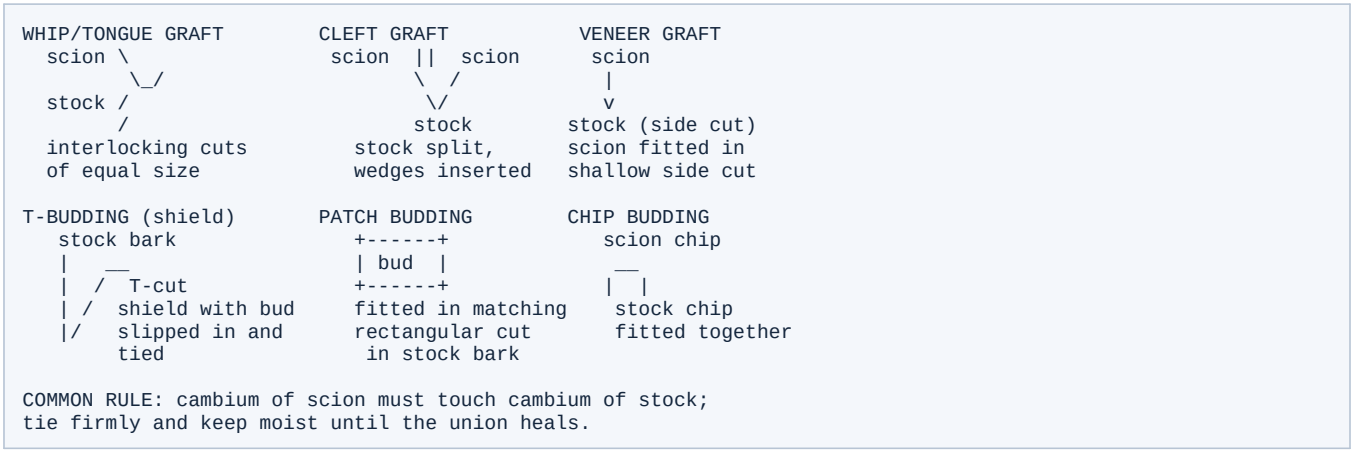


Fig. 11.1 Common grafting and budding methods.

11.6 Factors Affecting Success of Grafting and Budding

- Compatibility between scion and rootstock.
- Healthy, disease-free and vigorous stock and scion.
- Correct stage of maturity of scion wood (usually dormant or semi-mature).
- Close contact of cambium layers.
- Right season and favourable temperature and humidity.
- Quick, clean cuts and skilled workmanship.

- Protection of the union from drying, water and disease until it heals.

11.7 After-Care of Grafted and Budded Plants

- Remove sprouts (water shoots) from the rootstock below the union.
- Loosen the tying material after the union has healed to avoid strangling.
- Provide shade and regular watering to the young plant.
- Stake the plant to protect the union from wind.
- Protect the new growth from pests and diseases.

11.8 Rootstock-Scion Combinations Used in Practice

Crop	Scion	Rootstock	Note
Mango	Neelam, Banganapalli, Alphonso	Olour seedling	Olour gives a vigorous, well-anchored tree
Citrus	Sathgudi, acid lime	Rough lemon, Rangpur lime	Rough lemon is drought-tolerant
Sapota	PKM-1, Cricket Ball	Seedling sapota (rayan)	Adapted to dry conditions
Rose	Hybrid varieties	Seedling / cutting stock	Budded on a vigorous stock

Table 11.4 Common combinations in Tamil Nadu and India

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Key points to remember

- Grafting joins a scion to a rootstock; budding uses a single bud.
- Cambium contact, compatibility, moisture and right season are essential.
- Grafting methods: whip/tongue, cleft, side, veneer, approach, stone, bark, inarching.
- Budding methods: T-budding, inverted T, patch, ring, chip, Forkert.
- Budding needs less scion material and is faster than grafting.
- T-budding is standard in citrus and rose; veneer grafting in mango and cashew.
- Union must be protected and kept moist until healed.

Short-answer questions

Q1. Differentiate grafting from budding.

Model answer: Grafting uses a shoot piece with one or more buds; budding uses a single bud with a shield of bark and needs less material.

Q2. Name two methods of grafting and two of budding.

Model answer: Grafting: whip/tongue and veneer. Budding: T-budding and patch budding.

Q3. Why must the cambium layers of scion and stock touch?

Model answer: Because union takes place through the cambium; without cambium contact the graft fails.

Long-answer questions

Q1. Describe the important methods of grafting with examples.

Answer outline:

1. Principle of grafting.
2. Whip/tongue, cleft, side, veneer, approach, stone and bark grafting.
3. Examples and suitability of each.
4. Requirements for success.
5. Conclusion.

Q2. Explain budding, its methods and advantages over grafting.

Answer outline:

1. Definition of budding.
2. T-budding, inverted T, patch, ring, chip and Forkert budding.
3. Advantages: less material, faster, more plants per scion.
4. Examples - citrus, rose, jackfruit.
5. Comparison table with grafting.
6. Conclusion.

Lecture 12. Use of Specialized Plant Parts in Propagation

Learning Outcomes

- After studying this lecture the student should be able to:
- List the specialised plant parts used in propagation.
- Give crops for each type of propagule.
- Explain banana sucker types and tissue culture.
- State quality standards for planting material.

Many horticultural plants are propagated by specialised vegetative structures that are naturally modified for multiplication. These are easy to use, need no special skill, and give true-to-type plants. They are common in banana, pineapple, turmeric, ginger, onion, potato and many ornamentals.

12.1 Types of Specialized Propagules

Propagule	Description	Crops
Sucker	A shoot arising from the base or underground stem	Banana, pineapple, date palm, chrysanthemum
Offset / slip	A short lateral shoot from the base	Pineapple, date palm
Runner / stolon	A slender stem that roots at nodes	Strawberry, buffalo grass
Rhizome	An underground thickened stem with buds	Turmeric, ginger, banana (corm-like), lotus
Corm	A short swollen underground stem	Colocasia, gladiolus, banana (corm)
Bulb	A modified underground bud with fleshy scales	Onion, garlic, tulip, lily
Tuber	A swollen underground stem with eyes (buds)	Potato, sweet potato
Tuberous root	A swollen root that bears buds at the crown	Dahlia, sweet potato
Clove / clove-off	A segment of a compound bulb	Garlic
Crown / stem disc	The top part of the fruit with growing point	Pineapple

Table 12.1 Specialized plant parts used in propagation

12.2 Banana - Suckers and Tissue Culture

- **Sword sucker:** A narrow, sword-shaped sucker with small base; best for planting.
- **Water sucker:** A broad, leafy sucker with a small base; poor for planting.
- **Tissue-culture plants:** Now widely used in Tamil Nadu for uniform, disease-free planting; the standard in commercial banana.
- Suckers should be free from virus and nematode; paring and hot-water treatment are used before planting.

12.3 Other Important Propagules

- **Pineapple:** Propagated by slips, suckers, crowns and hapas.
- **Turmeric and ginger:** Rhizomes (seed material called 'mother' and 'finger' rhizomes) are planted.
- **Onion and garlic:** Bulbs and cloves; onion also by seed.
- **Potato:** Tubers (seed tubers); true potato seed is used in breeding.
- **Strawberry:** Runners.

- **Colocasia and gladiolus:** Corms and cormels.
- **Date palm:** Offsets and suckers.

12.4 Advantages of Propagating by Specialized Plant Parts

- Simple and cheap; needs little skill or special structures.
- Gives true-to-type, uniform plants.
- The propagule carries stored food, so establishment is quick.
- Large numbers can be handled easily in the field.
- Ideal for crops that are hard to propagate by seed or grafting.

12.5 Quality Standards for Planting Material

- Free from pests, diseases and viruses.
- True to type and of the recommended variety.
- Healthy, vigorous and of the correct size and age.
- For banana, use pared, treated sword suckers or certified tissue-culture plants.
- For turmeric and ginger, use sound, disease-free rhizomes with healthy buds.

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Key points to remember

- Specialized propagules are naturally modified structures for multiplication.
- Suckers: banana, pineapple, date palm; sword sucker is best in banana.
- Runners: strawberry. Rhizomes: turmeric, ginger. Tubers: potato.
- Bulbs and cloves: onion and garlic. Corms: colocasia, gladiolus.
- Tissue-culture plants are standard in commercial banana for uniformity and freedom from disease.
- Pineapple is propagated by slips, suckers, crowns and hapas.
- These methods need little skill and give true-to-type plants.

Short-answer questions

Q1. Name the best type of sucker for planting banana.

Model answer: The sword sucker, because it has a well-developed base and grows into a strong plant.

Q2. Which propagules are used in turmeric and potato?

Model answer: Turmeric - rhizomes; potato - tubers (seed tubers).

Q3. Name three crops propagated by suckers.

Model answer: Banana, pineapple and date palm.

Long-answer questions

Q1. Describe the use of specialized plant parts in propagation with examples.

Answer outline:

1. Meaning of specialized propagules.
2. Suckers, offsets, runners, rhizomes, corms, bulbs, tubers, tuberous roots, cloves, crowns.
3. Crops for each.
4. Advantages of these methods.
5. Banana sucker types and tissue culture.
6. Conclusion.

Q2. Write a note on the propagation of banana and turmeric.

Answer outline:

1. Banana: sucker types (sword and water), tissue culture, planting material treatment.
2. Turmeric: rhizome as seed material, mother and finger rhizomes.
3. Advantages and precautions.
4. Conclusion.

Lecture 13. Rootstock Influence - Stock Scion Relationship in Fruit Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Explain the stock-scion relationship.
- Describe the influence of rootstock on the scion.
- Give common rootstocks and their effects.
- Explain compatibility and interstock use.

In a grafted or budded plant, the rootstock is the lower part that provides the root system, and the scion is the upper part that provides the shoot and fruit. The two live as one plant but each influences the other. Understanding this stock-scion relationship is important because the rootstock controls many characters of the tree.

13.1 Influence of Rootstock on the Scion

- **Vigour and tree size:** A vigorous rootstock gives a large tree; a dwarfing rootstock reduces size, as in apple (M9) and in some mango and citrus rootstocks.
- **Precocity:** Some rootstocks induce early bearing; others delay it.
- **Yield and fruit quality:** Rootstock affects fruit size, colour, juice content and total yield.
- **Disease and pest resistance:** Resistant rootstocks protect against soil-borne diseases and nematodes (for example, citrus rootstocks against tristeza and phytophthora).
- **Soil adaptation:** Rootstocks differ in tolerance to salinity, alkalinity, drought and waterlogging.
- **Nutrient uptake:** Rootstock affects uptake of nutrients and hence micronutrient deficiency.
- **Anchorage and life span:** Strong root systems give better anchorage and longer life.

13.2 Influence of Scion on the Rootstock

- The scion supplies carbohydrates to the root system through the phloem.
- A very vigorous scion can increase root growth; a weak scion can reduce it.
- Scion characteristics (bearing habit, fruiting) influence the demand on roots.

13.3 Rootstocks Used in Important Fruit Crops

Crop	Rootstock	Important effect
Mango	Seedling of mango (polyembryonic types such as Olour)	Vigorous, well-anchored tree; Olour is a common rootstock in South India
Citrus	Rough lemon, Rangpur lime, trifoliate orange, sour orange	Rough lemon is vigorous and drought-tolerant; trifoliate orange is dwarfing and cold-hardy; sour orange is resistant to some diseases
Apple	M9 (dwarfing), M7, MM106, MM111	M9 is dwarfing and precocious; MM111 is vigorous and drought-tolerant
Grape	Dog ridge, 110R, 1613 (for some regions)	Nematode resistance and vigour control
Sapota	Seedling sapota (rayan)	Vigorous, adapted to dry conditions
Guava	Seedling guava, Psidium cattleianum (for some purposes)	Vigour and some disease tolerance

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Crop	Rootstock	Important effect
Pear	Quince (dwarfing), seedling pear	Quince induces dwarfing and early bearing

Table 13.1 Common rootstocks and their important effects

13.4 Stock-Scion Compatibility

- **Compatible:** Scion and stock unite and grow normally - usually within the same species or closely related species.
- **Incompatible:** Union fails or breaks later - shown by yellowing, poor growth, breakage at the union, or sudden death.
- Incompatibility may be localised (at the union) or translocated (whole-tree effect).
- Compatibility is a genetic and physiological character; it is tested before a rootstock is recommended.

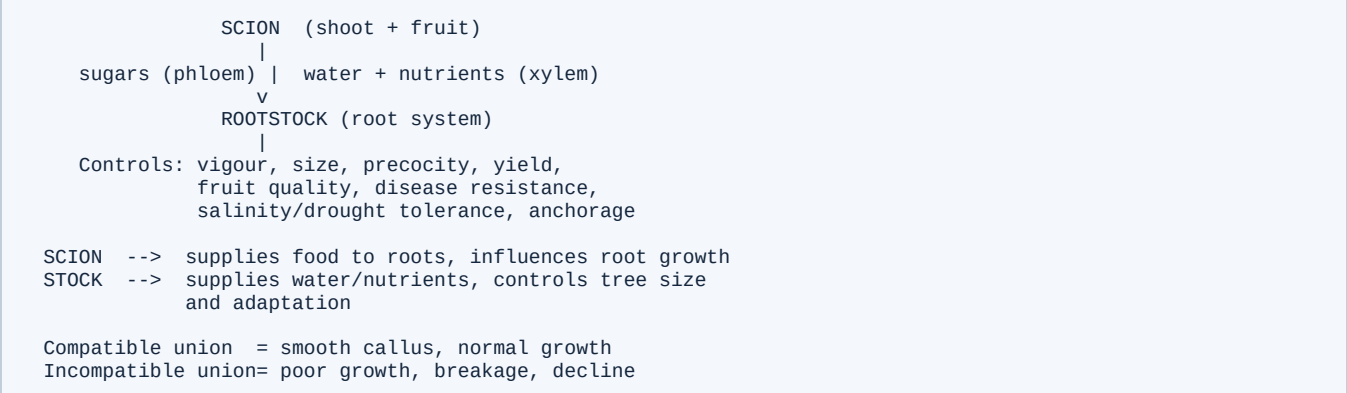


Fig. 13.1 Stock-scion relationship - two-way influence.

13.5 Interstock and Double Working

- **Interstock:** A piece inserted between the rootstock and the scion to combine characters of both.
- Used to get dwarfing plus a strong root system, or disease resistance plus good fruit quality.
- **Double working:** Two graftings - a rootstock and an interstock - used when a single stock cannot give all the needed characters.
- Interstock use adds cost and complexity but can solve specific problems.

13.6 Choice of Rootstock

- Match the rootstock to the soil and climate of the site.
- Consider disease resistance (for example, citrus rootstocks against tristeza).
- Consider the desired tree size and planting density.
- Consider precocity and the market requirement.
- Use a rootstock proven in the local area, such as Olour mango in Tamil Nadu.

13.7 Importance of Rootstock Selection in Orchards

- A wrong rootstock can cause poor growth, decline and loss of the whole orchard.
- Rootstock decides the planting density that can be used.
- It decides tolerance to soil problems such as salinity and free lime.
- It decides resistance to important diseases and nematodes.
- Always use a rootstock proven in the local soil and climate.

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Key points to remember

- Rootstock = lower (root) part; scion = upper (shoot) part of a grafted plant.
- Rootstock controls vigour, tree size, precocity, yield, fruit quality and disease resistance.
- Dwarfing rootstocks (M9 in apple, trifoliate orange in citrus) reduce tree size.
- Resistant rootstocks protect against soil-borne diseases and nematodes.
- Rootstock affects tolerance to salinity, drought and waterlogging.
- The scion supplies food to the roots and influences root growth.
- Compatibility between stock and scion is essential; incompatible unions fail or break.

Short-answer questions

Q1. What is a rootstock and what is a scion?

Model answer: The rootstock is the lower part providing the root system; the scion is the upper part providing the shoot and fruit.

Q2. Give one dwarfing rootstock each for apple and citrus.

Model answer: Apple - M9; citrus - trifoliate orange.

Q3. What are the signs of stock-scion incompatibility?

Model answer: Poor growth, yellowing, breakage at the union and eventual decline or death of the tree.

Long-answer questions

Q1. Discuss the influence of rootstock on scion in fruit crops.

Answer outline:

1. Meaning of stock and scion.
2. Influence on vigour and tree size.
3. Influence on precocity and yield.
4. Influence on fruit quality.
5. Influence on disease resistance and soil adaptation.
6. Examples from table.
7. Conclusion.

Q2. Explain stock-scion relationship and compatibility with examples.

Answer outline:

1. Two-way relationship: stock to scion and scion to stock.
2. Factors controlled by rootstock.
3. Compatible vs incompatible unions.
4. Signs of incompatibility.
5. Importance in rootstock selection.
6. Conclusion.

Lecture 14. Micropropagation of Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define micropropagation.
- Describe its steps and types.
- State its advantages and disadvantages.
- Explain its use in Tamil Nadu.

Micropropagation is the propagation of plants using tissue culture - growing small pieces of plant tissue (explants) on a sterile nutrient medium under controlled conditions to produce many identical plants. It is a rapid, large-scale method and is especially important for banana, ornamentals and disease-free planting material.

14.1 Principle and Steps

1. **Selection of mother plant:** Choose a healthy, true-to-type, disease-free plant.
2. **Preparation and sterilisation of explant:** Small pieces (shoot tip, axillary bud, leaf) are surface-sterilised.
3. **Inoculation:** The explant is placed on a sterile nutrient medium under aseptic conditions.
4. **Initiation of culture:** The explant starts to grow and multiply.
5. **Multiplication (proliferation):** Repeated sub-culturing produces many shoots.
6. **Rooting:** Shoots are transferred to a rooting medium.
7. **Hardening (acclimatisation):** Rooted plantlets are gradually adapted to field conditions.
8. **Field planting:** Hardened plants are planted in the field or nursery.

14.2 Types of Micropropagation

- **Axillary bud proliferation:** Most common; axillary buds are induced to grow into shoots.
- **Somatic embryogenesis:** Embryos formed from somatic cells.
- **Organogenesis:** Shoots or roots formed directly from tissue.
- **Meristem culture:** Using the tiny shoot meristem to obtain virus-free plants.
- **Callus culture:** Undifferentiated mass used for regeneration or suspension cultures.

14.3 Advantages and Disadvantages

Advantages	Disadvantages
Rapid, large-scale multiplication from a small explant	High initial cost and need for a laboratory
Produces disease-free (especially virus-free) plants	Requires trained technical staff
True-to-type, uniform plants	Risk of somaclonal variation (off-types)
Useful for crops that are difficult to propagate otherwise	Plants may need careful hardening
Enables year-round production of planting material	Contamination can destroy a batch
Helps in conservation of rare and endangered plants	Rooted plants are costlier than conventional ones

Table 14.1 Micropropagation - merits and demerits

14.4 Micropropagation in Tamil Nadu

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- Banana tissue-culture plants are widely used in Tamil Nadu; TNAU and private laboratories supply disease-free plantlets.
- Tissue culture is used in sugarcane, ornamentals, pineapple, and in the production of virus-free potato and citrus.
- Hardening units and shade net houses are used to acclimatise plantlets before field planting.

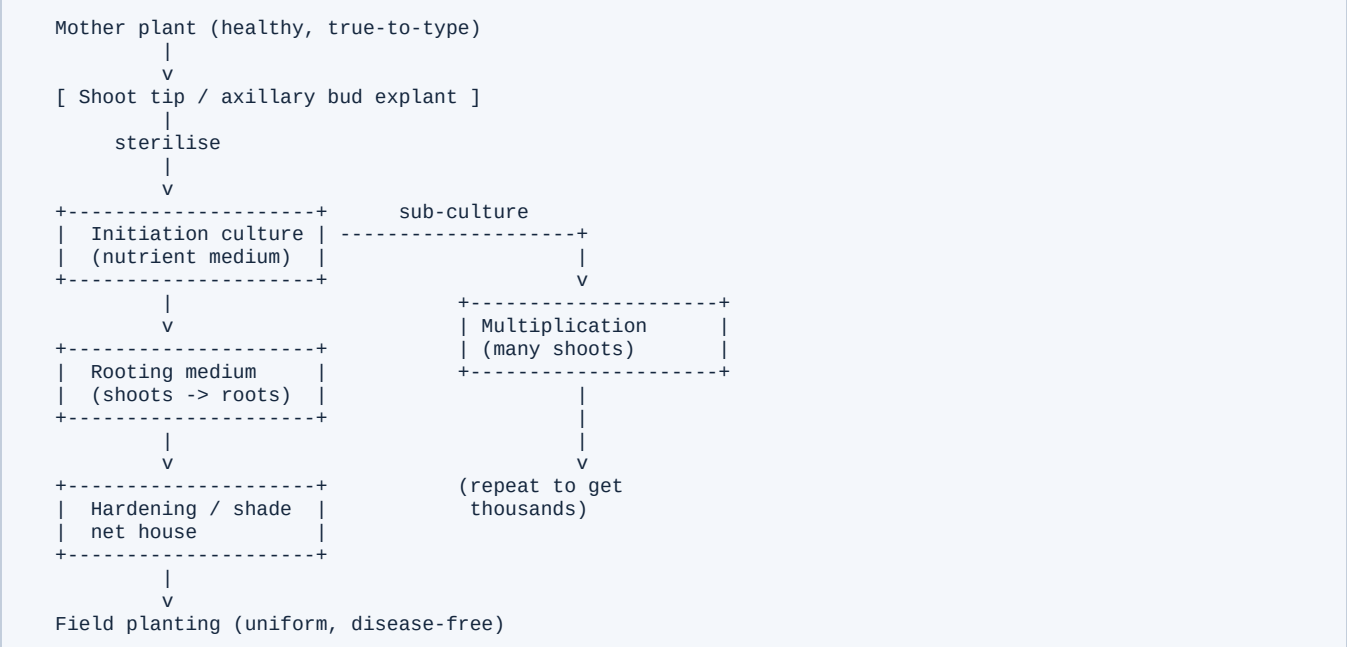


Fig. 14.1 Micropropagation cycle.

14.5 Culture Medium and Conditions

- A typical medium contains macronutrients, micronutrients, vitamins, a sugar source (sucrose), a gelling agent (agar) and growth regulators.
- The medium pH and the hormone balance (auxin to cytokinin ratio) decide whether shoots or roots form.
- Cultures are kept under controlled light (about 16 hours light) and temperature (about 25 deg C).
- Aseptic conditions are essential to prevent contamination.

14.6 Factors Affecting Micropropagation

- Genotype and species of the plant.
- Type and age of the explant.
- Medium composition and growth regulators.
- Sterilisation and aseptic technique.
- Light, temperature and sub-culturing schedule.
- Skill of the technician and laboratory discipline.

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Key points to remember

- Micropropagation is propagation using tissue culture on a sterile nutrient medium.
- Steps: mother plant selection, sterilisation, inoculation, initiation, multiplication, rooting, hardening, field planting.
- Types: axillary bud proliferation, somatic embryogenesis, organogenesis, meristem culture, callus culture.
- Meristem culture gives virus-free plants.
- Advantages: rapid, disease-free, uniform, year-round multiplication.
- Disadvantages: costly, needs a laboratory and skilled staff, risk of somaclonal variation.
- Widely used for banana in Tamil Nadu; TNAU and private labs supply plantlets.

Short-answer questions

Q1. Define micropropagation.

Model answer: It is the propagation of plants using tissue culture - growing small explants on a sterile nutrient medium to produce many identical plants.

Q2. Which type of culture gives virus-free plants?

Model answer: Meristem (shoot-tip) culture.

Q3. State two advantages of micropropagation.

Model answer: Rapid large-scale multiplication and production of disease-free, true-to-type plants.

Long-answer questions

Q1. Describe the steps and importance of micropropagation in horticulture.

Answer outline:

1. Definition of micropropagation.
2. Steps from explant to field planting.
3. Types of micropropagation.
4. Advantages and disadvantages.
5. Applications - banana, ornamentals, potato.
6. Conclusion.

Q2. Explain the advantages and limitations of tissue culture propagation.

Answer outline:

1. Rapid multiplication.
2. Disease-free planting material.
3. Uniformity and true-to-type plants.
4. Limitations: cost, skill, contamination, somaclonal variation.
5. Hardening requirement.
6. Conclusion.

Lecture 15. Vegetable Gardens - Nutrition and Kitchen Garden, Truck Garden, Riverbed Cultivation, Vegetable Forcing, Market Garden and Floating Garden

Learning Outcomes

- After studying this lecture the student should be able to:
- Describe the different systems of vegetable gardening.
- Explain the importance of a nutrition or kitchen garden.
- Describe riverbed cultivation, forcing and floating gardens.
- Compare the systems.

Vegetables can be grown in many systems depending on the purpose, the location and the market. These systems range from the small home kitchen garden for family nutrition to commercial truck gardens and specialised systems such as riverbed and floating gardens.

15.1 Nutrition Garden / Kitchen Garden

- A small garden near the house to supply fresh vegetables for the family throughout the year.
- Ensures a balanced supply of vitamins and minerals at low cost.
- Uses kitchen waste and household water; needs little space (as little as 20-50 sq m).
- Crops chosen for year-round supply: leafy vegetables (amaranthus, palak), tomato, brinjal, bhendi, chilli, beans, radish, carrot, coriander.
- Layout uses beds, borders and trellises; successive sowing keeps supply continuous.

15.2 Truck Garden

- A commercial vegetable garden located near a city or market and connected by good roads (originally where trucks could carry produce).
- Grows vegetables on a large scale for the urban market.
- Crops are chosen for high demand and quick turnover: tomato, cabbage, cauliflower, brinjal, bhendi, beans.
- Emphasis on high yield, grading and quick transport to market.

15.3 Market Garden

- A garden producing vegetables and fruits for sale in a nearby market.
- Usually a well-managed, medium-sized holding with a range of crops for steady market supply.
- Focus on quality, continuity of supply and profitability.

15.4 Riverbed Cultivation

- Cultivation of vegetables on the beds of rivers and streams during the dry season when water recedes.
- The soil is fertile (silt deposits), moist and easy to work; residual moisture supports the crop.
- Common crops: cucurbits (cucumber, watermelon, muskmelon), chilli, brinjal, tomato, leafy vegetables.
- Practised widely along riverbeds in Tamil Nadu (for example, along the Cauvery and Vaigai).
- Risk: sudden floods can destroy the crop.

15.5 Vegetable Forcing

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- Growing vegetables out of their normal season, or in a protected structure, to obtain early or off-season produce.
- Done by protected cultivation (polyhouse, shade net, greenhouse), by using the right varieties, or by soil warming.
- Gives high prices for early or off-season produce.
- Common crops: tomato, capsicum, cucumber, lettuce, exotic vegetables.

15.6 Floating Garden

- A garden grown on a floating mat of vegetation and soil on a water body.
- The classic example is the floating gardens (locally called phumdis) of Loktak Lake, Manipur, and similar systems in Kashmir (Dal Lake) and parts of Bengal.
- Crops: vegetables and some cereals grown on the floating biomass.
- Suited to areas that are waterlogged or flooded for long periods.

15.7 Comparison of Vegetable Growing Systems

System	Purpose	Scale	Typical crops
Kitchen / nutrition garden	Family nutrition	Very small	Leafy vegetables, tomato, brinjal, bhendi
Truck garden	Commercial urban market	Large	Tomato, cabbage, cauliflower, beans
Market garden	Sale in nearby market	Medium	Mixed vegetables and fruits
Riverbed cultivation	Dry-season cropping	Variable	Cucurbits, chilli, brinjal
Vegetable forcing	Off-season / early produce	Protected, small-medium	Tomato, capsicum, cucumber, lettuce
Floating garden	Waterlogged areas	Small, specialised	Vegetables on floating biomass

Table 15.1 Comparison of vegetable garden systems

15.8 A Model Nutrition Garden for a Family of Five

A plot of about 20 m x 10 m (200 sq m) can supply most vegetables for a family of five. The plot is divided into small beds and border rows, with a few fruit plants.

Crop	Area / place	Purpose
Amaranthus, palak, coriander	Successive small beds	Continuous leafy supply
Tomato, brinjal, bhendi, chilli	Main beds	Fruit vegetables
Beans, cowpea	Borders	Protein and support
Radish, carrot, beetroot	Small beds	Root vegetables
Banana, papaya, curry leaf	Corners / border	Fruit and flavouring
Marigold, jasmine	Along the fence	Flowers and pest deterrence

Table 15.2 Suggested crops in a model nutrition garden

15.9 Vegetable Forcing Structures

- **Polyhouse (greenhouse):** Controls temperature and humidity; for off-season and high-value crops.
- **Shade net house:** Reduces light and heat; suits the hot plains.
- **Low tunnel:** A low cover over beds for early crops and frost protection.
- **Mulch and row cover:** Combined with forcing to advance the crop.

15.10 Succession Cropping in the Kitchen Garden

- Sow small quantities of leafy vegetables every 15-20 days for a continuous supply.
- Follow one crop with another in the same bed (for example, radish followed by bhendi).
- Use border rows and trellises for beans and cucurbits.
- Keep a small nursery bed for raising seedlings of tomato, brinjal and chilli.

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Key points to remember

- Vegetable growing systems suit different purposes and locations.
- Kitchen/nutrition garden supplies fresh vegetables for the family at low cost.
- Truck garden is commercial and market-oriented, near cities.
- Riverbed cultivation uses fertile silt and residual moisture in the dry season.
- Vegetable forcing gives off-season or early produce, often under protected cultivation.
- Floating gardens (for example, Loktak Lake) suit waterlogged areas.
- The choice of system depends on purpose, market and location.

Short-answer questions

Q1. What is a nutrition garden?

Model answer: A small garden near the house that supplies fresh vegetables for the family throughout the year, ensuring a balanced diet at low cost.

Q2. What is riverbed cultivation and which crops suit it?

Model answer: Growing vegetables on fertile river beds in the dry season; suitable for cucurbits, chilli, brinjal and leafy vegetables.

Q3. What is vegetable forcing?

Model answer: Growing vegetables out of season or under protection to obtain early or off-season produce for high prices.

Long-answer questions

Q1. Describe the different systems of vegetable gardening.

Answer outline:

1. Kitchen/nutrition garden.
2. Truck garden.
3. Market garden.
4. Riverbed cultivation.
5. Vegetable forcing.
6. Floating garden.
7. Comparison and suitability.

Q2. Explain the importance of a kitchen garden for family nutrition.

Answer outline:

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1. Definition and size.
2. Crops for year-round supply.
3. Nutritional benefits.
4. Low cost, use of kitchen waste and water.
5. Layout and management.
6. Conclusion.

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Lecture 16. Principles, Planning and Layout of Orchards; Planting Systems - Square, Rectangular, Quincunx, Hexagonal, Contour Terracing

Learning Outcomes

- After studying this lecture the student should be able to:
- State the principles of orchard planning.
- Describe the planting systems and their merits.
- Calculate plant population for each system.
- Explain windbreaks and intercultivation.

An orchard is a systematic planting of fruit trees. Careful planning before planting saves labour, allows machinery and irrigation, ensures good light for every tree, and gives the best use of land. The arrangement of trees in an orchard is called the planting system or layout.

16.1 Principles of Orchard Planning

- Select the site with suitable soil, climate, water and drainage.
- Choose crops and varieties suited to the site and the market.
- Provide for roads, paths, irrigation channels, drainage and a windbreak.
- Plan the planting system so that every tree gets adequate light, air and nutrition.
- Keep space for a nursery, store, packing shed and future expansion.
- Align rows north-south where possible for even sunlight.
- Use a permanent layout plan and mark positions before digging pits.

16.2 Steps in Orchard Establishment

1. Selection of site and soil testing.
2. Clearing, levelling and lay-out of roads and irrigation.
3. Preparation of the layout plan on paper with spacing.
4. Marking of pit positions with pegs or lime.
5. Digging and filling of pits (usually 60 cm x 60 cm x 60 cm or larger, filled with topsoil and manure).
6. Planting of healthy, true-to-type plants at the right season.
7. Staking, watering and after-care.

16.3 Planting Systems

The main planting systems differ in the way trees are arranged and in how the land area is used.

System	Arrangement	Plant population formula (N = number of plants)
Square	Trees at corners of a square	N = Area / (spacing x spacing)
Rectangular	Rows with row spacing greater than plant spacing	N = Area / (row spacing x plant spacing)
Quincunx	Square planting with a filler tree in the centre of each square	N = (Area / s^2) + (Area / (s/2)^2)/... see worked example; about 2x the square system

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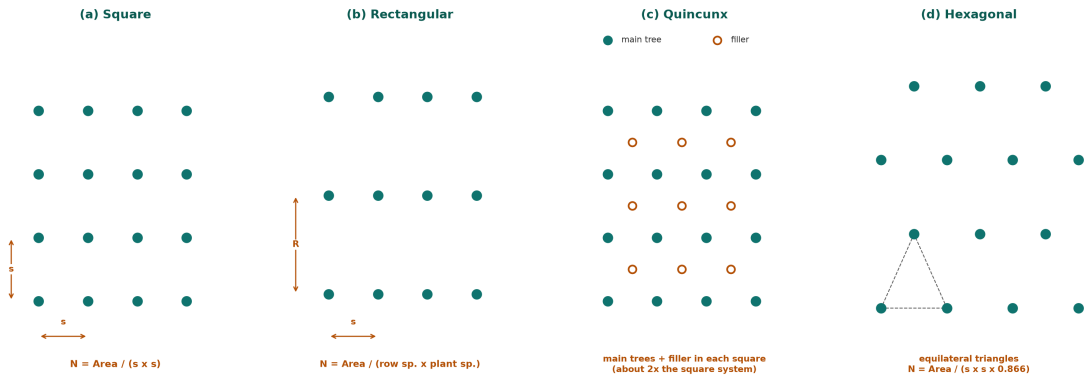


Figure 16.1 Planting systems

Square, rectangular, quincunx and hexagonal layouts with plant positions and formulas.

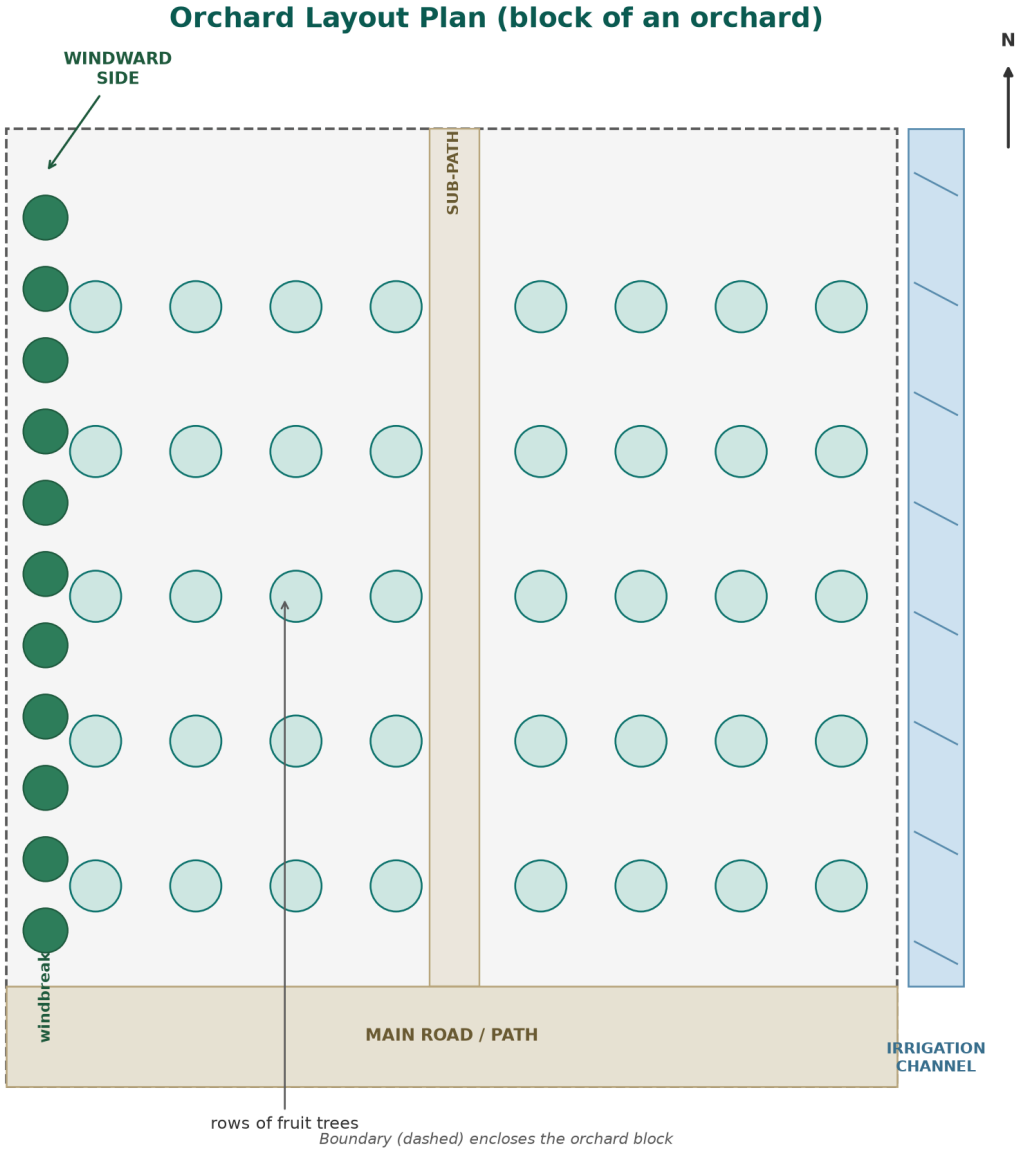


Figure 16.2 Orchard layout plan

A block of an orchard with rows, paths, windbreak and irrigation channel.

System	Arrangement	Plant population formula (N = number of plants)
Hexagonal / triangular	Trees at corners of equilateral triangles	N = Area / (spacing x spacing x 0.866)
Contour / terracing	Rows along contours on sloped land; terraces on steep land	N = Area / (spacing between rows along contour x spacing between plants)
Diagonal / filler	Square planting with fillers in between (temporary trees)	Main trees by square system plus temporary fillers

Table 16.1 Planting systems and plant population formulas

16.4 Comparison of Planting Systems

System	Merits	Demerits
Square	Simple, easy to lay out, equal spacing, intercultivation in both directions	Land between trees near the centre is less used; lower population than some systems
Rectangular	Good for mechanisation, more light along rows	Irregular spacing; less efficient use of some operations
Quincunx	Uses space efficiently with fillers; higher early income	Filler trees compete and must be removed later; management is complex
Hexagonal	About 15% more trees than square; good light and use of land	Laying out is difficult; intercultivation is awkward
Contour/terrace	Controls soil erosion on slopes; conserves moisture	Costly to make; difficult mechanisation; uneven spacing

Table 16.2 Merits and demerits of planting systems

16.5 Worked Example - Plant Population

Example 1 (Square system). A mango orchard of 1 hectare (10,000 sq m) is planted at 8 m x 8 m. Calculate the number of plants.

N = Area / (spacing x spacing) = 10,000 / (8 x 8) = 10,000 / 64 = **156 plants** (about 156).

Example 2 (Rectangular system). The same orchard is planted at 8 m (between rows) x 5 m (between plants). N = 10,000 / (8 x 5) = 10,000 / 40 = **250 plants**.

Example 3 (Hexagonal system). Spacing 8 m on the triangle. N = Area / (spacing x spacing x 0.866) = 10,000 / (8 x 8 x 0.866) = 10,000 / 55.4 = **180 plants** (about 15% more than the square system).

Example 4 (Quincunx system). Main mango trees at 8 m x 8 m, with a filler (for example, papaya) in the centre of each square. Main trees = 10,000 / 64 = 156. Each square of 8 m x 8 m is 64 sq m; a filler in each square means fillers = 10,000 / 64 = 156 as well, if each square gets one filler. Total = 156 + 156 = **312 plants** (156 main + 156 fillers). Fillers are removed when the main trees grow large.

16.6 Layout Diagrams

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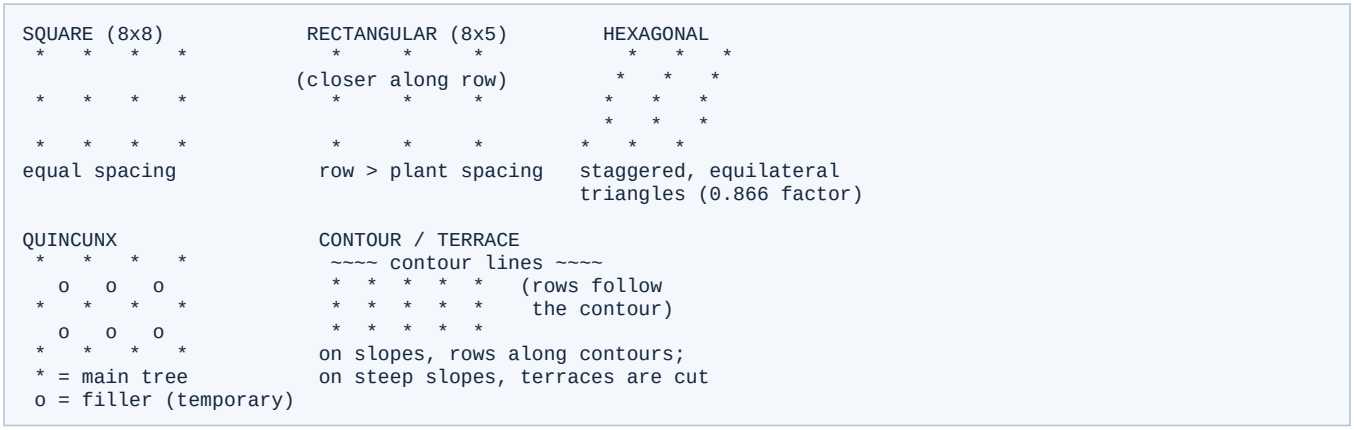


Fig. 16.1 Planting system layouts (dots are trees).

16.7 Windbreaks and Shelterbelts

- A windbreak is a row of trees planted across the direction of the prevailing wind.
- It reduces wind speed, prevents fruit drop, reduces erosion and protects young plants.
- Choose tall, hardy, fast-growing species that are not hosts of crop pests.
- Plant windbreaks before or with the orchard, on the windward side.

16.8 Intercultivation and Orchard Equipment

- Intercultivation keeps the orchard free of weeds and conserves moisture.
- Tools: spade, crowbar, pickaxe, pruning saw, secateurs, sprayer, wheelbarrow.
- Machines: tractor with cultivator, rotavator, power sprayer, drip system.
- Keep paths and drainage channels clear for easy movement and water flow.

16.10 Worked Example - Orchard Establishment Cost

Estimate the establishment cost of one hectare of a mango orchard planted at 5 m x 5 m. Plant population = 10,000 / (5 x 5) = 400 plants. The items below are illustrative; use the rates given in the question in an examination.

Item	Basis	Amount (Rs.)
Land preparation, levelling and layout	Lump sum	15,000
Digging and filling of pits	400 pits at Rs. 40	16,000
Planting material	400 plants at Rs. 120	48,000
FYM and basal fertilizer	Lump sum	20,000
Planting, staking and mulching	Lump sum	8,000
Drip irrigation system	Lump sum	60,000
Fencing and windbreak	Lump sum	40,000
First-year maintenance	Irrigation, weeding, plant protection, labour	30,000
Total establishment cost	Sum of items	2,37,000

Table 16.3 Illustrative establishment cost of a mango orchard (1 ha, 400 plants)

Cost per plant = 2,37,000 / 400 = about **Rs. 593 per plant** in the first year. If the orchard is kept as a cost for five years before full bearing, add the maintenance cost of the intervening years. Higher density raises the planting and pit cost but lowers the cost per unit of expected yield.

EXAM FOCUS

Key points to remember

- Orchard planning ensures good light, use of land, and space for operations.
- Principles: right site, right crop, roads, irrigation, windbreak, permanent layout plan.
- Planting systems: square, rectangular, quincunx, hexagonal and contour/terrace.
- Square: N = Area / (spacing x spacing).
- Rectangular: N = Area / (row spacing x plant spacing).
- Hexagonal: N = Area / (spacing x spacing x 0.866); gives about 15% more plants.
- Quincunx uses temporary fillers in the centre of squares for extra early income.

Short-answer questions

Q1. Write the formula for plant population in the square system.

Model answer: N = Area / (spacing x spacing).

Q2. Why is the hexagonal system preferred on level land?

Model answer: Because it accommodates about 15% more trees than the square system and gives good use of land and light.

Q3. What is a quincunx system?

Model answer: A square planting with a filler tree (usually temporary) in the centre of each square to use space and give early income.

Long-answer questions

Q1. Explain the principles of orchard planning and the planting systems.

Answer outline:

1. Principles of orchard planning.
2. Steps in orchard establishment.
3. Planting systems: square, rectangular, quincunx, hexagonal, contour/terrace.
4. Formulas and merits/demerits.
5. Worked example.
6. Conclusion.

Q2. Calculate the number of plants for 1 ha in square, rectangular and hexagonal systems at 8 m spacing and explain the differences.

Answer outline:

1. Square: 10,000/(8x8) = 156.
2. Rectangular at 8x5: 10,000/40 = 250.
3. Hexagonal: 10,000/(8x8x0.866) = 180.
4. Reason for differences (arrangement and land use).
5. Choice of system.

Lecture 17. High Density and Ultra High Density Planting Systems and Calculation of Planting Densities

Learning Outcomes

- After studying this lecture the student should be able to:
- Define HDP and UHDP.
- Calculate planting density for different spacings.
- Compare the economics of traditional and HDP orchards.
- State the management needs of HDP.

High density planting (HDP) is the planting of fruit trees closer than the traditional spacing so that more trees are accommodated per hectare. It aims to increase yield per unit area, achieve early and higher returns, and make better use of land, labour and inputs. Ultra high density planting (UHDP) uses even closer spacing and is common in crops such as mango, guava and citrus.

17.1 Concepts of Density

System	Spacing (indicative)	Plants per ha (approx.)	Aim
Traditional / low density	10 m x 10 m or wider	100 or fewer	Large trees, long life, low initial cost
High density planting (HDP)	5 m x 5 m to 6 m x 6 m	~275 to ~400	More plants, early returns, higher yield per ha
Ultra high density (UHDP)	2.5 m x 2.5 m to 4.5 m x 3 m	~1000 to ~1600	Very high early yield, quick returns, intensive management

Table 17.1 Traditional, high density and ultra high density planting

17.2 Advantages and Limitations of HDP/UHDP

Advantages	Limitations
Higher plant population and higher yield per hectare	High initial cost of plants and inputs
Early bearing and quick returns on investment	Needs intensive management (pruning, nutrition, water)
Better use of land, light and inputs	Greater competition for water and nutrients
Easier mechanisation and harvesting in some crops	Higher risk of pest and disease spread
Suited to dwarfing rootstocks and compact varieties	Trees may become unmanageable if not pruned
Good for replanting and short-cycle orchards	Requires careful choice of variety and rootstock

Table 17.2 Merits and demerits of high density planting

17.3 Planting Density Calculation

Planting density is the number of plants per unit area. For any system, density = 1 / (area per plant). For a rectangular spacing, plants per hectare = 10,000 / (row spacing x plant spacing) with spacing in metres.

- Square: N per ha = 10,000 / (s x s).
- Rectangular: N per ha = 10,000 / (a x b).
- Hexagonal: N per ha = 10,000 / (s x s x 0.866).
- Density is increased by reducing spacing; the yield per plant usually falls but yield per hectare rises.

17.4 Worked Example - Planting Density

Example 1. Calculate plants per hectare for mango at 6 m x 6 m (HDP), 3 m x 3 m (UHDP) and 10 m x 10 m (traditional).

At 10 m x 10 m: N = 10,000 / 100 = **100 plants/ha**.

At 6 m x 6 m: N = 10,000 / 36 = **278 plants/ha** (about 278).

At 3 m x 3 m: N = 10,000 / 9 = **1,111 plants/ha** (about 1,111).

So UHDP gives roughly eleven times the plant population of traditional planting.

Example 2. A farmer has 2 ha and wants to plant guava at 3 m x 3 m. Number of plants = 2 x 10,000 / (3 x 3) = 20,000 / 9 = **2,222 plants**.

17.5 Worked Example - High Density Planting Economics

Compare the economics of traditional (10 m x 10 m) and high density (5 m x 5 m) mango over the early years. The numbers below are illustrative; use the prices given in the question in an examination.

Item	Traditional (10 m x 10 m)	High density (5 m x 5 m)
Plants per ha	100	400
Cost per plant (plant + pit + planting)	Rs. 200	Rs. 200
Establishment cost (plants)	Rs. 20,000	Rs. 80,000
Yield per plant at year 5 (kg)	40	20
Total yield at year 5 (kg)	4,000	8,000
Price per kg	Rs. 30	Rs. 30
Gross income at year 5	Rs. 1,20,000	Rs. 2,40,000
Net gain over plant cost at year 5	Rs. 1,00,000	Rs. 1,60,000

Table 17.3 Illustrative comparison of traditional and HDP mango (per hectare)

Conclusion: although the establishment cost is higher, HDP gives a higher gross income because the total yield per hectare is greater even though yield per plant is lower. HDP pays back faster and gives higher returns per hectare.

17.6 Crops Suited to High Density Planting

Crop	HDP spacing	UHDP spacing	Note
Mango	5 m x 5 m to 6 m x 3 m	3 m x 3 m	Needs dwarfing or pruned varieties
Guava	3 m x 3 m	2 m x 2 m	Responds well to pruning; two crops a year
Citrus	4.5 m x 4.5 m	3 m x 3 m	Dwarfing rootstocks help
Banana	1.8 m x 1.8 m	1.5 m x 1.5 m	Already a high-density crop
Pineapple	45 cm x 30 cm	Closer with paired rows	Naturally dense

Table 17.4 Crops and typical HDP/UHDP spacings

17.7 Management Practices for HDP/UHDP

- Regular pruning and topping to keep trees compact and well-lit.
- Assured irrigation, preferably drip with fertigation.
- Balanced nutrition to support the high plant population.

- Careful plant protection; dense canopies favour pests and diseases.
- Use of dwarfing rootstocks and compact varieties.

EXAM FOCUS

Key points to remember

- HDP plants trees closer than traditional spacing to raise yield per hectare.
- UHDP uses very close spacing (about 1,000-1,600 plants/ha) for quick returns.
- Density = 10,000 / (spacing x spacing) plants per hectare for square planting.
- HDP gives early bearing, higher yield per hectare and better use of inputs.
- Limitations: high cost, intensive management, competition for water and nutrients.
- UHDP suits dwarfing rootstocks and compact varieties with pruning.
- HDP usually lowers yield per plant but raises yield per hectare.

Short-answer questions

Q1. Define high density planting.

Model answer: It is the planting of fruit trees closer than traditional spacing to accommodate more plants per hectare and increase yield per unit area.

Q2. Calculate plants per hectare at 4 m x 4 m spacing.

Model answer: N = 10,000 / (4 x 4) = 10,000 / 16 = 625 plants per hectare.

Q3. Give two limitations of UHDP.

Model answer: High initial cost and intensive management (pruning, nutrition, water) with greater competition among trees.

Long-answer questions

Q1. Explain high density and ultra high density planting with their merits and demerits.

Answer outline:

1. Concepts of density.
2. HDP and UHDP spacings and populations.
3. Advantages.
4. Limitations.
5. Suitable crops and rootstocks.
6. Conclusion.

Q2. Calculate the plant population for a 1 ha mango orchard under traditional, HDP and UHDP spacings and compare the economics.

Answer outline:

1. Spacings 10x10, 6x6, 3x3.
2. Populations 100, 278, 1,111.
3. Establishment cost comparison.
4. Yield per plant vs yield per hectare.
5. Returns and payback.
6. Conclusion.

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Lecture 18. Principles, Objectives and Methods of Training in Fruit Crops and their Merits and Demerits

Learning Outcomes

- After studying this lecture the student should be able to:
- Define training and state its objectives.
- Describe the systems of training.
- Explain the merits and demerits of training.
- State the time and tools of training.

Training is the manipulation of the plant to develop a strong framework of branches that can carry a heavy crop, receive good light and be managed easily. It is done mainly in the early years to give the tree a definite shape. Training and pruning are related: training gives the shape, pruning maintains it.

18.1 Objectives of Training

- To develop a strong framework of main branches that can bear the crop without breaking.
- To give the tree a convenient shape for spraying, harvesting and intercultivation.
- To expose the maximum leaf area to sunlight for good photosynthesis.
- To balance vegetative growth and fruit production.
- To remove weak, crowded and diseased branches early.
- To make the tree easier to manage and to increase its useful life.

18.2 Principles of Training

- Work with the natural growth habit of the species.
- Develop a central leader or an open centre according to the crop.
- Keep the main branches well spaced and well distributed around the trunk.
- Maintain a balance between the top and the root system.
- Train early, when branches are young and can be bent or removed easily.

18.3 Methods (Systems) of Training

System	Description	Suitable crops	Merits	Demerits
Central leader	A single main axis with branches in tiers	Apple, pear, mango	Strong tree, good light, high yield	Tall trees, difficult harvesting
Open centre / vase	Central stem removed; 3-5 main branches spread out	Peach, plum, guava	Good light in centre, easy harvesting	Weaker framework than central leader
Modified leader	Central leader cut after some years to open the top	Apple, pear, citrus	Combines strength with light	Needs careful pruning
Bush / bush form	Low, many-branched, no clear trunk	Gooseberry, pomegranate	Low, easy to manage	Poor light if crowded
Espalier / trellis	Branches trained on a flat frame or wire	Apple, pear, grape	Saves space, ornamental	Labour-intensive
Cordon / trellis (grape)	Single or double stem trained on wire	Grape	High yield, easy management	Needs support structure

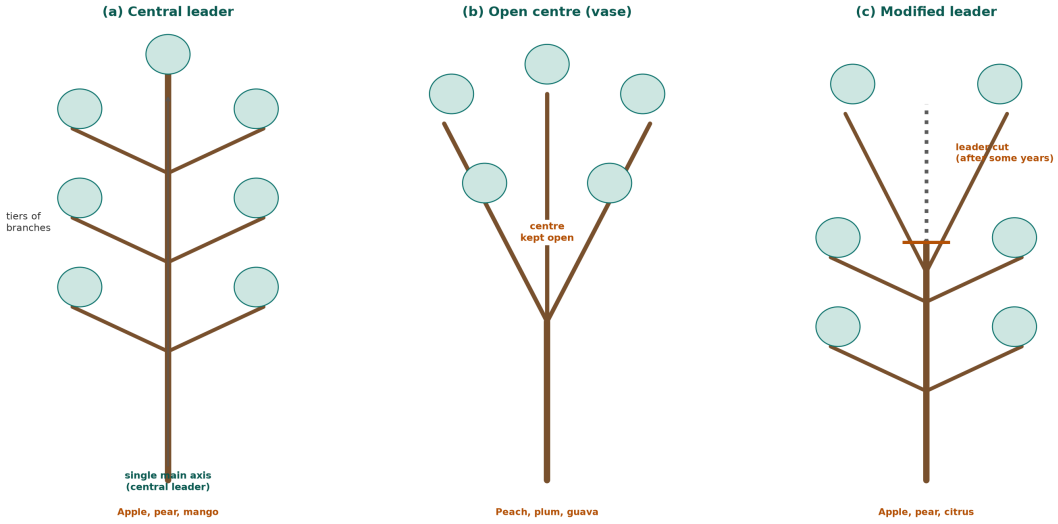


Figure 18.1 Training systems in fruit crops

Central leader, open centre (vase) and modified leader tree forms.

System	Description	Suitable crops	Merits	Demerits
Head and spread (banyan type)	Main branches spread wide, no central axis	Mango (some systems)	Wide, sturdy canopy	Large space needed

Table 18.1 Training systems in fruit crops

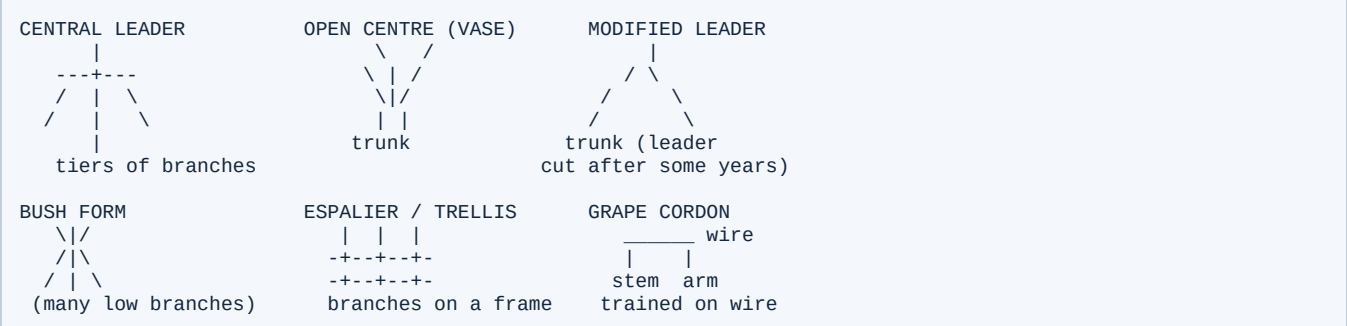


Fig. 18.1 Training systems (schematic).

18.4 Merits and Demerits of Training

Merits	Demerits
Strong framework that carries a heavy crop	Requires skill and labour, especially in the early years
Good light penetration and better fruit colour	Costly to establish and maintain certain systems
Easier spraying, harvesting and intercultivation	Wrong training can weaken the tree or delay bearing
Balances growth and fruiting; longer useful life	Some systems need supports (trellis, wire)
Removes weak and diseased branches early	Time-consuming in large orchards

Table 18.2 General merits and demerits of training

18.5 Time and Tools of Training

- Training is done mainly in the first two to four years after planting.
- The best time is the dormant season or just before active growth.
- Tools: secateurs, pruning saw, knife, and tying material for bending branches.
- Bending and tying branches spreads the canopy and improves light.
- Remove training aids once the branches hold their position.

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Key points to remember

- Training develops the shape and framework of the tree; pruning maintains it.
- Objectives: strong framework, good light, easy management, balanced growth.
- Systems: central leader, open centre (vase), modified leader, bush, espalier, cordon, head and spread.
- Central leader suits apple, pear and mango; open centre suits peach, plum and guava.
- Training should follow the natural growth habit and be done early.
- Merits: strong tree, good light, easy operations, longer life.
- Demerits: needs skill and labour; wrong training harms the tree.

Short-answer questions

Q1. Define training.

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Model answer: Training is the manipulation of the plant to develop a strong framework of branches of a desired shape for good light, easy management and heavy cropping.

Q2. Name three systems of training.

Model answer: Central leader, open centre (vase) and modified leader.

Q3. Which training system suits peach and guava?

Model answer: The open centre (vase) system.

Long-answer questions

Q1. Explain the objectives, principles and methods of training in fruit crops.

Answer outline:

1. Definition of training.
2. Objectives.
3. Principles.
4. Methods (systems) with suitable crops.
5. Merits and demerits.
6. Conclusion.

Q2. Describe the training systems suitable for different fruit crops with a table.

Answer outline:

1. Central leader - apple, pear, mango.
2. Open centre - peach, plum, guava.
3. Modified leader - apple, citrus.
4. Bush and espalier.
5. Grape cordon.
6. Merits and demerits of each.

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Lecture 19. Principles, Objectives, Types and Methods of Pruning and Canopy Management in Fruit Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define pruning and canopy management.
- State the objectives and principles of pruning.
- Describe the types of pruning.
- Explain crop regulation.

Pruning is the removal of a part of the plant - shoots, branches, roots or flowers - to influence growth, yield and quality. Canopy management is the planned manipulation of the tree canopy, mainly by pruning and training, to keep it productive, well-lit and manageable.

19.1 Objectives of Pruning

- To remove dead, diseased, weak and crowded branches.
- To maintain the desired shape and size of the tree.
- To regulate the balance between vegetative growth and fruiting.
- To improve light penetration and air movement inside the canopy.
- To increase fruit size, colour and quality.
- To remove water sprouts and root suckers.
- To rejuvenate old and declining trees.

19.2 Principles of Pruning

- Prune in the right season (usually dormant or after harvest as per crop).
- Do not remove more than necessary; heavy pruning reduces yield in the following season.
- Make clean cuts close to the branch collar and treat large wounds.
- Balance top pruning with the root system; heavy top pruning may cause excess water sprouts.
- Prune according to the bearing habit of the crop.

19.3 Types of Pruning

Type	Purpose	Remarks
Thinning out	Remove entire branches to open the canopy	Improves light; does not stimulate much regrowth
Heading back	Cut the tips of branches	Stimulates branching and controls size
Pinching / tipping	Removal of soft growing tips	Controls growth, encourages side shoots
Disbudding	Removal of excess flower buds	Improves size and quality of remaining fruit
De-blossoming	Removal of flowers	Avoids over-cropping in the off year
Root pruning	Cutting of roots	Controls vigour; used before transplanting or to induce flowering
Rejuvenation pruning	Heavy cutting back of old trees	Restores vigour and productivity

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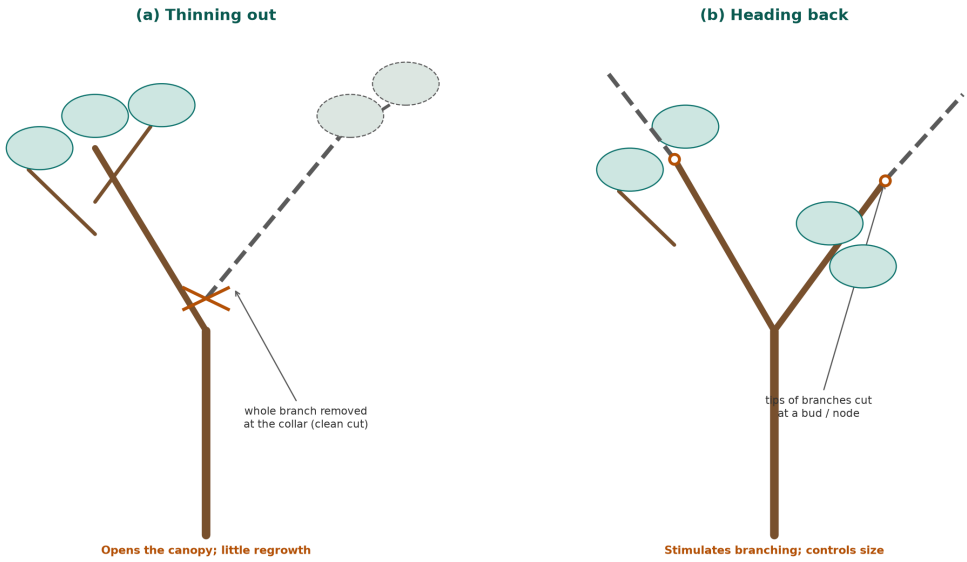


Figure 19.1 Pruning types

Thinning out compared with heading back on a labelled branch.

Type	Purpose	Remarks
Top working	Replacing the canopy with a new variety by grafting	Change of variety in an established orchard

Table 19.1 Types of pruning

19.4 Canopy Management

Canopy management aims at a canopy that is open enough for light to reach the inner branches, and compact enough to be manageable. It uses training, pruning, bending and de-horning.

- **Pruning and thinning** to remove crowding and open the centre.
- **Branch bending and training** to spread branches and improve light.
- **Topping and hedging** to control height in HDP orchards.
- **Removal of water sprouts and root suckers** to divert food to the crop.
- **Leaf removal** around bunches in grape to improve colour and reduce disease.

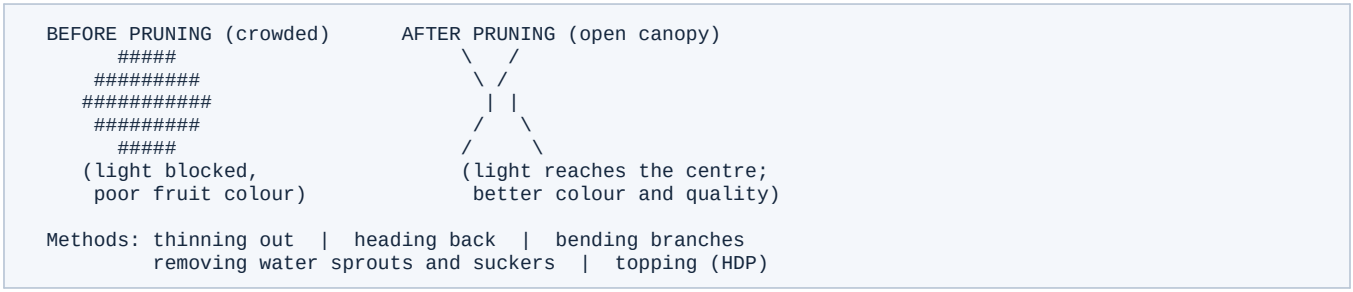


Fig. 19.1 Canopy management - opening the centre for light.

19.5 Time and Tools of Pruning

- Prune in the dormant season (temperate fruits) or after harvest (tropical fruits).
- Avoid pruning just before flowering or during heavy rain.
- Tools: secateurs, lopping shears, pruning saw, knife, wound dressing.
- Sterilise tools between trees to avoid spreading disease.
- Treat large cuts with a wound paste and remove the pruned material from the orchard.

19.6 Crop Regulation

- Crop regulation adjusts flowering and fruiting to a desired season.
- In mango, regulation is done by withholding water and applying paclobutrazol, or by de-blossoming to shift the crop.
- In guava, pruning after harvest regulates the next crop.
- Regulation helps to avoid glut, get better prices and reduce alternate bearing.

19.7 Merits and Demerits of Pruning

Merits	Demerits
Removes dead, diseased and weak wood	Heavy pruning reduces the next crop
Improves light and air in the canopy	Opens the tree to sunburn and disease
Controls tree size and shape	Needs skill and labour
Balances growth and fruiting	Wounds may allow entry of disease
Improves fruit size, colour and quality	Wrong timing can remove bearing wood

Table 19.2 Merits and demerits of pruning

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Key points to remember

- Pruning removes plant parts to influence growth, yield and quality.
- Canopy management keeps the canopy open, well-lit and productive.
- Objectives: remove dead/weak branches, control size, balance growth and fruiting, improve quality.
- Types: thinning out, heading back, pinching, disbudding, de-blossoming, root pruning, rejuvenation, top working.
- Prune in the right season and only as much as needed.
- Heavy pruning reduces the next crop; balance with the root system.
- Rejuvenation pruning restores old orchards; top working changes the variety.

Short-answer questions

Q1. Define pruning.

Model answer: Pruning is the removal of a part of the plant such as shoots, branches, roots or flowers to influence growth, yield and quality.

Q2. Differentiate thinning out from heading back.

Model answer: Thinning out removes whole branches to open the canopy; heading back cuts branch tips to stimulate branching and control size.

Q3. What is the aim of canopy management?

Model answer: To keep the tree canopy open, well-lit and compact so that it remains productive and easy to manage.

Long-answer questions

Q1. Explain the objectives, principles and types of pruning in fruit crops.

Answer outline:

1. Definition of pruning.
2. Objectives.
3. Principles.
4. Types with purpose and remarks.
5. Merits and demerits.
6. Conclusion.

Q2. What is canopy management? Describe its methods.

Answer outline:

1. Definition.
2. Importance of light in the canopy.
3. Methods: pruning, thinning, bending, topping, removal of water sprouts.
4. Use in HDP and rejuvenation.
5. Conclusion.

Lecture 20. Types and Use of Growth Regulators in Horticulture

Learning Outcomes

- After studying this lecture the student should be able to:
- Name the classes of growth regulators.
- State the uses of each class with crops.
- Calculate ppm concentrations.
- State the precautions in their use.

Growth regulators are natural or synthetic chemicals that control plant growth and development in very small amounts. They are used in horticulture to improve rooting, control size, induce flowering, prevent fruit drop, improve fruit size and quality, and delay or hasten ripening.

20.1 Classes of Growth Regulators

Class	Examples	Main effects
Auxins	IAA, IBA, NAA, 2,4-D	Rooting of cuttings, fruit set, prevent fruit drop, weed control at high dose
Gibberellins	GA3	Cell elongation, seed germination, break dormancy, increase fruit size, delay ripening in citrus
Cytokinins	Kinetin, BAP, zeatin	Cell division, shoot formation, delay senescence, increase branching
Abscisic acid (ABA)	ABA	Induces dormancy, closes stomata, promotes abscission, stress hormone
Ethylene	Ethephon (ethrel)	Ripening, fruit colour, abscission, sex expression in cucurbits
Growth retardants	CCC (chlormequat), MH (maleic hydrazide), paclobutrazol, daminozide	Reduce height, control vigour, induce flowering in mango

Table 20.1 Classes of plant growth regulators

20.2 Uses of Growth Regulators in Horticulture

Use	Regulator	Indicative concentration	Crop / note
Rooting of cuttings	IBA / NAA	500-5,000 ppm (quick dip); 100-500 ppm (soak)	Rose, guava, difficult cuttings
Prevent fruit drop	NAA, 2,4-D	10-20 ppm (NAA)	Mango, citrus, apple
Induce flowering	Paclobutrazol, ethephon	Soil drench of paclobutrazol per tree	Mango, litchi
Increase fruit size	GA3	10-50 ppm	Grape, citrus
Delay ripening / extend shelf life	GA3, 2,4-D	10-20 ppm	Citrus
Hasten ripening	Ethephon (ethrel)	250-1,000 ppm	Banana, mango
Control plant height	CCC, MH, paclobutrazol	As per crop	Ornamentals, mango

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Use	Regulator	Indicative concentration	Crop / note
Sex expression in cucurbits	Ethephon, GA3	As per crop	Cucumber, pumpkin (more female flowers)
Thinning of flowers/fruits	NAA, ethephon	As per crop	Apple, peach
Breaking dormancy	GA3, thiourea	As per crop	Potato, temperate fruits

Table 20.2 Uses of growth regulators with crops and indicative concentrations

20.3 Precautions in Use

- Use the correct concentration; too high a dose can damage the crop.
- Apply at the correct stage of growth.
- Prepare fresh solutions and use the correct solvent (many need alcohol or a little alkali to dissolve).
- Avoid spraying in hot midday or before rain.
- Follow label directions and safety precautions.

20.4 Worked Example - PPM Calculation

Parts per million (ppm) means milligrams of the substance per litre of solution (1 ppm = 1 mg/L). To prepare a solution of a given concentration, use: weight of chemical (mg) = ppm x volume of water (L).

Example 1. Prepare 10 litres of 50 ppm GA3 solution. Weight = 50 ppm x 10 L = 500 mg = **0.5 g GA3 in 10 litres of water**.

Example 2. Prepare 5 litres of 20 ppm NAA solution. Weight = 20 x 5 = 100 mg = **0.1 g NAA in 5 litres of water**.

Example 3. A stock solution is 1,000 ppm. How much is needed to make 20 litres of 25 ppm solution? Using C1V1 = C2V2: V1 = (25 x 20) / 1,000 = 0.5 L = **500 mL of stock in 20 litres of water**.

Note: since ppm solutions are very dilute, dissolve the small quantity of chemical first in a little alcohol or water, then make up to the final volume. Always label the solution and use it fresh.

20.5 Methods of Applying Growth Regulators

- **Foliar spray:** Most common; applied to leaves with a sprayer.
- **Dip / soak:** Used for cuttings and for treating harvested produce.
- **Soil drench:** Poured around the base; used for paclobutrazol in mango.
- **Injection:** Injected into the trunk or fruit stalk in some crops.
- **Seed treatment:** Soaking seed before sowing.

20.6 Growth Regulators in Crop Regulation and Post-Harvest Use

- Paclobutrazol is used to regulate flowering in mango and litchi.
- Ethephon is used to hasten ripening and degreening in citrus and banana.
- GA3 is used to delay ripening and extend shelf life in citrus.
- NAA is used to prevent pre-harvest fruit drop in mango and apple.
- These uses must follow correct concentration, stage and method.

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Key points to remember

- Growth regulators control growth and development in very small amounts.
- Classes: auxins, gibberellins, cytokinins, ABA, ethylene and growth retardants.
- Auxins (IBA, NAA) root cuttings and prevent fruit drop.
- GA3 breaks dormancy, increases fruit size and delays ripening.
- Ethephon hastens ripening and changes sex expression in cucurbits.
- Growth retardants (CCC, MH, paclobutrazol) control height and induce flowering in mango.
- 1 ppm = 1 mg/L; use the correct concentration and stage.

Short-answer questions

Q1. Name the five classes of plant growth regulators.

Model answer: Auxins, gibberellins, cytokinins, abscisic acid and ethylene (with growth retardants as a further group).

Q2. Which regulator is used to induce flowering in mango?

Model answer: Paclobutrazol (a growth retardant), applied as a soil drench.

Q3. How many milligrams are needed for 1 litre of a 20 ppm solution?

Model answer: 20 mg per litre (since 1 ppm = 1 mg/L).

Long-answer questions

Q1. Explain the types and uses of growth regulators in horticulture.

Answer outline:

1. Definition of growth regulators.
2. Classes with examples.
3. Uses: rooting, fruit set, size, ripening, flowering, height control.
4. Indicative concentrations.
5. Precautions.
6. Conclusion.

Q2. Write short notes on the use of auxins, gibberellins and ethylene in horticulture.

Answer outline:

1. Auxins: rooting, fruit drop prevention, 2,4-D uses.
2. Gibberellins: dormancy, fruit size, ripening delay.
3. Ethylene/ethephon: ripening, colour, sex expression.
4. Concentrations and crops.
5. Conclusion.

Lecture 21. Water Management - Types of Irrigation Methods for Horticultural Crops and their Merits and Demerits

Learning Outcomes

- After studying this lecture the student should be able to:
- State the factors affecting irrigation requirement.
- Describe the irrigation methods with merits and demerits.
- Explain drip irrigation and fertigation.
- Explain irrigation scheduling and critical stages.

Water is the most limiting input in horticulture. Fruit and vegetable crops need assured moisture at critical stages such as flowering, fruit set and fruit development. Efficient irrigation saves water, controls weeds, and improves yield and quality. Water management means applying the right quantity of water at the right time by the right method.

21.1 Factors Affecting Irrigation Requirement

- Crop and its rooting depth.
- Soil type and water-holding capacity.
- Climate: rainfall, temperature, humidity, wind.
- Stage of crop growth (critical stages need assured water).
- Method of irrigation and its efficiency.

21.2 Types of Irrigation Methods

Method	Description	Merits	Demerits
Surface - flooding	Water is let into the field and spreads	Simple, cheap; suits close-planted crops	Large water loss, waterlogging, disease
Furrow	Water runs in furrows between rows	Less water than flooding; suits row crops	Needs levelled land; furrows must be maintained
Basin	A basin around each tree	Good for fruit trees; directs water to the root zone	Labour to make basins; waterlogging if overfilled
Ring / pot	Ring basin around the trunk	Keeps water away from the trunk; suits trees	Basin must be renewed
Basin + furrow	Combination for tree crops	Efficient for orchards	More labour
Drip irrigation	Water applied drop by drop near the roots through emitters	Saves 40-60% water, less weed, fertigation possible, higher yield	High initial cost, emitters clog, needs maintenance
Sprinkler	Water sprayed like rain	Suits close crops and undulating land; frost protection	Wind affects distribution; leaf disease; high cost
Micro-sprinkler / micro-jet	Small sprinklers near the plant	Good for orchards and nurseries	Clogging; needs filtration
Sub-surface drip	Drip lines buried below the soil	Very efficient; less evaporation and weed	Costly; harder to inspect and repair

Table 21.1 Irrigation methods for horticultural crops

21.3 Drip Irrigation - Advantages

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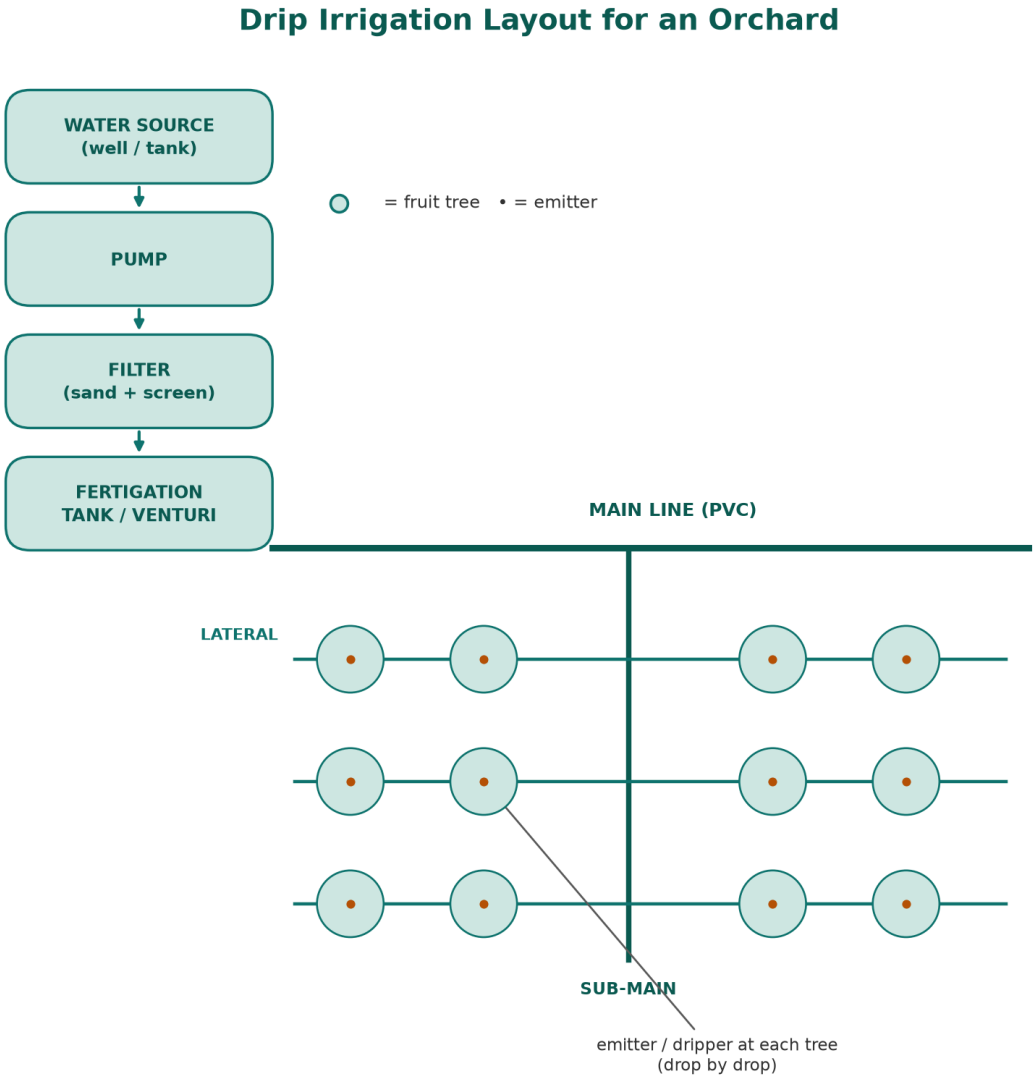


Figure 21.1 Drip irrigation layout for an orchard

Main line, sub-main, laterals and emitters around the trees.

- Saves water (about 40-60% compared with surface methods).
- Applies water directly to the root zone; less evaporation and runoff.
- Reduces weed growth between rows (dry surface).
- Allows fertigation - applying fertiliser through water, saving labour and improving efficiency.
- Improves yield and quality; reduces disease (dry foliage).
- Works on undulating land and where water is scarce.

21.4 Irrigation Layout Diagram

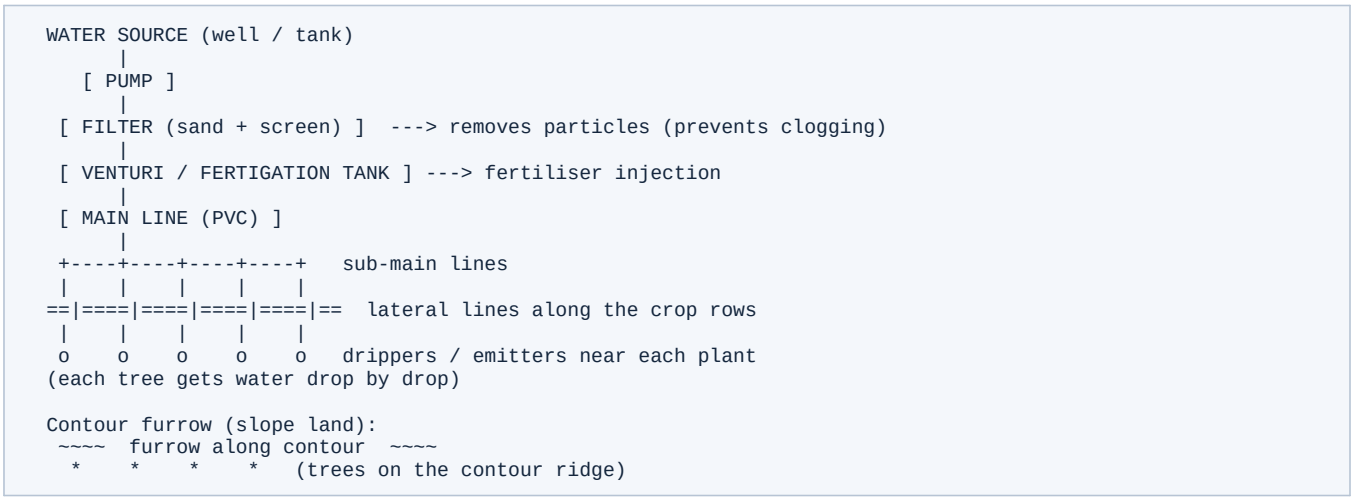


Fig. 21.1 Drip irrigation layout for an orchard.

21.5 Irrigation Scheduling and Critical Stages

- Irrigate based on soil moisture, crop stage and climate, not on a fixed calendar alone.
- Critical stages for assured water: flowering, fruit set and fruit development.
- Water stress at these stages causes heavy flower and fruit drop.
- Tools: tensiometer, soil moisture meter, and simple feel-and-see tests.
- Drip irrigation with fertigation is the most efficient for horticultural crops.

21.6 Water Requirement

- Water requirement depends on the crop, its canopy, the climate and the soil.
- Most fruit crops need about 600-1,500 mm of water per year including rainfall.
- Drip irrigation meets the requirement with much less water than surface methods.
- Excess water is as harmful as deficit water; provide drainage.

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Key points to remember

- Water management means the right quantity of water at the right time by the right method.
- Methods: surface (flooding, furrow, basin, ring), drip, sprinkler, micro-sprinkler, sub-surface drip.
- Drip saves 40-60% water, reduces weeds and allows fertigation.
- Furrow suits row crops; basin/ring suit fruit trees.
- Sprinkler suits close crops and undulating land but spreads leaf disease.
- Critical stages: flowering, fruit set and fruit development need assured water.
- Choice of method depends on crop, soil, water availability and cost.

Short-answer questions

Q1. Name any four irrigation methods.

Model answer: Furrow, basin, drip and sprinkler.

Q2. State two advantages of drip irrigation.

Model answer: It saves water (40-60%) and allows fertigation, improving fertiliser efficiency.

Q3. Why is a basin or ring method used for fruit trees?

Model answer: Because it directs water to the root zone of each tree and keeps water away from the trunk.

Long-answer questions

Q1. Describe the irrigation methods used for horticultural crops with merits and demerits.

Answer outline:

1. Importance of water management.
2. Surface methods: flooding, furrow, basin, ring.
3. Drip and micro-irrigation.
4. Sprinkler and sub-surface drip.
5. Merits and demerits of each.
6. Conclusion.

Q2. Write a note on drip irrigation and fertigation in horticulture.

Answer outline:

1. Principle of drip irrigation.
2. Components (filter, main, sub-main, laterals, emitters).
3. Advantages - water saving, weed control, yield.
4. Fertigation and its benefits.
5. Limitations.
6. Conclusion.

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Lecture 22. Weed Management - Types of Weeds and their Management in Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define a weed and state the harm it causes.
- Classify weeds with examples.
- Describe the methods of weed management.
- Explain herbicide types and integrated weed management.

A weed is any plant growing where it is not wanted. Weeds compete with the crop for water, nutrients, light and space, harbour pests and diseases, and reduce yield and quality. Weed management is therefore essential in horticulture, where the crop is grown at wide spacing and weeds get plenty of room.

22.1 Types of Weeds

Basis	Types	Examples
Life cycle	Annual	Echinochloa (barnyard grass), Amaranthus, Trianthema
	Biennial	Wild carrot, some thistles
	Perennial	Cynodon dactylon (hariali), Cyperus rotundus (nutgrass), Sorghum halepense
Morphology	Grasses	Cynodon, Echinochloa, Digitaria
	Sedges	Cyperus rotundus (nutgrass)
	Broad-leaved	Amaranthus, Parthenium, Euphorbia
Habitat	Terrestrial	Most field weeds
	Aquatic	Water hyacinth, Pistia
	Parasitic	Striga, Orobanche, Cuscuta (dodder)

Table 22.1 Classification of weeds

Nutgrass (*Cyperus rotundus*) and hariali (*Cynodon dactylon*) are among the most troublesome perennial weeds of Indian horticultural fields because they spread by underground tubers and stolons.

22.2 Methods of Weed Management

Method	Description	Remarks
Hand weeding / hoeing	Removal by hand or hoe	Effective but labour-intensive and costly
Mechanical	Tractor-drawn cultivators, mowers	Suits large orchards; not between close plants
Cultural	Crop rotation, closer spacing, competitive crops, timely planting	Reduces weed build-up; low cost
Mulching	Covering soil with organic or plastic mulch	Smothers weeds and conserves moisture
Solarisation	Covering moist soil with transparent plastic in summer	Kills weed seeds and soil pests by heat
Biological	Use of natural enemies (insects, pathogens)	For example, against water hyacinth; slow but safe

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Method	Description	Remarks
Chemical (herbicides)	Application of herbicides	Quick and effective; must be selective and safe

Table 22.2 Methods of weed management

22.3 Herbicides

- **Pre-emergence herbicides:** Applied to the soil before weeds emerge - for example, pendimethalin, atrazine, diuron.
- **Post-emergence herbicides:** Applied to growing weeds - for example, glyphosate (non-selective), 2,4-D (broad-leaved weeds).
- **Selective herbicides:** Kill certain weeds without harming the crop.
- **Non-selective herbicides:** Kill all vegetation; used in non-crop areas.
- Herbicides must be chosen for the crop, the weed and the stage; wrong use damages the crop.
- Integrated weed management (IWM) combines cultural, mechanical, mulching and chemical methods for lasting control.

22.4 Weed Management in Orchards and Vegetable Fields

- **Orchards:** Clean basins around trees, mow or mulch the inter-row area, use cover crops in the alleys.
- **Vegetable fields:** Timely hand weeding and hoeing, mulching, and pre-emergence herbicides.
- **Nursery:** Fine hand weeding and sterile media to avoid weeds.
- Integrated weed management gives lasting control at lower cost.

22.5 Aquatic and Parasitic Weeds

- **Water hyacinth** (*Eichhornia crassipes*) and **Pistia** clog irrigation channels and ponds.
- They are controlled by manual removal, biological agents and, in channels, by mechanical means.
- **Parasitic weeds** such as *Striga*, *Orobanche* and *Cuscuta* draw food from the host crop and are controlled by crop rotation and resistant varieties.

22.6 Losses Caused by Weeds

- Competition for water, nutrients, light and space reduces yield.
- Weeds harbour insect pests and disease-causing organisms.
- They increase the cost of cultivation and labour.
- Some weeds are poisonous or allelopathic to crops.
- Aquatic weeds clog irrigation and drainage channels.

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Key points to remember

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- A weed is a plant growing where it is not wanted.
- Weeds compete for water, nutrients, light and space and harbour pests.
- Types by life cycle: annual, biennial, perennial; by morphology: grasses, sedges, broad-leaved.
- Nutgrass and hariali are the worst perennial weeds.
- Methods: hand weeding, mechanical, cultural, mulching, solarisation, biological, chemical.
- Pre-emergence herbicides (pendimethalin) act before weeds emerge; post-emergence (glyphosate) act after.
- Integrated weed management is the best long-term approach.

Short-answer questions

Q1. Define a weed.

Model answer: A weed is any plant growing where it is not wanted, competing with the crop for water, nutrients, light and space.

Q2. Name two perennial weeds of horticultural fields.

Model answer: Nutgrass (Cyperus rotundus) and hariali (Cynodon dactylon).

Q3. Differentiate pre-emergence from post-emergence herbicides.

Model answer: Pre-emergence herbicides are applied to the soil before weeds emerge; post-emergence herbicides are applied to growing weeds.

Long-answer questions

Q1. Explain the types of weeds and their management in horticultural crops.

Answer outline:

1. Definition of weed and harm caused.
2. Classification by life cycle, morphology and habitat with examples.
3. Methods of management.
4. Herbicides - types and examples.
5. Integrated weed management.
6. Conclusion.

Q2. Write a note on chemical weed control in horticulture.

Answer outline:

1. Role of herbicides.
2. Pre-emergence and post-emergence.
3. Selective and non-selective.
4. Examples and crops.
5. Precautions and integrated approach.
6. Conclusion.

Lecture 23. Fertility Management, Manures and Fertilizers and Methods of Application for Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Name the essential plant nutrients.
- Compare manures and fertilizers.
- Calculate fertilizer doses.
- Describe methods of application and deficiency symptoms.

Fertility management means supplying the nutrients that a crop needs to give a good yield and quality without harming the soil. Horticultural crops remove large quantities of nutrients because of their high yields, so both organic manures and chemical fertilizers are needed, applied by the right method at the right time.

23.1 Essential Plant Nutrients

- **Primary macronutrients:** Nitrogen (N), phosphorus (P), potassium (K).
- **Secondary macronutrients:** Calcium (Ca), magnesium (Mg), sulphur (S).
- **Micronutrients:** Iron (Fe), manganese (Mn), zinc (Zn), copper (Cu), boron (B), molybdenum (Mo), chlorine (Cl), nickel (Ni).
- Carbon, hydrogen and oxygen come from air and water.

23.2 Manures and Fertilizers

Point	Manures (organic)	Fertilizers (inorganic)
Source	Plant and animal waste, compost, green manure	Chemically manufactured salts
Nutrient content	Low and variable	High and definite
Effect on soil	Improves structure, water holding, microbial activity	Little effect on structure; supplies nutrients
Bulk	Bulky; needed in large quantities	Concentrated; small quantities
Availability of nutrients	Slow release	Quick release
Examples	FYM, compost, vermicompost, green manure, oil cakes, bone meal	Urea, SSP, DAP, MOP, complex fertilizers

Table 23.1 Comparison of manures and fertilizers

23.3 Common Fertilizers and their Nutrient Content

Fertilizer	Nutrient supplied	Approximate content
Urea	Nitrogen	46% N
Ammonium sulphate	Nitrogen	20-21% N
Single super phosphate (SSP)	Phosphorus	16% P2O5
Di-ammonium phosphate (DAP)	N and P	18% N, 46% P2O5
Muriate of potash (MOP)	Potassium	60% K2O

Fertilizer	Nutrient supplied	Approximate content
Sulphate of potash (SOP)	Potassium	50% K2O
Complex fertilizers (for example, 19:19:19)	N, P, K	19% each of N, P2O5, K2O

Table 23.2 Common fertilizers and approximate nutrient content

23.4 Methods of Fertilizer Application

Method	Description	Use
Broadcasting	Spreading over the whole field	Close-planted crops, basal dose
Band placement	Placing fertilizer in bands near the row	Row crops; reduces loss
Ring / trench placement	Placing in a ring or trench around the tree	Fruit trees
Pit method	Placing in pits around the plant	Perennial crops
Foliar spray	Spraying a dilute solution on leaves	Quick correction of deficiency
Fertigation	Applying through drip irrigation	Orchards and protected crops; efficient
Injection / soil drench	Injecting into soil or drenching around the trunk	Trees; some growth regulators

Table 23.3 Methods of application

23.5 Worked Example - Fertilizer Dose Calculation

The recommended dose for a crop is usually given as kg of N, P2O5 and K2O per hectare or per tree. Convert this to actual fertilizer quantity using the nutrient content.

Example 1 (per hectare). A vegetable crop is recommended 120 kg N, 60 kg P2O5 and 60 kg K2O per hectare. Calculate the quantity of urea (46% N), SSP (16% P2O5) and MOP (60% K2O).

Urea = 120 x 100 / 46 = **261 kg urea/ha**.

SSP = 60 x 100 / 16 = **375 kg SSP/ha**.

MOP = 60 x 100 / 60 = **100 kg MOP/ha**.

Example 2 (per tree). A mature mango tree is recommended 1 kg N, 0.5 kg P2O5 and 1 kg K2O per year. Fertilizer per tree:

Urea = 1 x 100 / 46 = **2.17 kg**; SSP = 0.5 x 100 / 16 = **3.13 kg**; MOP = 1 x 100 / 60 = **1.67 kg per tree per year**. If the orchard has 100 trees, multiply by 100 for the total requirement.

Example 3 (split application). If the nitrogen is to be applied in three equal splits, each split = 261 / 3 = **87 kg urea per hectare per split**.

23.6 Nutrient Cycle in an Orchard

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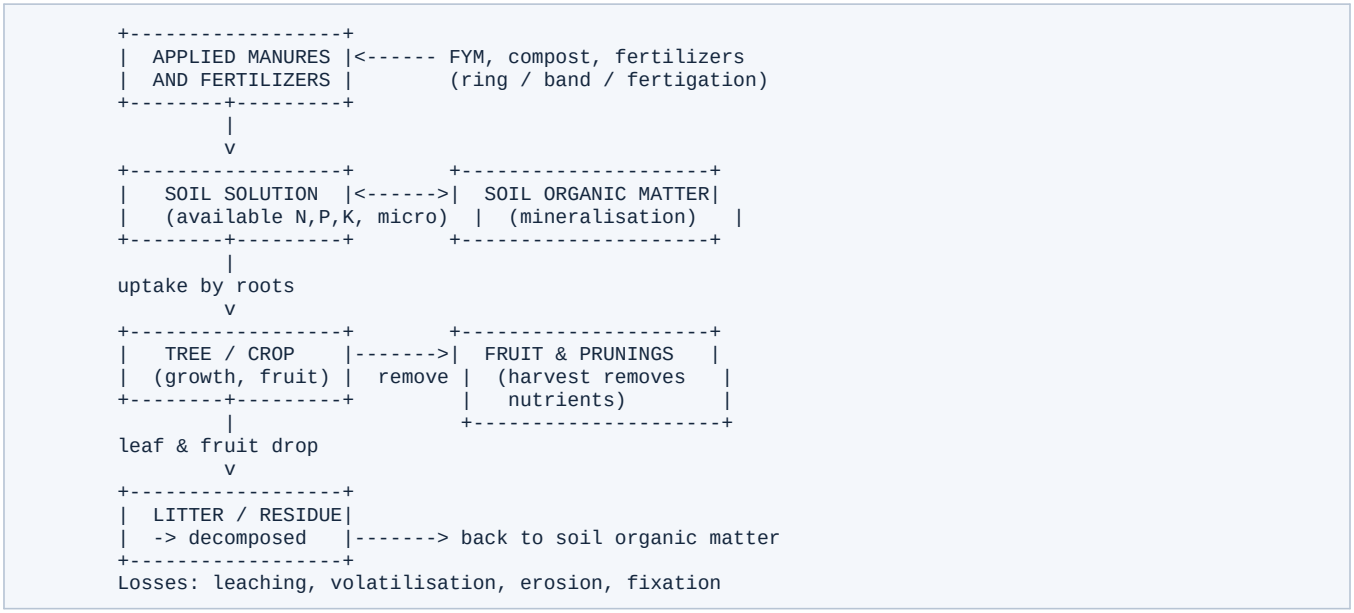


Fig. 23.1 Nutrient cycle in an orchard.

23.7 Integrated Nutrient Management (INM) and Organic Farming

- **INM** combines organic manures, chemical fertilizers and biofertilizers to maintain soil fertility and crop yield.
- **Biofertilizers:** *Azospirillum*, *Azotobacter* (nitrogen), *Phosphobacteria* (phosphorus), mycorrhiza, and *Rhizobium* for legumes.
- **Organic farming** relies on compost, vermicompost, green manure, oil cakes and biofertilizers, with no synthetic chemicals.
- INM improves soil health, reduces cost and is more sustainable than fertilizer alone.

23.8 Deficiency Symptoms of Important Nutrients

Nutrient	Deficiency symptom
Nitrogen	Overall pale green to yellow leaves; poor growth
Phosphorus	Dull, dark green leaves with purple tinge; poor root and flower development
Potassium	Leaf margin scorching and yellowing; poor fruit quality
Calcium	Death of growing tips; poor fruit keeping quality
Magnesium	Interveinal yellowing of older leaves
Iron	Interveinal yellowing of young leaves (common on high-lime soils)
Zinc	Small leaves (little leaf), shortened internodes (common in citrus)
Boron	Death of growing points, cracking of fruits, poor fruit set

Table 23.4 Common deficiency symptoms

23.9 Time and Split Application of Fertilizers

- Apply the basal dose (P and K, part of N) at planting or at the start of growth.
- Apply the remaining N in one or two splits during active growth and after flowering.
- In fruit trees, apply fertilizer in a ring or trench around the trunk, in the active root zone.
- Fertigation through drip gives small, frequent doses matched to crop demand.
- Avoid applying fertilizer to dry soil; water after application.

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Key points to remember

- Fertility management supplies the nutrients a crop needs without harming the soil.
- Nutrients: primary (N, P, K), secondary (Ca, Mg, S) and micronutrients (Fe, Zn, B, etc.).
- Manures are bulky, low in nutrients and improve soil; fertilizers are concentrated and quick-acting.
- Urea has 46% N, SSP 16% P2O5, MOP 60% K2O, DAP 18% N and 46% P2O5.
- Fertilizer quantity = nutrient recommended x 100 / nutrient content of fertilizer.
- Methods: broadcasting, band, ring/trench, pit, foliar spray, fertigation.
- Fertigation through drip is efficient; foliar spray corrects deficiency quickly.

Short-answer questions

Q1. Name the primary and secondary nutrients.

Model answer: Primary: nitrogen, phosphorus, potassium. Secondary: calcium, magnesium, sulphur.

Q2. How much urea is needed for 100 kg N?

Model answer: 100 x 100 / 46 = 217 kg urea (approximately).

Q3. Give two advantages of fertigation.

Model answer: It saves fertiliser and labour and applies nutrients directly to the root zone through drip irrigation.

Long-answer questions

Q1. Explain fertility management in horticultural crops, including manures and fertilizers.

Answer outline:

1. Importance of fertility management.
2. Essential nutrients.
3. Manures vs fertilizers - table.
4. Common fertilizers and their nutrient content.
5. Methods of application.
6. Conclusion.

Q2. Calculate the fertilizer requirement for a crop recommended 150 kg N, 75 kg P2O5 and 75 kg K2O per hectare and explain the method.

Answer outline:

1. Urea = 150 x 100/46 = 326 kg.
2. SSP = 75 x 100/16 = 469 kg.
3. MOP = 75 x 100/60 = 125 kg.
4. Explanation of the formula.
5. Split application and method.
6. Conclusion.

Lecture 24. Cropping Systems, Intercropping and Multi-Tier Cropping in Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define cropping systems.
- Explain intercropping and its benefits.
- Describe multi-tier cropping with a model.
- Explain crop rotation benefits.

Cropping systems are the patterns in which crops are grown on a piece of land over time and space. In horticulture, where perennial trees occupy land for many years, cropping systems that use the space between and below the trees give extra income and better use of resources.

24.1 Cropping Systems

System	Description	Example
Monocropping	One crop on the same land year after year	Continuous banana or coconut
Crop rotation	Different crops in sequence on the same land	Tomato - cowpea - maize
Intercropping	Two or more crops grown together in the same field at the same time	Coconut + banana; mango + pulses
Mixed cropping	Several crops sown together without a definite row pattern	Vegetables mixed in a home garden
Multi-tier / multi-storey cropping	Crops of different heights grown in layers	Coconut (top) + banana (middle) + pineapple (ground)
Sequential cropping	One crop after another in the same year	Paddy - vegetable
Relay cropping	A second crop sown before the first is harvested	Sowing cowpea in standing tomato

Table 24.1 Important cropping systems

24.2 Intercropping

- Growing two or more crops together in the same field at the same time in definite rows.
- Benefits: better use of land, light, water and labour; extra income; insurance against failure of one crop; weed suppression.
- Legume intercrops (cowpea, groundnut) add nitrogen to the soil.
- Choice of intercrop: should not compete with the main crop for light, water and nutrients.
- In orchards, intercrops are grown in the early years before the canopy closes.

24.3 Multi-Tier (Multi-Storey) Cropping

Multi-tier cropping grows crops of different heights and light requirements in layers so that the available light, space and nutrients are used fully. It is common in coconut and arecanut gardens and in home gardens.

```
   SUNLIGHT
      | | |
      v v v
   ==================  TIER 1 (top, full light)
     coconut palms
   ==================  TIER 2 (medium light)
     banana / papaya / mango
   ==================  TIER 3 (partial shade)
     black pepper (on coconut), cocoa, clove
   ==================  TIER 4 (ground, shade-tolerant)
     pineapple, turmeric, ginger, elephant foot yam
   ==================  GROUND COVER
     fodder / legume cover crops

  Uses light, water and nutrients at every level -> higher total income
```

Fig. 24.1 Multi-tier cropping model (coconut-based).

24.4 Benefits of Crop Rotation

- Maintains soil fertility (legumes fix nitrogen).
- Breaks the life cycle of pests, diseases and weeds.
- Improves soil structure and organic matter.
- Balances the use of nutrients from different depths.
- Reduces the risk of crop failure and stabilises income.

24.5 Intercropping in Coconut Gardens

- Coconut is a classic multi-tier crop.
- Intercrops: banana, papaya, pineapple, turmeric, ginger, elephant foot yam, black pepper, cocoa.
- Legume cover crops between the rows improve soil and suppress weeds.
- Intercropping gives extra income and uses light, water and nutrients at every level.

24.6 Choice of an Intercrop

- Choose a crop that does not compete with the main crop for light, water and nutrients.
- Prefer a legume (cowpea, groundnut) to add nitrogen.
- Choose a short-duration crop so it is harvested before the main crop needs the space.
- Avoid a crop that shares pests and diseases with the main crop.
- In orchards, intercrop only in the early years before the canopy closes.

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Key points to remember

- Cropping systems describe the pattern of crops over time and space.
- Intercropping grows two or more crops together in definite rows.
- Multi-tier cropping grows crops in layers of different heights.
- Benefits: better use of land, light, water and labour; extra income; weed control.
- Legume intercrops add nitrogen to the soil.
- Intercrops should not compete with the main crop.
- In orchards, intercropping is practised in the early years.

Short-answer questions

Q1. Define intercropping.

Model answer: Intercropping is the growing of two or more crops together in the same field at the same time in definite rows.

Q2. Give two advantages of intercropping.

Model answer: Better use of land, light, water and labour, and extra income with insurance against failure of one crop.

Q3. Give an example of multi-tier cropping.

Model answer: Coconut (top tier) + banana (middle) + pineapple (ground tier).

Long-answer questions

Q1. Explain the cropping systems followed in horticulture with examples.

Answer outline:

1. Definition of cropping systems.
2. Monocropping and rotation.
3. Intercropping and mixed cropping.
4. Multi-tier cropping.
5. Sequential and relay cropping.
6. Benefits and examples.
7. Conclusion.

Q2. Write a note on multi-tier cropping with a suitable model.

Answer outline:

1. Concept of multi-tier cropping.
2. Choice of crops for each tier.
3. Coconut-based model.
4. Benefits - light, water, nutrient use and income.
5. Precautions.
6. Conclusion.

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Lecture 25. Mulching - Objectives, Types, Merits and Demerits

Learning Outcomes

- After studying this lecture the student should be able to:
- Define mulching and state its objectives.
- Describe the types of mulching materials.
- Explain the merits and demerits of mulching.
- State the use of mulching in different crops.

Mulching is the practice of covering the soil surface around plants with a material to conserve moisture, control weeds and improve growing conditions. The material used is called mulch. It is one of the simplest and most useful practices in horticulture.

25.1 Objectives of Mulching

- To conserve soil moisture by reducing evaporation.
- To control weeds by blocking light.
- To moderate soil temperature (cooler in summer, warmer in winter).
- To reduce soil erosion and crusting.
- To improve soil structure and add organic matter (organic mulches).
- To keep fruit clean and reduce fruit rot (fruit crops).
- To improve yield and quality.

25.2 Types of Mulching Materials

Material	Type	Merits	Demerits
Straw / hay	Organic	Cheap, adds organic matter, cools soil	May carry weed seed, harbours pests, fire risk
Dry leaves / grass	Organic	Locally available, improves soil	Decomposes fast, may blow away
Compost / FYM	Organic	Adds nutrients and organic matter	Bulky, may contain weed seed
Sawdust / wood shavings	Organic	Long-lasting, good moisture retention	Ties up nitrogen during decomposition
Coir pith	Organic	Excellent moisture retention, locally available in TN	Slow decomposition; needs nitrogen supplement
Green manure / cover crop	Living	Adds nitrogen, improves soil	Competes with crop if not managed
Black polythene	Inorganic	Very good weed and moisture control, warms soil	Costly, not biodegradable, disposal problem
Silver/black polythene	Inorganic	Reflects light (less pest), controls weeds	Costly, disposal
Transparent polythene (solarisation)	Inorganic	Kills weed seed and soil pests by heat	Not for weed control during crop

Table 25.1 Types of mulching materials compared

25.3 Merits and Demerits of Mulching

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Merits	Demerits
Conserves soil moisture; saves irrigation water	Organic mulch may harbour pests and diseases
Controls weeds effectively (especially plastic mulch)	Plastic mulch is costly and causes disposal problems
Moderates soil temperature	Some organic mulches tie up nitrogen
Reduces erosion and improves soil	Improper mulch can cause waterlogging or root disease
Improves yield and fruit quality; keeps fruit clean	Mulch may need to be replaced every season

Table 25.2 General merits and demerits of mulching

In Tamil Nadu, coir pith is a widely used and locally available organic mulch, and black polythene mulch is common in vegetables and banana. Plastic mulching is promoted under state horticulture schemes for its water-saving and weed-control benefits.

25.4 Mulching in Different Crops

- **Vegetables:** Black polythene mulch for tomato, capsicum, cucumber; saves water and controls weeds.
- **Banana:** Coir pith or dry leaves around the base to conserve moisture.
- **Fruit orchards:** Organic mulch in basins and cover crops in the alleys.
- **Flowers:** Mulch to keep beds clean and moist.
- **Nursery:** Light mulch to protect young seedlings.

EXAM FOCUS

Key points to remember

- Mulching is covering the soil surface with a material called mulch.
- Objectives: conserve moisture, control weeds, moderate temperature, reduce erosion, improve soil and fruit quality.
- Organic mulches: straw, leaves, compost, sawdust, coir pith, cover crops.
- Inorganic mulches: black, silver/black and transparent polythene.
- Plastic mulch controls weeds and moisture best but is costly and non-biodegradable.
- Coir pith is a common organic mulch in Tamil Nadu.
- Mulching improves yield and quality and saves water.

Short-answer questions

Q1. Define mulching.

Model answer: Mulching is the covering of the soil surface around plants with a material (mulch) to conserve moisture, control weeds and improve growing conditions.

Q2. Name three organic and two inorganic mulching materials.

Model answer: Organic: straw, coir pith, compost. Inorganic: black polythene, silver/black polythene.

Q3. State two merits of plastic mulching.

Model answer: It controls weeds very well and conserves soil moisture (and warms the soil).

Long-answer questions

Q1. Explain the objectives, types, merits and demerits of mulching.

Answer outline:

1. Definition and objectives.
2. Organic and inorganic mulches.
3. Comparison of materials.
4. Merits and demerits.
5. Uses in Tamil Nadu (coir pith, plastic mulch).
6. Conclusion.

Q2. Write a note on the use of mulching in horticultural crops.

Answer outline:

1. Concept of mulching.
2. Effect on moisture, weeds and temperature.
3. Types of mulch.
4. Effect on yield and quality.
5. Precautions.
6. Conclusion.

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Lecture 26. Fruitfulness and Unfruitfulness - Factors Influencing Fruitfulness and Unfruitfulness

Learning Outcomes

- After studying this lecture the student should be able to:
- Define fruitfulness and unfruitfulness.
- Explain the external and internal causes of unfruitfulness.
- Describe alternate bearing and its management.
- Explain the types and control of fruit drop.

Fruitfulness is the ability of a plant to produce flowers and set fruit regularly and in good quantity. Unfruitfulness is the failure to set fruit, or the setting of very little fruit, even though the plant grows well. Understanding the causes of unfruitfulness helps in taking corrective measures.

26.1 Causes of Unfruitfulness

The causes are broadly grouped into external (environmental and nutritional) and internal (genetic and physiological) factors.

Group	Factor	Explanation
External	Temperature	Very high or low temperature injures flowers and pollen, reducing set
	Rain and humidity	Washes away pollen, prevents pollination, encourages disease
	Wind	Strong wind causes flower and fruit drop; dry wind desiccates flowers
	Light	Poor light reduces flowering and fruit set
	Nutrition	Excess nitrogen gives only vegetative growth; deficiency of boron/zinc reduces set
	Water	Water stress or waterlogging at flowering causes heavy drop
Internal (genetic)	Self-incompatibility	Pollen of the same flower/variety fails to fertilise the ovule
	Sterility	Male or female sterility (no viable pollen or ovules)
	Abortion of embryo	Embryo dies after fertilisation (often in seedless fruits)
Internal (physiological)	Poor pollination / pollinizer absent	No cross pollen available; bees absent
	Imbalance of hormones	Deficiency of auxin/gibberellin causes drop; poor flowering
	Alternate bearing	Heavy crop one year is followed by a light crop the next
	Premature fruit drop	Fruit drops before maturity due to hormonal or nutritional causes

Table 26.1 Factors causing unfruitfulness

26.2 Fruitfulness and its Improvement

- Plant suitable varieties with a proven bearing record.
- Provide pollinizers and encourage pollinators (honey bees).
- Maintain balanced nutrition (avoid excess nitrogen; correct boron and zinc).

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- Give assured irrigation, especially at flowering and fruit set.
- Use growth regulators (NAA, GA3) to improve set and reduce drop.
- Prune and train to give good light and air.
- Protect flowers from frost, wind and rain where possible.

26.3 Alternate Bearing

Alternate (biennial) bearing is the tendency of a tree to bear a heavy crop in one year (the 'on' year) and a light or no crop in the next (the 'off' year). Mango is a classic example. It is caused by depletion of stored food, hormonal imbalance and the effect of the heavy crop on the formation of next year's flowering shoots. Regulation of the crop by thinning flowers/fruits, balanced nutrition and pruning helps to reduce the swing.

26.4 Fruit Drop - Types and Control

- **Types:** Post-bloom drop, June drop (natural thinning) and pre-harvest drop.
- Causes: hormonal imbalance, poor nutrition, water stress, pests and diseases, and adverse weather.
- Control: balanced nutrition (avoid excess nitrogen), assured irrigation, plant protection, and growth regulators (NAA, GA3).
- Thinning of excess fruit reduces competition and improves the size of the remaining fruit.

26.5 Improving Fruit Set in Practice

- Provide pollinizers and honey bee hives at flowering.
- Avoid excess nitrogen; keep nutrition balanced.
- Correct boron and zinc deficiency before flowering.
- Give assured irrigation at flowering and fruit set.
- Use NAA or GA3 to improve set and reduce drop where recommended.
- Protect the crop from frost, wind and rain where possible.

EXAM FOCUS

Key points to remember

- Fruitfulness is the ability to flower and set fruit regularly and well.
- Unfruitfulness is failure or poor fruit set despite good growth.
- External causes: temperature, rain, wind, light, nutrition, water.
- Internal causes: self-incompatibility, sterility, embryo abortion, poor pollination, hormonal imbalance, alternate bearing.
- Alternate bearing is heavy crop in the 'on' year and light crop in the 'off' year (mango).
- Improvement by right varieties, pollinizers, balanced nutrition, irrigation and growth regulators.
- Correcting boron and zinc deficiency improves fruit set.

Short-answer questions

Q1. Define unfruitfulness.

Model answer: Unfruitfulness is the failure of a plant to set fruit, or the setting of very little fruit, even when growth is satisfactory.

Q2. What is self-incompatibility?

Model answer: It is the failure of pollen of the same flower or variety to fertilise the ovule, so no fruit is set without cross pollen.

Q3. What is alternate bearing? Give an example.

Model answer: Bearing a heavy crop in one year and a light or no crop in the next; mango is a classic example.

Long-answer questions

Q1. Explain the factors causing unfruitfulness in fruit crops and their remedies.

Answer outline:

1. Definition of fruitfulness and unfruitfulness.
2. External factors: climate, nutrition, water, light.
3. Internal factors: self-incompatibility, sterility, embryo abortion, poor pollination.
4. Hormonal and alternate bearing.
5. Remedies.
6. Conclusion.

Q2. Write a note on alternate bearing and its management.

Answer outline:

1. Definition and example.
2. Causes: food depletion, hormones, heavy crop effect.
3. Effects on grower income.
4. Management: crop regulation, thinning, nutrition, pruning.
5. Conclusion.

Lecture 27. Bearing Habits of Fruit Trees with Examples

Learning Outcomes

- After studying this lecture the student should be able to:
- Define bearing habit.
- Describe the types of bearing habit with examples.
- Explain the bearing habits of important crops.
- Explain crop regulation based on bearing habit.

Bearing habit is the manner and place in which a fruit tree produces its flowers and fruits. It differs from crop to crop and is important for pruning, training and crop management. Knowing where the flowers are borne tells the grower which wood to keep and which to remove.

27.1 Types of Bearing Habits

Type of bearing habit	Where flowers/fruits are borne	Examples
Terminal bearing	At the tips of shoots of the current season	Mango (panicles at the tip of shoots)
Lateral / axillary bearing	On the sides of shoots, in leaf axils	Guava, citrus, sapota
On previous year's wood	On shoots that grew in the previous season	Apple, pear, peach (spurs and shoots)
On spurs	Short, slow-growing spur shoots	Apple, pear, plum
On current season's wood	On shoots of the same season	Grape, fig, ber
On old wood / main branches	Directly on thick branches and trunk	Jackfruit, cocoa, durian (cauliflory)
Continuous / ever-bearing	Flowers and fruits almost throughout the year	Banana, papaya, guava (some types)
Periodic / seasonal	One or two definite flowering seasons	Mango, litchi, most temperate fruits

Table 27.1 Bearing habits of fruit trees

27.2 Bearing Habits of Important Crops

- **Mango:** Terminal bearing; flowers in terminal panicles; one main season (with some off-season flowering).
- **Banana:** Bears once in its life (monocarpic); the bunch comes from the main pseudostem, and suckers continue the crop.
- **Citrus:** Flowers on lateral shoots of the previous and current season; often flowers in flushes.
- **Guava:** Axillary bearing; can flower almost throughout the year; two or three crops possible.
- **Grape:** Bears on current season's shoots; pruning regulates the crop.
- **Apple and pear:** Bear mostly on spurs and previous year's wood.
- **Jackfruit:** Bears on the trunk and thick branches (cauliflory).
- **Papaya:** Flowers and fruits continuously on the trunk in leaf axils.

27.3 Crop Regulation Based on Bearing Habit

- Knowing the bearing habit decides which wood to prune and which to keep.

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- Terminal-bearing crops (mango) are pruned after harvest to encourage new terminal shoots for the next crop.
- Lateral-bearing crops (guava, citrus) can be pruned more freely without losing the bearing wood.
- Spur-bearing crops (apple) need careful pruning to preserve the spurs.
- **Bahar treatment** in mango regulates flowering to a desired season by water and hormone management.

EXAM FOCUS

Key points to remember

- Bearing habit is the manner and place of flowering and fruiting.
- Types: terminal, lateral, on previous year's wood, on spurs, on current season's wood, on old wood, continuous and seasonal.
- Mango bears terminally; guava and citrus bear laterally.
- Apple and pear bear on spurs and previous year's wood.
- Grape and fig bear on the current season's shoots.
- Jackfruit bears on the trunk and thick branches (cauliflory).
- Knowing the bearing habit guides pruning and crop regulation.

Short-answer questions

Q1. What is bearing habit?

Model answer: Bearing habit is the manner and place in which a fruit tree produces its flowers and fruits.

Q2. Where does mango bear its flowers?

Model answer: At the tips of shoots, in terminal panicles (terminal bearing).

Q3. Name two crops that bear on spurs.

Model answer: Apple and pear.

Long-answer questions

Q1. Describe the bearing habits of fruit trees with examples.

Answer outline:

1. Definition of bearing habit.
2. Types of bearing habit.
3. Crop examples for each type.
4. Importance in pruning and crop regulation.
5. Conclusion.

Q2. Explain the bearing habits of mango, banana, citrus and guava.

Answer outline:

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1. Mango - terminal bearing.
2. Banana - monocarpic, one bunch per pseudostem.
3. Citrus - lateral bearing in flushes.
4. Guava - axillary, nearly continuous.
5. Management implications.
6. Conclusion.

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Lecture 28. Pollination, Pollinizers and Pollinators in Horticulture

Learning Outcomes

- After studying this lecture the student should be able to:
- Define pollination and its types.
- Name the agents of pollination.
- Explain pollinizers and their requirements.
- Explain how pollination is improved.

Pollination is the transfer of pollen from the anther to the stigma. It is the first step in sexual reproduction and fruit set. In horticulture, many crops need a different variety nearby (a pollinizer) and a carrier of pollen (a pollinator) for good fruit set.

28.1 Types of Pollination

Type	Description	Examples
Self-pollination (autogamy)	Pollen transferred within the same flower or plant	Tomato, pea, most self-pollinated crops
Cross-pollination (allogamy)	Pollen transferred between different plants	Mango, apple, most fruit trees
Geitonogamy	Pollination between flowers of the same plant	Common in many fruit trees
Xenogamy	Pollination between flowers of different plants of the same species	Apple, pear
Cleistogamy	Pollination within a closed, unopened flower	Some self-pollinated species

Table 28.1 Types of pollination

28.2 Agents of Pollination (Pollinators)

- **Biotic:** Insects (bees, butterflies, moths, beetles, flies), birds, bats and some mammals.
- **Abiotic:** Wind (anemophily) and water (hydrophily).
- Honey bees (*Apis* species) are the most important pollinators for horticultural crops.
- Pollinator activity is reduced by pesticide sprays during flowering and by loss of habitat.

28.3 Pollinizers and Pollination Requirements

- **Pollinizer:** A variety planted to supply compatible pollen to the main variety.
- A pollinizer must flower at the same time as the main variety, be compatible, and produce plenty of good pollen.
- It is planted in a definite proportion, for example, one pollinizer row for every few rows of the main variety.
- **Self-unfruitful crops** (apple, pear, many plums) need a pollinizer.
- **Self-fruitful crops** (mango in many cases, guava, most citrus) can set fruit with their own pollen, but cross-pollination often improves set.
- Some crops (for example, banana) set fruit without pollination (vegetatively propagated, parthenocarpic).

28.4 Improving Pollination

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1. Plant suitable pollinizers in the orchard.
2. Keep honey bee hives in or near the orchard during flowering.
3. Avoid insecticide sprays at flowering time.
4. Provide irrigation and nutrition to keep flowers healthy.
5. Hand pollination in crops where it is practicable.

28.5 Floral Biology and Pollination in Mango

- The mango inflorescence is a terminal panicle with both male and hermaphrodite flowers.
- Flowers are small, and pollination is mainly by insects, especially flies and honey bees.
- Only a small percentage of hermaphrodite flowers set fruit; most flowers drop.
- Fruit set improves with a dry, warm spell and good pollinator activity.
- Rain, high humidity and low temperature at flowering reduce set.

28.6 Pollination Problems and Their Remedies

- Absence of a pollinizer in a self-unfruitful crop - plant a suitable pollinizer.
- Poor pollinator activity - keep bee hives and avoid insecticide sprays at flowering.
- Rain or wind at flowering - no direct control; choose the right season and variety.
- Poor pollen viability due to heat - provide shade and assured irrigation.

EXAM FOCUS

Key points to remember

- Pollination is the transfer of pollen from anther to stigma.
- Types: self-pollination, cross-pollination, geitonogamy, xenogamy, cleistogamy.
- Pollinators are agents of pollen transfer - insects, birds, wind, water; honey bees are most important.
- A pollinizer is a variety planted to supply compatible pollen.
- Self-unfruitful crops (apple, pear) need a pollinizer; self-fruitful crops may not.
- Pollinizer must flower at the same time and be compatible.
- Avoid insecticide sprays at flowering to protect pollinators.

Short-answer questions

Q1. Define pollination.

Model answer: Pollination is the transfer of pollen grains from the anther to the stigma of a flower.

Q2. What is a pollinizer?

Model answer: A pollinizer is a variety planted in the orchard to supply compatible pollen to the main variety, ensuring good fruit set.

Q3. Name the most important pollinator of horticultural crops.

Model answer: The honey bee (Apis species).

Long-answer questions

Q1. Explain pollination, pollinizers and pollinators with examples.

Answer outline:

1. Definition of pollination.
2. Types of pollination.
3. Agents of pollination.
4. Pollinizer - requirements and planting.
5. Self-fruitful vs self-unfruitful crops.
6. Improving pollination.
7. Conclusion.

Q2. Write a note on the role of honey bees in horticultural production.

Answer outline:

1. Honey bee as a pollinator.
2. Crops benefited.
3. Effect on fruit set and yield.
4. Need for hives during flowering.
5. Protection from pesticides.
6. Conclusion.

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Lecture 29. Fertilization and Parthenocarpy in Horticulture

Learning Outcomes

- After studying this lecture the student should be able to:
- Explain the process of fertilization.
- Describe fruit development and the role of growth regulators.
- Define parthenocarpy and its types.
- Explain apomixis with examples.

Fertilization is the fusion of the male gamete (from the pollen tube) with the female gamete (egg) inside the ovule, forming the zygote. It follows pollination and leads to the development of seed and fruit. Parthenocarpy is the development of fruit without fertilization, giving seedless fruit.

29.1 Fertilization

1. Pollen lands on the stigma and germinates, forming a pollen tube.
2. The pollen tube grows down the style and enters the ovule.
3. The male gamete fuses with the egg (syngamy) to form the zygote.
4. A second male gamete fuses with the polar nuclei (triple fusion) to form the endosperm (double fertilization).
5. The zygote develops into the embryo; the ovule becomes the seed; the ovary becomes the fruit.

If fertilization does not occur, the flower usually drops. Fruit set therefore depends on successful pollination and fertilization (except in parthenocarpic fruits).

29.2 Fruit Development and Growth

- After fertilization, the ovary grows rapidly; this is the fruit set phase.
- Fruit growth follows a sigmoid (S-shaped) or double sigmoid curve depending on the crop.
- Stages: cell division, cell enlargement, maturation and ripening.
- Auxins, gibberellins and cytokinins control fruit growth; ethylene and ABA control ripening.

29.3 Parthenocarpy

- **Definition:** Development of fruit without fertilization of the ovule, resulting in seedless fruit.
- **Natural parthenocarpy:** Occurs naturally - banana, pineapple, some citrus and grape varieties, seedless watermelon.
- **Induced parthenocarpy:** Brought about by applying growth regulators such as GA3 or auxins to unpollinated flowers - tomato, brinjal, grape, citrus.
- **Stenospermocarpy:** Fertilization occurs but the embryo aborts, leaving a small soft seed - seedless grapes.
- **Advantages:** Seedless fruit has better quality and consumer preference; it also avoids the effect of poor pollination.
- **Related term:** Parthenocarpic fruits are seedless but are not always completely without seed traces.

29.4 Vegetative Parthenocarpy and Apomixis

- **Vegetative parthenocarpy:** Seedless fruit development under conditions unfavourable for pollination (for example, some pear varieties in cold weather).

- **Apomixis:** Formation of seed without fertilization; the seed is genetically identical to the mother (nucellar embryony in mango and citrus gives true-to-type seedlings).
- Nucellar seedlings of mango (for example, Olour) are used as rootstocks because they are true to type.

29.5 Fruit Set, Fruit Drop and Thinning

- Fruit set is the retention and growth of the ovary after pollination and fertilization.
- Only a small fraction of flowers become mature fruit; the rest drop.
- Natural fruit drop acts as thinning; excess fruit is removed to improve size and quality.
- Growth regulators and balanced nutrition improve set and reduce drop.
- Fruit thinning is done by hand or chemically in apple, peach and grape.

EXAM FOCUS

Key points to remember

- Fertilization is the fusion of male and female gametes to form the zygote.
- It involves syngamy and triple fusion (double fertilization).
- The ovary becomes the fruit, the ovule becomes the seed.
- Without fertilization, flowers usually drop (except in parthenocarpic fruits).
- Parthenocarpy is fruit development without fertilization, giving seedless fruit.
- Natural parthenocarpy: banana, pineapple, seedless watermelon; induced: GA3 in tomato, grape.
- Apomixis (nucellar embryony) gives true-to-type seed in mango and citrus.

Short-answer questions

Q1. Define fertilization.

Model answer: Fertilization is the fusion of the male gamete from the pollen tube with the female egg inside the ovule, forming the zygote.

Q2. Define parthenocarpy.

Model answer: Parthenocarpy is the development of fruit without fertilization of the ovule, giving seedless fruit.

Q3. Give two examples of natural parthenocarpy.

Model answer: Banana and pineapple (also seedless watermelon).

Long-answer questions

Q1. Explain the process of fertilization and fruit development.

Answer outline:

1. Pollination and pollen tube growth.
2. Syngamy and triple fusion.
3. Development of embryo, seed and fruit.
4. Fruit growth curve and stages.
5. Role of growth regulators.
6. Conclusion.

Q2. What is parthenocarpy? Describe its types and importance.

Answer outline:

1. Definition.
2. Natural and induced parthenocarpy.
3. Stenospermocarpy.
4. Examples - banana, grape, tomato.
5. Advantages of seedless fruit.
6. Conclusion.

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Lecture 30. Rejuvenation of Old Orchards

Learning Outcomes

- After studying this lecture the student should be able to:
- State the causes of decline of old orchards.
- Describe the methods of rejuvenation.
- Explain the steps in a rejuvenation programme.
- Give a case example.

Rejuvenation is the revival of old, unproductive or declining orchards to restore their vigour, yield and quality. Old orchards often become crowded, shaded, diseased and low-yielding; with proper rejuvenation they can be made productive again at much less cost than replanting.

30.1 Reasons for Decline of Old Orchards

- Crowding and shading after years of growth.
- Senility and exhaustion of the trees.
- Accumulation of pests and diseases in the soil and canopy.
- Poor nutrition and lack of management.
- Soil problems: erosion, waterlogging, salinity.
- Injury from storms, drought or neglect.
- Uneven and unproductive varieties in the orchard.

30.2 Methods of Rejuvenation

Method	Description	Use
Pruning of dead and diseased wood	Removal of dead, diseased and crowded branches	First step in every rejuvenation programme
Heading back / heavy pruning	Cutting back main branches to reduce the canopy and force new growth	Old, senile trees
De-horning	Complete removal of the canopy to force new shoots	Very old, unproductive trees
Top working	Grafting improved varieties on the existing rootstock	Changing an unproductive variety
Root pruning and soil treatment	Cutting back old roots and improving the soil	Declining trees with poor root activity
Nutrition and irrigation	Manuring and assured irrigation to support new growth	All rejuvenated trees
Pest and disease control	Cleaning the orchard and treating problems	Before and after pruning
Thinning of trees	Removal of some trees to reduce crowding	Over-crowded orchards

Table 30.1 Methods of rejuvenation

30.3 Steps in Rejuvenation

1. Survey the orchard and select trees worth rejuvenating.
2. Clean the orchard; remove dead wood and infected material.
3. Prune or de-horn to reduce the canopy and force new shoots.

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4. Treat wounds and apply a protective paste.
5. Apply manure, fertilizer and irrigation to support regrowth.
6. Train the new shoots and select the best ones for the new canopy.
7. Control pests and diseases on the new growth.
8. Top work if a variety change is wanted.

30.4 Case Example - Rejuvenation of a Declining Mango Orchard

- A neglected, crowded mango orchard with low yield is selected.
- Dead, diseased and crowded branches are removed; the canopy is headed back.
- Wounds are treated and the orchard is cleaned.
- FYM, fertilizer and irrigation are applied to force new growth.
- New shoots are trained and the best are retained to form a new framework.
- Pests and diseases on the new growth are controlled.
- The orchard returns to good production within a few years at far less cost than replanting.

30.5 Precautions in Rejuvenation

- Do not de-horn a healthy, productive tree.
- Rejuvenate in the right season, not just before flowering.
- Support the tree with nutrition and irrigation before and after pruning.
- Treat all wounds to prevent disease entry.
- Top work only with a compatible, recommended variety.

EXAM FOCUS

Key points to remember

- Rejuvenation restores vigour, yield and quality to old orchards.
- Causes of decline: crowding, senility, pests, poor nutrition, soil problems.
- Methods: pruning, heading back, de-horning, top working, root pruning, nutrition.
- De-horning removes the whole canopy to force new shoots.
- Top working changes the variety on the existing rootstock.
- Rejuvenation is cheaper than replanting.
- After rejuvenation, train new shoots and support with nutrition and plant protection.

Short-answer questions

Q1. Define rejuvenation of an orchard.

Model answer: Rejuvenation is the revival of old, declining or unproductive orchards to restore their vigour, yield and quality.

Q2. What is de-horning?

Model answer: De-horning is the complete removal of the canopy of an old tree to force a new framework of shoots.

Q3. State two reasons for the decline of old orchards.

Model answer: Crowding and shading, and accumulation of pests and diseases.

Long-answer questions

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Q1. Explain the causes of decline and the methods of rejuvenation of old orchards.

Answer outline:

1. Meaning of rejuvenation.
2. Causes of decline.
3. Methods: pruning, de-horning, top working, root pruning.
4. Steps in rejuvenation.
5. Advantages over replanting.
6. Conclusion.

Q2. Describe a rejuvenation programme for an old mango orchard.

Answer outline:

1. Survey and selection of trees.
2. Cleaning and removal of dead wood.
3. Heading back / de-horning.
4. Wound treatment.
5. Manuring and irrigation.
6. Training new shoots and top working.
7. Plant protection and conclusion.

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Lecture 31. Maturity; Methods for Judging Maturity in Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Define maturity, physiological and commercial maturity.
- Describe the methods of judging maturity.
- State the maturity indices of important crops.
- Explain changes during ripening.

Maturity is the stage at which a fruit or vegetable has reached full development and is ready for harvest for a particular purpose. Judging maturity correctly is very important: immature produce is poor in quality and does not ripen well, while over-mature produce spoils quickly and cannot be transported or stored.

31.1 Maturity and Ripening

- **Horticultural maturity:** The stage at which the produce is ready for harvest for the intended use (fresh market, storage, processing).
- **Physiological maturity:** The stage when the fruit completes its development and can ripen normally after harvest.
- **Commercial maturity:** The stage at which the produce is harvested for a specific market or transport distance.
- **Ripening:** The changes after maturity - colour, softening, sugar development, aroma - making the fruit edible.

31.2 Methods of Judging Maturity

Method	Basis	Examples
Visual	Colour, size, shape, surface features	Colour change in mango, tomato; size in banana
Physical	Firmness, ease of separation, specific gravity	Firmness in mango; abscission layer in guava
Chemical	Total soluble solids (TSS), acidity, sugar:acid ratio, starch	TSS in grape and mango; starch in apple
Physiological	Days from flowering/fruit set; heat units	Days to maturity in banana, pineapple
Sensory / organoleptic	Taste, aroma, feel	Field judgement by experienced growers
Instrumental	Penetrometer, refractometer, colour charts	Penetrometer for firmness; refractometer for TSS

Table 31.1 Methods of judging maturity

31.3 Maturity Indices for Important Crops

Crop	Maturity index	Indicative value
Mango	Colour, TSS, specific gravity, days from flowering	TSS about 12-15% (varies); shoulders fill out; some colour
Banana	Days from shooting, angularity of fingers, colour	Fingers lose angularity; about 75-80% maturity for distant market

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Crop	Maturity index	Indicative value
Guava	Colour change, firmness, TSS	Skin turns light green to yellowish; TSS about 10-12%
Citrus (acid lime)	Colour, juice content, TSS:acid ratio	Juice content and colour change
Grape	TSS, colour, sugar:acid ratio	TSS about 16-20% depending on variety
Sapota	Scratching the skin (brownish colour under skin), TSS	Brown streaks under skin indicate maturity
Tomato	Colour (breaker to red), firmness	Breaker stage for transport; red for local market
Onion	Tops falling over, neck drying, skin colour	50% tops fallen and dried
Potato	Skin set, top drying	Skin does not peel off; haulms dry
Cabbage	Head firmness and size	Firm, well-filled head
Carrot	Root size and colour	Root reaches marketable size before becoming woody
Cauliflower	Curd size, compactness	Compact, unopened curd

Table 31.2 Maturity indices for major horticultural crops

Values given are indicative ranges for typical varieties; the exact index depends on variety, season and market. In an examination, quote the index and its approximate value, noting that it varies.

31.4 Changes During Ripening and Factors Affecting Maturity

- Ripening changes: colour development, softening, sugar increase, acidity decrease, aroma and vitamin changes.
- Climacteric fruits (mango, banana) ripen after harvest with a rise in respiration; non-climacteric fruits (citrus, grape) do not.
- Factors affecting maturity: variety, climate, nutrition, irrigation, crop load and light.
- Correct harvest maturity decides storage life, transport ability and eating quality.

31.5 Maturity and Storage Life

- Produce harvested at the right maturity stores and travels best.
- Mature-green tomato travels well; fully red tomato does not.
- Banana for distant markets is harvested at about 75-80% maturity.
- Over-mature fruit softens quickly and spoils in transit.
- Under-mature fruit is poor in quality and may not ripen properly.

EXAM FOCUS

Key points to remember

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- Maturity is the stage at which produce is ready for harvest for a purpose.
- Horticultural, physiological and commercial maturity are different concepts.
- Methods: visual, physical, chemical, physiological, sensory and instrumental.
- TSS (refractometer) and firmness (penetrometer) are common objective measures.
- Mango: TSS and colour; banana: angularity and days; sapota: brown streaks under skin.
- Immature produce is poor; over-mature produce spoils quickly.
- Maturity index depends on variety, season and market.

Short-answer questions

Q1. Define maturity.

Model answer: Maturity is the stage of development at which a fruit or vegetable is ready for harvest for a particular purpose.

Q2. How is maturity judged in sapota?

Model answer: By scratching the skin - brownish streaks under the skin indicate maturity (and by TSS).

Q3. Name two instruments used to judge maturity.

Model answer: Refractometer (for TSS) and penetrometer (for firmness).

Long-answer questions

Q1. Explain the methods of judging maturity in horticultural crops with examples.

Answer outline:

1. Definition of maturity and ripening.
2. Visual, physical, chemical, physiological and instrumental methods.
3. Maturity indices of important crops.
4. Importance of correct maturity.
5. Conclusion.

Q2. Write a note on maturity indices for mango, banana, tomato and onion.

Answer outline:

1. Mango - colour, TSS, specific gravity.
2. Banana - days, angularity.
3. Tomato - colour stages.
4. Onion - tops falling and neck drying.
5. Effect of maturity on storage and market.
6. Conclusion.

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Lecture 32. Post Harvest Operations - Harvesting, Grading and Packing in Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Describe good harvesting practices.
- Explain grading and its bases.
- Describe packing materials and methods.
- Explain the post-harvest chain.

Post-harvest operations are all the steps taken after harvest until the produce reaches the consumer. Horticultural produce is perishable, so losses can be very high (often 20-40% in fruits and vegetables) if these steps are not done well. Careful harvesting, grading, packing and cooling reduce losses and improve price.

32.1 Harvesting

- Harvest at the correct maturity for the intended market.
- Harvest in the cool part of the day (early morning or evening).
- Use the right tools and handle produce gently; avoid bruising.
- Keep harvested produce in shade; avoid delay before cooling.
- Use clean, ventilated containers for field collection.

32.2 Grading

Grading is the sorting of produce into uniform lots according to size, colour, shape, maturity, freedom from defects and quality. Grading gives uniformity, better price, easier packing and better market acceptance.

Basis	What is considered	Example
Size	Diameter, length, weight	Mango, citrus, onion graded by size
Colour	Stage of ripeness and colour	Tomato, apple
Shape and form	Uniformity of shape	Banana fingers, cucumber
Freedom from defects	Bruises, blemishes, disease, insect damage	All crops
Maturity	Uniform ripeness	Banana, mango
Varietal purity	True-to-type produce	Export consignments

Table 32.1 Bases of grading

32.3 Packing

- Packing protects produce during handling, transport and storage.
- Materials: wooden boxes, corrugated fibre-board boxes, plastic crates, bamboo baskets, gunny bags (for hardy crops).
- Cushioning material (paper, straw, foam) reduces bruising.
- Ventilation holes allow air movement and reduce heat build-up.
- Correct filling and stacking prevent crushing; avoid over-packing.
- Labelling gives variety, grade, weight, grower and destination.

32.4 Post-Harvest Chain

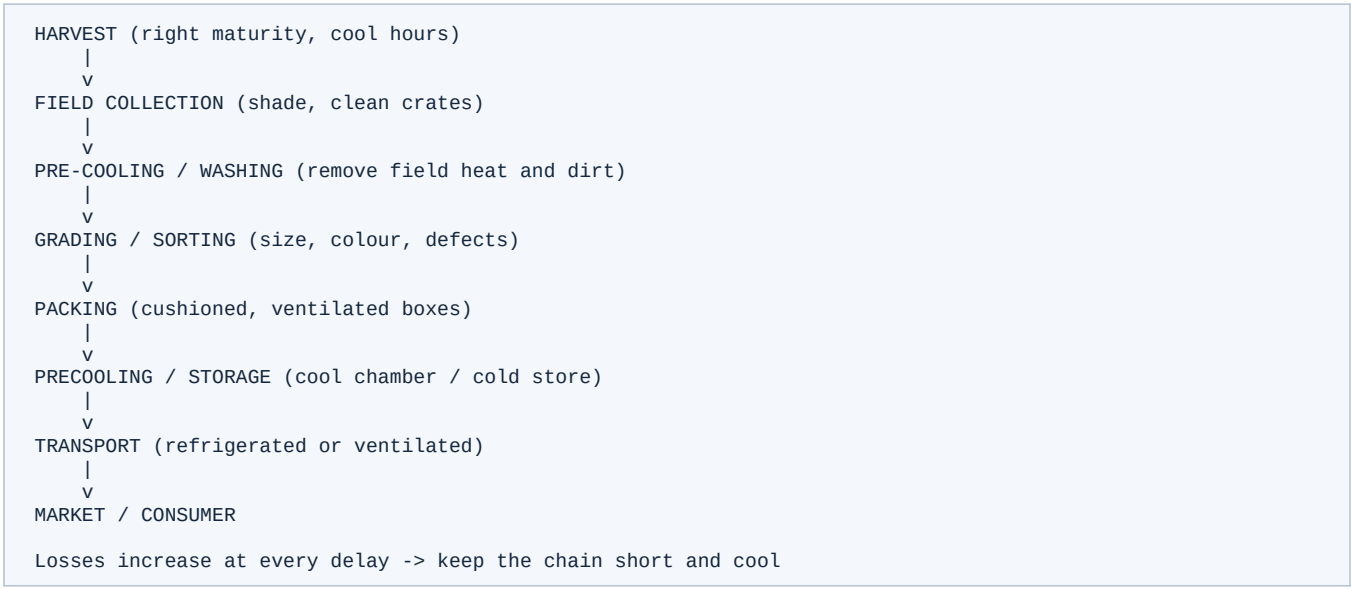


Fig. 32.1 Post-harvest handling chain for fruits and vegetables.

32.5 Value Addition and Processing

- Value addition raises income and reduces waste of surplus and low-grade produce.
- Products: fruit pulp and juice, jam, jelly, squash, pickles, dried and frozen vegetables, spice oils.
- Processing units need clean, graded raw material and good hygiene.
- Tamil Nadu has fruit and vegetable processing units and food parks that support growers.

32.6 Causes of Post-Harvest Losses and Their Reduction

- Causes: rough handling, delay, high temperature, poor packing, disease and pests.
- Reduce losses by gentle handling, shade, quick cooling, grading and proper packing.
- Maintain the cold chain from farm to market.
- Use clean, ventilated containers and avoid over-packing.
- Train harvest labour in careful handling.

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Key points to remember

- Post-harvest operations run from harvest to the consumer.
- Fruits and vegetables are perishable; losses can be 20-40%.
- Harvest at correct maturity and in cool hours; handle gently.
- Grading sorts produce by size, colour, shape, defects and maturity.
- Grading gives uniformity, better price and market acceptance.
- Packing protects produce with cushioning and ventilation.
- A short, cool post-harvest chain reduces losses.

Short-answer questions

Q1. Define grading.

Model answer: Grading is the sorting of produce into uniform lots according to size, colour, shape, maturity, freedom from defects and quality.

Q2. Why should harvesting be done in the cool part of the day?

Model answer: Because lower temperature reduces field heat and respiration, slowing deterioration and reducing post-harvest loss.

Q3. Name three packing materials used for horticultural produce.

Model answer: Corrugated fibre-board boxes, plastic crates and wooden boxes.

Long-answer questions

Q1. Describe the post-harvest operations in horticultural crops.

Answer outline:

1. Importance of post-harvest handling.
2. Harvesting practices.
3. Grading - bases and benefits.
4. Packing - materials and methods.
5. Post-harvest chain.
6. Conclusion.

Q2. Explain the importance of grading and packing in the marketing of fruits and vegetables.

Answer outline:

1. Meaning of grading and packing.
2. Bases of grading.
3. Benefits of grading - uniformity, price, market.
4. Packing materials and cushioning.
5. Reduction of losses.
6. Conclusion.

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Lecture 33. Transport and Storage Methods for Horticultural Crops

Learning Outcomes

- After studying this lecture the student should be able to:
- Describe the modes of transport and their suitability.
- Explain storage methods for horticultural produce.
- State the storage conditions of important crops.
- Explain the cold chain.

Transport and storage are the last links in the post-harvest chain. Because horticultural produce is perishable and often produced far from the market, good transport and storage are essential to reach consumers in good condition and to reduce losses and price fluctuations.

33.1 Transport of Horticultural Produce

Mode	Features	Suitability
Road (truck, tempo)	Flexible, door-to-door, most common	Short and medium distance; most fruits and vegetables
Rail	Bulk, cheaper for long distance	Long distance; bulk movement
Air	Fast, costly	High-value perishables - cut flowers, exotic fruits
Sea (reefer)	Cheap per unit, slow	Export of bulk perishables with refrigeration

Table 33.1 Modes of transport and their suitability

- Use refrigerated (reefer) transport for long distances and for highly perishable crops.
- For short distances, use well-ventilated vehicles and avoid the hottest part of the day.
- Handle gently and stack properly to avoid bruising and crushing.
- Pre-cool before loading; maintain the cold chain without breaks.
- Avoid mixing ethylene-producing fruits (banana, apple) with ethylene-sensitive vegetables (leafy greens).

33.2 Storage of Horticultural Produce

- **Refrigerated storage (cold storage):** Lowers temperature and slows respiration, ripening and spoilage; the main method for perishables.
- **Controlled atmosphere (CA) storage:** Low oxygen and high carbon dioxide with low temperature; extends storage life of apple, etc.
- **Modified atmosphere packaging (MAP):** A modified atmosphere created inside a package or film.
- **Zero-energy cool chamber:** A low-cost evaporative cool chamber for on-farm short storage.
- **Ventilated storage:** Cool, well-ventilated rooms for hardy crops (onion, potato, garlic).
- **Traditional storage:** Pits, clamps and heaps for potato, yam and similar crops in rural areas.

33.3 Storage Conditions for Important Crops

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Crop	Temperature	Relative humidity	Approximate storage life
Apple	0-4 deg C	90-95%	Several months (CA storage longer)
Mango	10-13 deg C	85-90%	2-4 weeks
Banana (green)	13-15 deg C	85-95%	2-4 weeks
Guava	5-10 deg C	85-90%	2-3 weeks
Citrus	3-10 deg C	85-90%	4-8 weeks
Grape	0-2 deg C	90-95%	1-3 months
Tomato (mature green)	10-13 deg C	85-90%	1-2 weeks
Onion (dry)	0-4 deg C (or cool dry)	65-70%	Several months
Potato	4-8 deg C	90-95%	Several months
Leafy vegetables	0-2 deg C	95%+	1-2 weeks

Table 33.2 Approximate storage conditions for important horticultural crops

Values are typical ranges; the best conditions vary with variety, maturity and intended storage period. Chilling injury occurs when tropical fruits such as banana and mango are stored below their safe temperature. Onion and garlic need low humidity and good ventilation, unlike most fruits.

33.4 Cold Chain and Cold Storage in Tamil Nadu

- A cold chain links pre-cooling, cold storage, refrigerated transport and cold retail.
- Tamil Nadu has cold storage capacity for fruits, vegetables and flowers, with more being added under MIDH and state schemes.
- Zero-energy cool chambers are promoted for on-farm short-term storage.
- Cold chain investment reduces post-harvest losses and stabilises prices for growers.

33.5 Factors Affecting Storage Life

- Crop and variety - some keep far longer than others.
- Maturity at harvest - over-mature produce stores poorly.
- Temperature and humidity - the most important storage factors.
- Freedom from injury and disease.
- The gas atmosphere around the produce (ethylene and carbon dioxide levels).
- Care and speed at every step from harvest to store.

EXAM FOCUS

Key points to remember

- Transport and storage complete the post-harvest chain.
- Road transport is most common; air is used for high-value perishables; sea reefer for bulk export.
- Refrigerated transport maintains the cold chain for long distances.
- Storage methods: cold storage, CA storage, MAP, zero-energy cool chamber, ventilated storage.
- Cold storage slows respiration, ripening and spoilage.
- Onion and garlic need low humidity; most fruits need high humidity.
- Tropical fruits suffer chilling injury if stored too cold (banana below about 13 deg C).

Short-answer questions

Q1. Which mode of transport is used for cut flowers and why?

Model answer: Air transport, because cut flowers are highly perishable and high in value, so speed is essential.

Q2. What is controlled atmosphere storage?

Model answer: A storage method using low temperature together with low oxygen and high carbon dioxide to extend the storage life of produce.

Q3. What is a zero-energy cool chamber?

Model answer: A low-cost evaporative cool chamber that keeps produce cool without electricity, used for on-farm short storage.

Long-answer questions

Q1. Explain the methods of transport and storage of horticultural crops.

Answer outline:

1. Importance of transport and storage.
2. Modes of transport and suitability.
3. Cold chain and pre-cooling.
4. Storage methods: cold storage, CA, MAP, zero-energy cool chamber.
5. Storage conditions of important crops.
6. Conclusion.

Q2. Write a note on cold storage and controlled atmosphere storage of fruits and vegetables.

Answer outline:

1. Need for cold storage.
2. Principle of refrigeration.
3. Controlled atmosphere - low O2, high CO2.
4. Crops suited to each.
5. Chilling injury and precautions.
6. Conclusion.

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Glossary of Horticultural Terms

Term	Meaning
Abscission	Natural separation of leaves, flowers or fruits from the plant
Apomixis	Formation of seed without fertilization, giving true-to-type progeny (for example, nucellar embryony in mango)
Arboriculture	Culture of trees and shrubs, especially woody ornamentals
Auxin	A class of growth regulator (for example, IAA, IBA, NAA) controlling rooting, growth and fruit set
Biennial	A plant that completes its life cycle in two seasons
Budding	A form of grafting using a single bud with a shield of bark
Callus	Undifferentiated tissue formed at a cut surface, through which grafts unite
Canopy management	Planned manipulation of the tree canopy for light, yield and ease of management
Cauliflory	Bearing of flowers and fruits directly on the trunk or thick branches (jackfruit)
Chilling injury	Damage to tropical produce when stored below its safe temperature
Clone	A group of genetically identical plants produced by vegetative propagation
Corm	A short swollen underground stem used in propagation (colocasia, gladiolus)
Cutting	A detached vegetative part made to form roots and shoots
Dormancy	A resting condition in which a viable seed or bud does not grow under favourable conditions
Ethephon	An ethylene-releasing growth regulator used to hasten ripening and change sex expression
Fertigation	Application of fertilizers through irrigation water, especially drip
Fruitfulness	The ability of a plant to flower and set fruit regularly
Gibberellin (GA3)	A growth regulator that promotes elongation, breaks dormancy and increases fruit size
Grading	Sorting produce into uniform lots by size, colour, quality and other characters
Grafting	Joining a scion to a rootstock so that they unite and grow as one plant
HDP	High density planting - planting trees closer than traditional spacing
Layering	Forming roots on a stem while it is still attached to the parent plant
Micropropagation	Propagation of plants by tissue culture
Mulching	Covering the soil surface with a material to conserve moisture and control weeds
Parthenocarpy	Development of fruit without fertilization, giving seedless fruit
Pollinizer	A variety planted to supply compatible pollen to the main variety
Pomology	The branch of horticulture dealing with fruit crops
Precocity	Early bearing habit of a plant
Pruning	Removal of plant parts to influence growth, yield and quality
Rootstock	The lower part of a grafted plant, providing the root system
Scion	The upper part of a grafted plant, providing the shoot and fruit
Self-incompatibility	Failure of a plant's own pollen to fertilise its ovules
Stratification	Keeping seed moist at low temperature to break dormancy

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Term	Meaning
Sucker	A shoot arising from the base or underground stem of a plant
TSS	Total soluble solids - a measure of sugars and dissolved solids, used as a maturity index
Training	Manipulation of a plant to develop a desired shape and framework
Unfruitfulness	Failure to set fruit, or setting of very little fruit, despite good growth

Common horticultural terms and their meanings

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Reference List and Suggested Reading

The following standard texts and course materials cover the topics of HOR 101 - Fundamentals of Horticulture. Students are advised to consult the current edition of each book and the latest TNAU / ICAR e-course material for up-to-date statistics.

- **Kumar, N.** - *Introduction to Horticulture*. Oxford & IBH Publishing Co., New Delhi. (Standard introductory text.)
- **Bose, T. K. and Mitra, S. K.** - *Fruits of India: Tropical and Subtropical*. Naya Prokash, Calcutta. (Detailed crop-wise reference.)
- **Bose, T. K., Mitra, S. K. and Sanyal, D.** - *Fruits: Tropical and Subtropical* (volumes).
- **Chattopadhyay, T. K.** - *A Textbook on Pomology* (volumes). Kalyani Publishers. (Fruit science.)
- **Singh, S. P.** - *Commercial Fruits*. Kalyani Publishers, Ludhiana.
- **Singh, N. P., Sankaran, M. and others** - ICAR e-course material on horticulture and pomology.
- **Chadha, K. L.** - *Handbook of Horticulture*. ICAR, New Delhi. (Ready reference.)
- **Tamil Nadu Agricultural University (TNAU)** - Course material and e-courses for HOR 101 - Fundamentals of Horticulture, and crop production guides and packages of practices, Coimbatore.
- **TNAU Agritech Portal** - Crop-wise horticulture technology and packages of practices.
- **ICAR** - *Handbook of Agriculture* and ICAR e-Krishi Shiksha course material on fruits, vegetables and floriculture.
- **National Horticulture Board (NHB)** - Horticultural statistics at a glance (latest edition) for area, production and export data.
- **Horticulture and Plantation Crops Department, Government of Tamil Nadu** - State schemes and statistics.
- **Mission for Integrated Development of Horticulture (MIDH)** - Scheme guidelines and technical material.
- **APEDA** - Export statistics and market information for horticultural products.

Note on figures: area, production, export and storage values change with year, variety and source. Where a precise value is not certain, a standard range has been given. For the examination and for field use, always quote the latest official statistics with the year and source.

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